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1  * SYNTAX for "Contributions of lifestyle risk factors to individual- and area-level socio-economic
disparities in active and total life expectancy among older adults: Evidence from the Precipitating
Events Project cohort study"
2
3  * STATA version: StataNow 19.5, MP-Parallel Edition
4
5  * STATA citation: StataCorp. 2025. Stata Statistical Software: Release 19. College Station, TX:
StataCorp LLC.
6
7  * Data access statement: Requests for access to data from the Precipitating Events Project for
meritorious analyses from qualified investigators should be directed to Thomas M. Gill
(thomas.gill@yale.edu). The syntax files are openly available at
https://github.com/OliviaMalkowski/ALE-disparities-lifestyle.git.
8
9  * Import master dataset
10 import sas using "P:\projects\PEP\Master\f2f_master_analysisvar306.sas7bdat"
11 * Count total number of participants and observations
12 unique StudyID
13 * 754 individuals, 5215 observations
14 * Assigns a number in ascending order to each row of observations
15 gen ascnr = _n
16 * Generate a variable that assigns the observation number (i.e., 1 for first data collection
timepoint, 2 for second data collection timepoint) to each row by participant ID
17 bysort StudyID(intdateF2F): gen obsnr = _n
18 * Generate a variable that assigns the number of total observations to each row of data for a given
participant
19 bysort StudyID: gen obscount = _N
20 * Check outcome code for comprehensive assessments that were refused
21 tab fout_fu if respondent==.
22 * Drop comprehensive assessments that were refused
23 drop if respondent==.
24 * Count total number of participants and observations
25 unique StudyID
26 * 754 individuals, 5168 observations
27 * Change working directory
28 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
29 * Save master dataset with a new name
30 save master.dta
31 clear
32
33 * Import baseline data
34 import sas using "P:\projects\PEP\Master\baseface.sas7bdat"
35 * Keep relevant variables
36 keep STUDYID fc5_pasebl age_b GDI1 INC female white edu_cont edu7cat age5cat BMI_bl highbp mi chf
stroke cancer diabetes hipfx otherfx arthrit parkinsn legamput liver lung_d ccsumbl1 ccsumbl2 diff4adl
diff7adl dep3adl diff3adl walk_dev hp_walk ds_walk hp_stair ds_stair hp_carry ds_carry hp_shop
ds_shop hp_clean ds_clean hp_meal ds_meal hp_meds ds_meds hp_bills ds_bills iadl_dep iadl_dis hp_mobil
ds_mobil bloc_cat blocksbl adl24_bl adl12ms badl14bl iadl10bl mobdisbl intdtebl mpase1w mpase2w
mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9cw mpasebl alc_bl
smoker_bl
37 * Rename variables to ensure consistency across intervals
38 rename * raw_*
39 * Generate a new variable called interval and assign the number 1 to each observation
40 gen interval = 1
41 * Rename raw_STUDYID variable to ensure consistency with master dataset
42 rename raw_STUDYID STUDYID
43 * Change working directory
44 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
45 * Save baseline dataset with a new name
46 save other_vrb1.dta
47 clear
48
49 * Import 18-month follow-up data

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50 import sas using "P:\projects\PEP\Master\fu18face.sas7bdat"
51 * Keep relevant variables
52 keep STUDYID fc5_pase18 age_18 female white edu_cont agecat18 BMI_18 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_18 mi_18 chf_18 strok_18 cancr_18 diab_18 hipfx_18 lungd_18 arth_18
ccsum18a ccsum18b ccsum18c d_4adl18 d_7adl18 d_3adl18 hp3adl18 walkd_18 hpwalk18 dswalk18 hpstir18
dsstir18 hpcary18 dscary18 hpshop18 dsshop18 hpclen18 dsclen18 hpmeal18 dsmeal18 hpmeds18 dsmeds18
hpbill18 dsbill18 iadlhp18 iadlds18 hp_mob18 ds_mob18 bloc_c18 blocks18 adl24_18 adl12ms badl1418
iadl1018 mobdis18 intdate18 mpase1w mpase2w mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw
mpase8cw mpase8dw mpase9bw mpase18 alc_18 smoker_18
53 * Rename variables to ensure consistency across intervals
54 rename fc5_pase18 fc5_paseb1
55 rename BMI_18 BMI_b1
56 rename agecat18 age5cat
57 rename age_18 age_b
58 rename ccsum18a ccsumb11
59 rename ccsum18c ccsumb12
60 rename Cancer cancer
61 rename d_4adl18 diff4adl
62 rename d_7adl18 diff7adl
63 rename d_3adl18 diff3adl
64 rename hp3adl18 dep3adl
65 rename walkd_18 walk_dev
66 rename hpwalk18 hp_walk
67 rename dswalk18 ds_walk
68 rename hpstir18 hp_stair
69 rename dsstir18 ds_stair
70 rename hpcary18 hp_carry
71 rename dscary18 ds_carry
72 rename hpshop18 hp_shop
73 rename dsshop18 ds_shop
74 rename hpclen18 hp_clean
75 rename dsclen18 ds_clean
76 rename hpmeal18 hp_meal
77 rename dsmeal18 ds_meal
78 rename hpmeds18 hp_meds
79 rename dsmeds18 ds_meds
80 rename hpbill18 hp_bills
81 rename dsbill18 ds_bills
82 rename iadlhp18 iadl_dep
83 rename iadlds18 iadl_dis
84 rename hp_mob18 hp_mobil
85 rename ds_mob18 ds_mobil
86 rename bloc_c18 bloc_cat
87 rename blocks18 blocksb1
88 rename adl24_18 adl24_b1
89 rename badl1418 badl14b1
90 rename iadl1018 iadl10b1
91 rename mobdis18 mobdisb1
92 rename mpase9bw mpase9cw
93 rename mpase18 mpaseb1
94 rename alc_18 alc_b1
95 rename smoker_18 smoker_b1
96 rename * raw_*
97 * Generate a new variable called interval and assign the number 2 to each observation
98 gen interval = 2
99 * Rename raw_STUDYID variable to ensure consistency with master dataset
100 rename raw_STUDYID STUDYID
101 * Change working directory
102 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
103 * Save 18-month follow-up dataset with a new name
104 save other_vrb2.dta
105 clear
106

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107 * Import 36-month follow-up data
108 import sas using "P:\projects\PEP\Master\fu36face.sas7bdat"
109 * Keep relevant variables
110 keep StudyID fc5_pase36 age_36 female white edu_cont agecat36 BMI_36 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_36 mi_36 chf_36 strok_36 cancr_36 diab_36 hipfx_36 lungd_36 arth_36
ccsum36a ccsum36b ccsum36c d_4adl36 d_7adl36 d_3adl36 hp3adl36 walkd_36 hpwalk36 dswalk36 hpstir36
dsstir36 hpcary36 dscary36 hpshop36 dsshop36 hpclen36 dsclen36 hpmeal36 dsmeal36 hpmeds36 dsmeds36
hpbill36 dsbill36 iadlhp36 iadlds36 hp_mob36 ds_mob36 bloc_c36 blocks36 adl24_36 adl12ms badl1436
iadl1036 mobdis36 intdate36 mpase1w mpase2w mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw
mpase8cw mpase8dw mpase9bw mpase36 smoker_36
111 * Rename variables to ensure consistency across intervals
112 rename fc5_pase36 fc5_paseb1
113 rename BMI_36 BMI_b1
114 rename agecat36 age5cat
115 rename age_36 age_b
116 rename ccsum36a ccsumb11
117 rename ccsum36c ccsumb12
118 rename Cancer cancer
119 rename d_4adl36 diff4adl
120 rename d_7adl36 diff7adl
121 rename d_3adl36 diff3adl
122 rename hp3adl36 dep3adl
123 rename walkd_36 walk_dev
124 rename hpwalk36 hp_walk
125 rename dswalk36 ds_walk
126 rename hpstir36 hp_stair
127 rename dsstir36 ds_stair
128 rename hpcary36 hp_carry
129 rename dscary36 ds_carry
130 rename hpshop36 hp_shop
131 rename dsshop36 ds_shop
132 rename hpclen36 hp_clean
133 rename dsclen36 ds_clean
134 rename hpmeal36 hp_meal
135 rename dsmeal36 ds_meal
136 rename hpmeds36 hp_meds
137 rename dsmeds36 ds_meds
138 rename hpbill36 hp_bills
139 rename dsbill36 ds_bills
140 rename iadlhp36 iadl_dep
141 rename iadlds36 iadl_dis
142 rename hp_mob36 hp_mobil
143 rename ds_mob36 ds_mobil
144 rename bloc_c36 bloc_cat
145 rename blocks36 blocksb1
146 rename adl24_36 adl24_b1
147 rename badl1436 badl14b1
148 rename iadl1036 iadl10b1
149 rename mobdis36 mobdisb1
150 rename mpase9bw mpase9cw
151 rename mpase36 mpaseb1
152 rename smoker_36 smoker_b1
153 rename * raw_*
154 * Generate a new variable called interval and assign the number 3 to each observation
155 gen interval = 3
156 * Rename raw_StudyID variable to ensure consistency with master dataset
157 rename raw_StudyID STUDYID
158 * Change working directory
159 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
160 * Save 36-month follow-up dataset with a new name
161 save other_vrb3.dta
162 clear
163

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164 * Import 54-month follow-up data
165 import sas using "P:\projects\PEP\Master\fu54face.sas7bdat"
166 * Keep relevant variables
167 keep StudyID fc5_pase54 age_54 female white edu_cont agecat54 BMI_54 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_54 mi_54 chf_54 strok_54 cancr_54 diab_54 hipfx_54 lungd_54 arth_54
ccsum54a ccsum54b ccsum54c d_4adl54 d_7adl54 d_3adl54 hp3adl54 walkd_54 hpwalk54 dswalk54 hpstir54
dsstir54 hpcary54 dscary54 hpshop54 dsshop54 hpclen54 dsclen54 hpmeal54 dsmeal54 hpmeds54 dsmeds54
hpbill54 dsbill54 iadlhp54 iadlds54 hp_mob54 ds_mob54 bloc_c54 blocks54 adl24_54 adl12ms badl1454
iadl1054 mobdis54 intdate54 mpase1w mpase2w mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw
mpase8cw mpase8dw mpase9bw mpase54 smoker_54
168 * Rename variables to ensure consistency across intervals
169 rename fc5_pase54 fc5_paseb1
170 rename BMI_54 BMI_b1
171 rename agecat54 age5cat
172 rename age_54 age_b
173 rename ccsum54a ccsumb11
174 rename ccsum54c ccsumb12
175 rename Cancer cancer
176 rename d_4adl54 diff4adl
177 rename d_7adl54 diff7adl
178 rename d_3adl54 diff3adl
179 rename hp3adl54 dep3adl
180 rename walkd_54 walk_dev
181 rename hpwalk54 hp_walk
182 rename dswalk54 ds_walk
183 rename hpstir54 hp_stair
184 rename dsstir54 ds_stair
185 rename hpcary54 hp_carry
186 rename dscary54 ds_carry
187 rename hpshop54 hp_shop
188 rename dsshop54 ds_shop
189 rename hpclen54 hp_clean
190 rename dsclen54 ds_clean
191 rename hpmeal54 hp_meal
192 rename dsmeal54 ds_meal
193 rename hpmeds54 hp_meds
194 rename dsmeds54 ds_meds
195 rename hpbill54 hp_bills
196 rename dsbill54 ds_bills
197 rename iadlhp54 iadl_dep
198 rename iadlds54 iadl_dis
199 rename hp_mob54 hp_mobil
200 rename ds_mob54 ds_mobil
201 rename bloc_c54 bloc_cat
202 rename blocks54 blocksb1
203 rename adl24_54 adl24_b1
204 rename badl1454 badl14b1
205 rename iadl1054 iadl10b1
206 rename mobdis54 mobdisb1
207 rename mpase9bw mpase9cw
208 rename mpase54 mpaseb1
209 rename smoker_54 smoker_b1
210 rename * raw_*
211 * Generate a new variable called interval and assign the number 4 to each observation
212 gen interval = 4
213 * Rename raw_StudyID variable to ensure consistency with master dataset
214 rename raw_StudyID STUDYID
215 * Change working directory
216 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
217 * Save 54-month follow-up dataset with a new name
218 save other_vrb4.dta
219 clear
220

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221 * Import 72-month follow-up data
222 import sas using "P:\projects\PEP\Master\fu72face.sas7bdat"
223 * Keep relevant variables
224 keep studyID fc5_pase72 age_72 female white edu_cont agecat72 BMI_72 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_72 mi_72 chf_72 strok_72 cancr_72 diab_72 hipfx_72 lungd_72 arth_72
ccsum72a ccsum72b ccsum72c d_4adl72 d_7adl72 d_3adl72 hp3adl72 walkd_72 hpwalk72 dswalk72 hpstir72
dsstir72 hpcary72 dscary72 hpshop72 dsshop72 hpclen72 dsclen72 hpmeal72 dsmeal72 hpmeds72 dsmeds72
hpbill72 dsbill72 iadlhp72 iadlds72 hp_mob72 ds_mob72 bloc_c72 blocks72 adl24_72 adl12ms badl1472
iadl1072 mobdis72 intdate72 mpase1w mpase2w mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw
mpase8cw mpase8dw mpase9bw mpase72 smoker_72
225 * Rename variables to ensure consistency across intervals
226 rename fc5_pase72 fc5_paseb1
227 rename BMI_72 BMI_b1
228 rename agecat72 age5cat
229 rename age_72 age_b
230 rename ccsum72a ccsumb11
231 rename ccsum72c ccsumb12
232 rename Cancer cancer
233 rename d_4adl72 diff4adl
234 rename d_7adl72 diff7adl
235 rename d_3adl72 diff3adl
236 rename hp3adl72 dep3adl
237 rename walkd_72 walk_dev
238 rename hpwalk72 hp_walk
239 rename dswalk72 ds_walk
240 rename hpstir72 hp_stair
241 rename dsstir72 ds_stair
242 rename hpcary72 hp_carry
243 rename dscary72 ds_carry
244 rename hpshop72 hp_shop
245 rename dsshop72 ds_shop
246 rename hpclen72 hp_clean
247 rename dsclen72 ds_clean
248 rename hpmeal72 hp_meal
249 rename dsmeal72 ds_meal
250 rename hpmeds72 hp_meds
251 rename dsmeds72 ds_meds
252 rename hpbill72 hp_bills
253 rename dsbill72 ds_bills
254 rename iadlhp72 iadl_dep
255 rename iadlds72 iadl_dis
256 rename hp_mob72 hp_mobil
257 rename ds_mob72 ds_mobil
258 rename bloc_c72 bloc_cat
259 rename blocks72 blocksb1
260 rename adl24_72 adl24_b1
261 rename badl1472 badl14b1
262 rename iadl1072 iadl10b1
263 rename mobdis72 mobdisb1
264 rename mpase9bw mpase9cw
265 rename mpase72 mpaseb1
266 rename smoker_72 smoker_b1
267 rename * raw_*
268 * Generate a new variable called interval and assign the number 5 to each observation
269 gen interval = 5
270 * Rename raw_studyID variable to ensure consistency with master dataset
271 rename raw_studyID STUDYID
272 * Change working directory
273 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
274 * Save 72-month follow-up dataset with a new name
275 save other_vrb5.dta
276 clear
277

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278 * Import 90-month follow-up data
279 import sas using "P:\projects\PEP\Master\fu90face.sas7bdat"
280 * Keep relevant variables
281 keep studyID fc5_pase90 age_90 female white edu_cont agecat90 BMI_90 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_90 mi_90 chf_90 strok_90 cancr_90 diab_90 hipfx_90 lungd_90 arth_90
ccsum90a ccsum90b ccsum90c d_4adl90 d_7adl90 d_3adl90 hp3adl90 walkd_90 hpwalk90 dswalk90 hpstir90
dsstir90 hpcary90 dscary90 hpshop90 dsshop90 hpclen90 dsclen90 hpmeal90 dsmeal90 hpmeds90 dsmeds90
hpbill90 dsbill90 iadlhp90 iadlds90 hp_mob90 ds_mob90 bloc_c90 blocks90 adl24_90 adl12ms badl1490
iadl1090 mobdis90 intdate90 mpase1w mpase2w mpase3w mpase4w mpase5w mpase6w mpase7w mpase8aw mpase8bw
mpase8cw mpase8dw mpase9bw mpase90 smoker_90
282 * Rename variables to ensure consistency across intervals
283 rename fc5_pase90 fc5_paseb1
284 rename BMI_90 BMI_b1
285 rename agecat90 age5cat
286 rename age_90 age_b
287 rename ccsum90a ccsumb11
288 rename ccsum90c ccsumb12
289 rename Cancer cancer
290 rename d_4adl90 diff4adl
291 rename d_7adl90 diff7adl
292 rename d_3adl90 diff3adl
293 rename hp3adl90 dep3adl
294 rename walkd_90 walk_dev
295 rename hpwalk90 hp_walk
296 rename dswalk90 ds_walk
297 rename hpstir90 hp_stair
298 rename dsstir90 ds_stair
299 rename hpcary90 hp_carry
300 rename dscary90 ds_carry
301 rename hpshop90 hp_shop
302 rename dsshop90 ds_shop
303 rename hpclen90 hp_clean
304 rename dsclen90 ds_clean
305 rename hpmeal90 hp_meal
306 rename dsmeal90 ds_meal
307 rename hpmeds90 hp_meds
308 rename dsmeds90 ds_meds
309 rename hpbill90 hp_bills
310 rename dsbill90 ds_bills
311 rename iadlhp90 iadl_dep
312 rename iadlds90 iadl_dis
313 rename hp_mob90 hp_mobil
314 rename ds_mob90 ds_mobil
315 rename bloc_c90 bloc_cat
316 rename blocks90 blocksb1
317 rename adl24_90 adl24_b1
318 rename badl1490 badl14b1
319 rename iadl1090 iadl10b1
320 rename mobdis90 mobdisb1
321 rename mpase9bw mpase9cw
322 rename mpase90 mpaseb1
323 rename smoker_90 smoker_b1
324 rename * raw_*
325 * Generate a new variable called interval and assign the number 6 to each observation
326 gen interval = 6
327 * Rename raw_studyID variable to ensure consistency with master dataset
328 rename raw_studyID STUDYID
329 * Change working directory
330 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
331 * Save 90-month follow-up dataset with a new name
332 save other_vrb6.dta
333 clear
334

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335 * Import 108-month follow-up data
336 import sas using "P:\projects\PEP\Master\fu108face.sas7bdat"
337 * Keep relevant variables
338 keep studyID fc5_pase108 age_108 female white edu_cont agecat108 BMI_108 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_108 mi_108 chf_108 strok_108 cancr_108 diab_108 hipfx_108 lungd_108
arth_108 ccsum108a ccsum108b ccsum108c d_4adl108 d_7adl108 d_3adl108 hp3adl108 walkd_108 hpwalk108
dswalk108 hpstir108 dsstir108 hpcary108 dscary108 hpshop108 dsshop108 hpclen108 dsclen108 hpmeal108
dsmeal108 hpmeds108 dsmeds108 hpbill108 dsbill108 iadlhp108 iadlds108 hp_mob108 ds_mob108 bloc_c108
blocks108 adl24_108 adl12ms badl14108 iadl10108 mobdis108 intdate108 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase108 smoker_108
339 * Rename variables to ensure consistency across intervals
340 rename fc5_pase108 fc5_pasebl
341 rename BMI_108 BMI_bl
342 rename agecat108 age5cat
343 rename age_108 age_b
344 rename ccsum108a ccsumbl1
345 rename ccsum108c ccsumbl2
346 rename Cancer cancer
347 rename d_4adl108 diff4adl
348 rename d_7adl108 diff7adl
349 rename d_3adl108 diff3adl
350 rename hp3adl108 dep3adl
351 rename walkd_108 walk_dev
352 rename hpwalk108 hp_walk
353 rename dswalk108 ds_walk
354 rename hpstir108 hp_stair
355 rename dsstir108 ds_stair
356 rename hpcary108 hp_carry
357 rename dscary108 ds_carry
358 rename hpshop108 hp_shop
359 rename dsshop108 ds_shop
360 rename hpclen108 hp_clean
361 rename dsclen108 ds_clean
362 rename hpmeal108 hp_meal
363 rename dsmeal108 ds_meal
364 rename hpmeds108 hp_meds
365 rename dsmeds108 ds_meds
366 rename hpbill108 hp_bills
367 rename dsbill108 ds_bills
368 rename iadlhp108 iadl_dep
369 rename iadlds108 iadl_dis
370 rename hp_mob108 hp_mobil
371 rename ds_mob108 ds_mobil
372 rename bloc_c108 bloc_cat
373 rename blocks108 blocksbl
374 rename adl24_108 adl24_bl
375 rename badl14108 badl14bl
376 rename iadl10108 iadl10bl
377 rename mobdis108 mobdisbl
378 rename mpase9bw mpase9cw
379 rename mpase108 mpasebl
380 rename smoker_108 smoker_bl
381 rename * raw_*
382 * Generate a new variable called interval and assign the number 7 to each observation
383 gen interval = 7
384 * Rename raw_studyID variable to ensure consistency with master dataset
385 rename raw_studyID STUDYID
386 * Change working directory
387 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
388 * Save 108-month follow-up dataset with a new name
389 save other_vrb7.dta
390 clear
391

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392 * Import 144-month follow-up data
393 import sas using "P:\projects\PEP\Master\fu144face.sas7bdat"
394 * Keep relevant variables
395 keep studyID fc5_pase144 age_144 female white edu_cont agecat144 BMI_144 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_144 mi_144 chf_144 strok_144 cancr_144 diab_144 hipfx_144 lungd_144
arth_144 ccsum144a ccsum144b ccsum144c d_4adl144 d_7adl144 d_3adl144 hp3adl144 walkd_144 hpwalk144
dswalk144 hpstir144 dsstir144 hpcary144 dscary144 hpshop144 dsshop144 hpclen144 dsclen144 hpmeal144
dsmeal144 hpmeds144 dsmeds144 hpbill144 dsbill144 iadlhp144 iadlds144 hp_mob144 ds_mob144 bloc_c144
blocks144 adl24_144 adl12ms badl14144 iadl10144 mobdis144 intdate144 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase144 smoker_144
396 * Rename variables to ensure consistency across intervals
397 rename fc5_pase144 fc5_pasebl
398 rename BMI_144 BMI_bl
399 rename agecat144 age5cat
400 rename age_144 age_b
401 rename ccsum144a ccsumbl1
402 rename ccsum144c ccsumbl2
403 rename Cancer cancer
404 rename d_4adl144 diff4adl
405 rename d_7adl144 diff7adl
406 rename d_3adl144 diff3adl
407 rename hp3adl144 dep3adl
408 rename walkd_144 walk_dev
409 rename hpwalk144 hp_walk
410 rename dswalk144 ds_walk
411 rename hpstir144 hp_stair
412 rename dsstir144 ds_stair
413 rename hpcary144 hp_carry
414 rename dscary144 ds_carry
415 rename hpshop144 hp_shop
416 rename dsshop144 ds_shop
417 rename hpclen144 hp_clean
418 rename dsclen144 ds_clean
419 rename hpmeal144 hp_meal
420 rename dsmeal144 ds_meal
421 rename hpmeds144 hp_meds
422 rename dsmeds144 ds_meds
423 rename hpbill144 hp_bills
424 rename dsbill144 ds_bills
425 rename iadlhp144 iadl_dep
426 rename iadlds144 iadl_dis
427 rename hp_mob144 hp_mobil
428 rename ds_mob144 ds_mobil
429 rename bloc_c144 bloc_cat
430 rename blocks144 blocksbl
431 rename adl24_144 adl24_bl
432 rename badl14144 badl14bl
433 rename iadl10144 iadl10bl
434 rename mobdis144 mobdisbl
435 rename mpase9bw mpase9cw
436 rename mpase144 mpasebl
437 rename smoker_144 smoker_bl
438 rename * raw_*
439 * Generate a new variable called interval and assign the number 9 to each observation
440 gen interval = 9
441 * Rename raw_studyID variable to ensure consistency with master dataset
442 rename raw_studyID STUDYID
443 * Change working directory
444 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
445 * Save 144-month follow-up dataset with a new name
446 save other_vrb9.dta
447 clear
448

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449 * Import 162-month follow-up data
450 import sas using "P:\projects\PEP\Master\fu162face.sas7bdat"
451 * Keep relevant variables
452 keep studyID fc5_pase162 age_162 female white edu_cont agecat162 BMI_162 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_162 mi_162 chf_162 strok_162 cancr_162 diab_162 hipfx_162 lungd_162
arth_162 ccsum162a ccsum162b ccsum162c d_4adl162 d_7adl162 d_3adl162 hp3adl162 walkd_162 hpwalk162
dswalk162 hpstir162 dsstir162 hpcary162 dscary162 hpshop162 dsshop162 hpclen162 dsclen162 hpmeal162
dsmeal162 hpmeds162 dsmeds162 hpbill162 dsbill162 iadlhp162 iadlds162 hp_mob162 ds_mob162 bloc_c162
blocks162 adl24_162 adl12ms badl14162 iadl10162 mobdis162 intdate162 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase162 smoker_162
453 * Rename variables to ensure consistency across intervals
454 rename fc5_pase162 fc5_pasebl
455 rename BMI_162 BMI_bl
456 rename agecat162 age5cat
457 rename age_162 age_b
458 rename ccsum162a ccsumbl1
459 rename ccsum162c ccsumbl2
460 rename Cancer cancer
461 rename d_4adl162 diff4adl
462 rename d_7adl162 diff7adl
463 rename d_3adl162 diff3adl
464 rename hp3adl162 dep3adl
465 rename walkd_162 walk_dev
466 rename hpwalk162 hp_walk
467 rename dswalk162 ds_walk
468 rename hpstir162 hp_stair
469 rename dsstir162 ds_stair
470 rename hpcary162 hp_carry
471 rename dscary162 ds_carry
472 rename hpshop162 hp_shop
473 rename dsshop162 ds_shop
474 rename hpclen162 hp_clean
475 rename dsclen162 ds_clean
476 rename hpmeal162 hp_meal
477 rename dsmeal162 ds_meal
478 rename hpmeds162 hp_meds
479 rename dsmeds162 ds_meds
480 rename hpbill162 hp_bills
481 rename dsbill162 ds_bills
482 rename iadlhp162 iadl_dep
483 rename iadlds162 iadl_dis
484 rename hp_mob162 hp_mobil
485 rename ds_mob162 ds_mobil
486 rename bloc_c162 bloc_cat
487 rename blocks162 blocksbl
488 rename adl24_162 adl24_bl
489 rename badl14162 badl14bl
490 rename iadl10162 iadl10bl
491 rename mobdis162 mobdisbl
492 rename mpase9bw mpase9cw
493 rename mpase162 mpasebl
494 rename smoker_162 smoker_bl
495 rename * raw_*
496 * Generate a new variable called interval and assign the number 10 to each observation
497 gen interval = 10
498 * Rename raw_studyID variable to ensure consistency with master dataset
499 rename raw_studyID STUDYID
500 * Change working directory
501 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
502 * Save 162-month follow-up dataset with a new name
503 save other_vrb10.dta
504 clear
505

```

```

506 * Import 180-month follow-up data
507 import sas using "P:\projects\PEP\Master\fu180face.sas7bdat"
508 * Keep relevant variables
509 keep studyID fc5_pase180 age_180 female white edu_cont agecat180 BMI_180 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_180 mi_180 chf_180 strok_180 cancr_180 diab_180 hipfx_180 lungd_180
arth_180 ccsum180a ccsum180b ccsum180c d_4adl180 d_7adl180 d_3adl180 hp3adl180 walkd_180 hpwalk180
dswalk180 hpstir180 dsstir180 hpcary180 dscary180 hpshop180 dsshop180 hpclen180 dsclen180 hpmeal180
dsmeal180 hpmeds180 dsmeds180 hpbill180 dsbill180 iadlhp180 iadlds180 hp_mob180 ds_mob180 bloc_c180
blocks180 adl24_180 adl12ms badl14180 iadl10180 mobdis180 intdate180 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase180 smoker_180

510 * Rename variables to ensure consistency across intervals
511 rename fc5_pase180 fc5_paseb1
512 rename BMI_180 BMI_b1
513 rename agecat180 age5cat
514 rename age_180 age_b
515 rename ccsum180a ccsumb11
516 rename ccsum180c ccsumb12
517 rename Cancer cancer
518 rename d_4adl180 diff4adl
519 rename d_7adl180 diff7adl
520 rename d_3adl180 diff3adl
521 rename hp3adl180 dep3adl
522 rename walkd_180 walk_dev
523 rename hpwalk180 hp_walk
524 rename dswalk180 ds_walk
525 rename hpstir180 hp_stair
526 rename dsstir180 ds_stair
527 rename hpcary180 hp_carry
528 rename dscary180 ds_carry
529 rename hpshop180 hp_shop
530 rename dsshop180 ds_shop
531 rename hpclen180 hp_clean
532 rename dsclen180 ds_clean
533 rename hpmeal180 hp_meal
534 rename dsmeal180 ds_meal
535 rename hpmeds180 hp_meds
536 rename dsmeds180 ds_meds
537 rename hpbill180 hp_bills
538 rename dsbill180 ds_bills
539 rename iadlhp180 iadl_dep
540 rename iadlds180 iadl_dis
541 rename hp_mob180 hp_mobil
542 rename ds_mob180 ds_mobil
543 rename bloc_c180 bloc_cat
544 rename blocks180 blocksb1
545 rename adl24_180 adl24_b1
546 rename badl14180 badl14b1
547 rename iadl10180 iadl10b1
548 rename mobdis180 mobdisb1
549 rename mpase9bw mpase9cw
550 rename mpase180 mpaseb1
551 rename smoker_180 smoker_b1
552 rename * raw_*
553 * Generate a new variable called interval and assign the number 11 to each observation
554 gen interval = 11
555 * Rename raw_studyID variable to ensure consistency with master dataset
556 rename raw_studyID STUDYID
557 * Change working directory
558 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
559 * Save 180-month follow-up dataset with a new name
560 save other_vrb11.dta
561 clear
562

```

```

563 * Import 198-month follow-up data
564 import sas using "P:\projects\PEP\Master\fu198face.sas7bdat"
565 * Keep relevant variables
566 keep studyID fc5_pase198 age_198 female white edu_cont agecat198 BMI_198 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_198 mi_198 chf_198 strok_198 cancr_198 diab_198 hipfx_198 lungd_198
arth_198 ccsum198a ccsum198b ccsum198c d_4adl198 d_7adl198 d_3adl198 hp3adl198 walkd_198 hpwalk198
dswalk198 hpstir198 dsstir198 hpcary198 dscary198 hpshop198 dsshop198 hpclen198 dsclen198 hpmeal198
dsmeal198 hpmeds198 dsmeds198 hpbill198 dsbill198 iadlhp198 iadlds198 hp_mob198 ds_mob198 bloc_c198
blocks198 adl24_198 adl12ms badl14198 iadl10198 mobdis198 intdate198 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase198 smoker_198

567 * Rename variables to ensure consistency across intervals
568 rename fc5_pase198 fc5_paseb1
569 rename BMI_198 BMI_b1
570 rename agecat198 age5cat
571 rename age_198 age_b
572 rename ccsum198a ccsumb11
573 rename ccsum198c ccsumb12
574 rename Cancer cancer
575 rename d_4adl198 diff4adl
576 rename d_7adl198 diff7adl
577 rename d_3adl198 diff3adl
578 rename hp3adl198 dep3adl
579 rename walkd_198 walk_dev
580 rename hpwalk198 hp_walk
581 rename dswalk198 ds_walk
582 rename hpstir198 hp_stair
583 rename dsstir198 ds_stair
584 rename hpcary198 hp_carry
585 rename dscary198 ds_carry
586 rename hpshop198 hp_shop
587 rename dsshop198 ds_shop
588 rename hpclen198 hp_clean
589 rename dsclen198 ds_clean
590 rename hpmeal198 hp_meal
591 rename dsmeal198 ds_meal
592 rename hpmeds198 hp_meds
593 rename dsmeds198 ds_meds
594 rename hpbill198 hp_bills
595 rename dsbill198 ds_bills
596 rename iadlhp198 iadl_dep
597 rename iadlds198 iadl_dis
598 rename hp_mob198 hp_mobil
599 rename ds_mob198 ds_mobil
600 rename bloc_c198 bloc_cat
601 rename blocks198 blocksb1
602 rename adl24_198 adl24_b1
603 rename badl14198 badl14b1
604 rename iadl10198 iadl10b1
605 rename mobdis198 mobdisb1
606 rename mpase9bw mpase9cw
607 rename mpase198 mpaseb1
608 rename smoker_198 smoker_b1
609 rename * raw_*
610 * Generate a new variable called interval and assign the number 12 to each observation
611 gen interval = 12
612 * Rename raw_studyID variable to ensure consistency with master dataset
613 rename raw_studyID STUDYID
614 * Change working directory
615 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
616 * Save 198-month follow-up dataset with a new name
617 save other_vrb12.dta
618 clear
619

```

```

620 * Import 216-month follow-up data
621 import sas using "P:\projects\PEP\Master\fu216face.sas7bdat"
622 * Keep relevant variables
623 keep studyID fc5_pase216 age_216 female white edu_cont agecat216 BMI_216 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_216 mi_216 chf_216 strok_216 cancr_216 diab_216 hipfx_216 lungd_216
arth_216 ccsum216a ccsum216b ccsum216c d_4adl216 d_7adl216 d_3adl216 hp3adl216 walkd_216 hpwalk216
dswalk216 hpstir216 dsstir216 hpcary216 dscary216 hpshop216 dsshop216 hpclen216 dsclen216 hpmeal216
dsmeal216 hpmeds216 dsmeds216 hpbill216 dsbill216 iadlhp216 iadlds216 hp_mob216 ds_mob216 bloc_c216
blocks216 adl24_216 adl12ms badl14216 iadl10216 mobdis216 intdate216 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase216 smoker_216

624 * Rename variables to ensure consistency across intervals
625 rename fc5_pase216 fc5_paseb1
626 rename BMI_216 BMI_b1
627 rename agecat216 age5cat
628 rename age_216 age_b
629 rename ccsum216a ccsumb11
630 rename ccsum216c ccsumb12
631 rename Cancer cancer
632 rename d_4adl216 diff4adl
633 rename d_7adl216 diff7adl
634 rename d_3adl216 diff3adl
635 rename hp3adl216 dep3adl
636 rename walkd_216 walk_dev
637 rename hpwalk216 hp_walk
638 rename dswalk216 ds_walk
639 rename hpstir216 hp_stair
640 rename dsstir216 ds_stair
641 rename hpcary216 hp_carry
642 rename dscary216 ds_carry
643 rename hpshop216 hp_shop
644 rename dsshop216 ds_shop
645 rename hpclen216 hp_clean
646 rename dsclen216 ds_clean
647 rename hpmeal216 hp_meal
648 rename dsmeal216 ds_meal
649 rename hpmeds216 hp_meds
650 rename dsmeds216 ds_meds
651 rename hpbill216 hp_bills
652 rename dsbill216 ds_bills
653 rename iadlhp216 iadl_dep
654 rename iadlds216 iadl_dis
655 rename hp_mob216 hp_mobil
656 rename ds_mob216 ds_mobil
657 rename bloc_c216 bloc_cat
658 rename blocks216 blocksb1
659 rename adl24_216 adl24_b1
660 rename badl14216 badl14b1
661 rename iadl10216 iadl10b1
662 rename mobdis216 mobdisb1
663 rename mpase9bw mpase9cw
664 rename mpase216 mpaseb1
665 rename smoker_216 smoker_b1
666 rename * raw_*
667 * Generate a new variable called interval and assign the number 13 to each observation
668 gen interval = 13
669 * Rename raw_studyID variable to ensure consistency with master dataset
670 rename raw_studyID STUDYID
671 * Change working directory
672 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
673 * Save 216-month follow-up dataset with a new name
674 save other_vrb13.dta
675 clear
676

```

```

677 * Import 234-month follow-up data
678 import sas using "P:\projects\PEP\Master\fu234face.sas7bdat"
679 * Keep relevant variables
680 keep studyID fc5_pase234 age_234 female white edu_cont agecat234 BMI_234 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_234 mi_234 chf_234 strok_234 cancr_234 diab_234 hipfx_234 lungd_234
arth_234 ccsum234a ccsum234b ccsum234c d_4adl234 d_7adl234 d_3adl234 hp3adl234 walkd_234 hpwalk234
dswalk234 hpstir234 dsstir234 hpcary234 dscary234 hpshop234 dsshop234 hpclen234 dsclen234 hpmeal234
dsmeal234 hpmeds234 dsmeds234 hpbill234 dsbill234 iadlhp234 iadlds234 hp_mob234 ds_mob234 bloc_c234
blocks234 adl24_234 adl12ms badl14234 iadl10234 mobdis234 intdate234 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase234 smoker_234
681 * Rename variables to ensure consistency across intervals
682 rename fc5_pase234 fc5_paseb1
683 rename BMI_234 BMI_b1
684 rename agecat234 age5cat
685 rename age_234 age_b
686 rename ccsum234a ccsumb11
687 rename ccsum234c ccsumb12
688 rename Cancer cancer
689 rename d_4adl234 diff4adl
690 rename d_7adl234 diff7adl
691 rename d_3adl234 diff3adl
692 rename hp3adl234 dep3adl
693 rename walkd_234 walk_dev
694 rename hpwalk234 hp_walk
695 rename dswalk234 ds_walk
696 rename hpstir234 hp_stair
697 rename dsstir234 ds_stair
698 rename hpcary234 hp_carry
699 rename dscary234 ds_carry
700 rename hpshop234 hp_shop
701 rename dsshop234 ds_shop
702 rename hpclen234 hp_clean
703 rename dsclen234 ds_clean
704 rename hpmeal234 hp_meal
705 rename dsmeal234 ds_meal
706 rename hpmeds234 hp_meds
707 rename dsmeds234 ds_meds
708 rename hpbill234 hp_bills
709 rename dsbill234 ds_bills
710 rename iadlhp234 iadl_dep
711 rename iadlds234 iadl_dis
712 rename hp_mob234 hp_mobil
713 rename ds_mob234 ds_mobil
714 rename bloc_c234 bloc_cat
715 rename blocks234 blocksb1
716 rename adl24_234 adl24_b1
717 rename badl14234 badl14b1
718 rename iadl10234 iadl10b1
719 rename mobdis234 mobdisb1
720 rename mpase9bw mpase9cw
721 rename mpase234 mpaseb1
722 rename smoker_234 smoker_b1
723 rename * raw_*
724 * Generate a new variable called interval and assign the number 14 to each observation
725 gen interval = 14
726 * Rename raw_studyID variable to ensure consistency with master dataset
727 rename raw_studyID STUDYID
728 * Change working directory
729 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
730 * Save 234-month follow-up dataset with a new name
731 save other_vrb14.dta
732 clear
733

```

```

734 * Import 252-month follow-up data
735 import sas using "P:\projects\PEP\Master\fu252face.sas7bdat"
736 * Keep relevant variables
737 keep studyID fc5_pase252 age_252 female white edu_cont agecat252 BMI_252 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_252 mi_252 chf_252 strok_252 cancr_252 diab_252 hipfx_252 lungd_252
arth_252 ccsum252a ccsum252b ccsum252c d_4adl252 d_7adl252 d_3adl252 hp3adl252 walkd_252 hpwalk252
dswalk252 hpstir252 dsstir252 hpcary252 dscary252 hpshop252 dsshop252 hpclen252 dsclen252 hpmeal252
dsmeal252 hpmeds252 dsmeds252 hpbill252 dsbill252 iadlhp252 iadlds252 hp_mob252 ds_mob252 bloc_c252
blocks252 adl24_252 adl12ms badl14252 iadl10252 mobdis252 intdate252 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase252 smoker_252

738 * Rename variables to ensure consistency across intervals
739 rename fc5_pase252 fc5_paseb1
740 rename BMI_252 BMI_b1
741 rename agecat252 age5cat
742 rename age_252 age_b
743 rename ccsum252a ccsumb11
744 rename ccsum252c ccsumb12
745 rename Cancer cancer
746 rename d_4adl252 diff4adl
747 rename d_7adl252 diff7adl
748 rename d_3adl252 diff3adl
749 rename hp3adl252 dep3adl
750 rename walkd_252 walk_dev
751 rename hpwalk252 hp_walk
752 rename dswalk252 ds_walk
753 rename hpstir252 hp_stair
754 rename dsstir252 ds_stair
755 rename hpcary252 hp_carry
756 rename dscary252 ds_carry
757 rename hpshop252 hp_shop
758 rename dsshop252 ds_shop
759 rename hpclen252 hp_clean
760 rename dsclen252 ds_clean
761 rename hpmeal252 hp_meal
762 rename dsmeal252 ds_meal
763 rename hpmeds252 hp_meds
764 rename dsmeds252 ds_meds
765 rename hpbill252 hp_bills
766 rename dsbill252 ds_bills
767 rename iadlhp252 iadl_dep
768 rename iadlds252 iadl_dis
769 rename hp_mob252 hp_mobil
770 rename ds_mob252 ds_mobil
771 rename bloc_c252 bloc_cat
772 rename blocks252 blocksb1
773 rename adl24_252 adl24_b1
774 rename badl14252 badl14b1
775 rename iadl10252 iadl10b1
776 rename mobdis252 mobdisb1
777 rename mpase9bw mpase9cw
778 rename mpase252 mpaseb1
779 rename smoker_252 smoker_b1
780 rename * raw_*
781 * Generate a new variable called interval and assign the number 15 to each observation
782 gen interval = 15
783 * Rename raw_studyID variable to ensure consistency with master dataset
784 rename raw_studyID STUDYID
785 * Change working directory
786 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
787 * Save 252-month follow-up dataset with a new name
788 save other_vrb15.dta
789 clear
790

```

```

791 * Import 270-month follow-up data
792 import sas using "P:\projects\PEP\Master\fu270face.sas7bdat"
793 * Keep relevant variables
794 keep studyID fc5_pase270 age_270 female white edu_cont agecat270 BMI_270 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_270 mi_270 chf_270 strok_270 cancr_270 diab_270 hipfx_270 lungd_270
arth_270 ccsum270a ccsum270b ccsum270c d_4adl270 d_7adl270 d_3adl270 hp3adl270 walkd_270 hpwalk270
dswalk270 hpstir270 dsstir270 hpcary270 dscary270 hpshop270 dsshop270 hpclen270 dsclen270 hpmeal270
dsmeal270 hpmeds270 dsmeds270 hpbill270 dsbill270 iadlhp270 iadlds270 hp_mob270 ds_mob270 bloc_c270
blocks270 adl24_270 adl12ms badl14270 iadl10270 mobdis270 intdate270 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase270 smoker_270

795 * Rename variables to ensure consistency across intervals
796 rename fc5_pase270 fc5_paseb1
797 rename BMI_270 BMI_b1
798 rename agecat270 age5cat
799 rename age_270 age_b
800 rename ccsum270a ccsumb11
801 rename ccsum270c ccsumb12
802 rename Cancer cancer
803 rename d_4adl270 diff4adl
804 rename d_7adl270 diff7adl
805 rename d_3adl270 diff3adl
806 rename hp3adl270 dep3adl
807 rename walkd_270 walk_dev
808 rename hpwalk270 hp_walk
809 rename dswalk270 ds_walk
810 rename hpstir270 hp_stair
811 rename dsstir270 ds_stair
812 rename hpcary270 hp_carry
813 rename dscary270 ds_carry
814 rename hpshop270 hp_shop
815 rename dsshop270 ds_shop
816 rename hpclen270 hp_clean
817 rename dsclen270 ds_clean
818 rename hpmeal270 hp_meal
819 rename dsmeal270 ds_meal
820 rename hpmeds270 hp_meds
821 rename dsmeds270 ds_meds
822 rename hpbill270 hp_bills
823 rename dsbill270 ds_bills
824 rename iadlhp270 iadl_dep
825 rename iadlds270 iadl_dis
826 rename hp_mob270 hp_mobil
827 rename ds_mob270 ds_mobil
828 rename bloc_c270 bloc_cat
829 rename blocks270 blocksb1
830 rename adl24_270 adl24_b1
831 rename badl14270 badl14b1
832 rename iadl10270 iadl10b1
833 rename mobdis270 mobdisb1
834 rename mpase9bw mpase9cw
835 rename mpase270 mpaseb1
836 rename smoker_270 smoker_b1
837 rename * raw_*
838 * Generate a new variable called interval and assign the number 16 to each observation
839 gen interval = 16
840 * Rename raw_studyID variable to ensure consistency with master dataset
841 rename raw_studyID STUDYID
842 * Change working directory
843 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
844 * Save 270-month follow-up dataset with a new name
845 save other_vrb16.dta
846 clear
847

```

```

848 * Import 288-month follow-up data
849 import sas using "P:\projects\PEP\Master\fu288face.sas7bdat"
850 * Keep relevant variables
851 keep studyID fc5_pase288 age_288 female white edu_cont agecat288 BMI_288 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_288 mi_288 chf_288 strok_288 cancr_288 diab_288 hipfx_288 lungd_288
arth_288 ccsum288a ccsum288b ccsum288c d_4adl288 d_7adl288 d_3adl288 hp3adl288 walkd_288 hpwalk288
dswalk288 hpstir288 dsstir288 hpcary288 dscary288 hpshop288 dsshop288 hpclen288 dsclen288 hpmeal288
dsmeal288 hpmeds288 dsmeds288 hpbill288 dsbill288 iadlhp288 iadlds288 hp_mob288 ds_mob288 bloc_c288
blocks288 adl24_288 adl12ms badl14288 iadl10288 mobdis288 intdate288 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase288 smoker_288
852 * Rename variables to ensure consistency across intervals
853 rename fc5_pase288 fc5_paseb1
854 rename BMI_288 BMI_b1
855 rename agecat288 age5cat
856 rename age_288 age_b
857 rename ccsum288a ccsumb11
858 rename ccsum288c ccsumb12
859 rename Cancer cancer
860 rename d_4adl288 diff4adl
861 rename d_7adl288 diff7adl
862 rename d_3adl288 diff3adl
863 rename hp3adl288 dep3adl
864 rename walkd_288 walk_dev
865 rename hpwalk288 hp_walk
866 rename dswalk288 ds_walk
867 rename hpstir288 hp_stair
868 rename dsstir288 ds_stair
869 rename hpcary288 hp_carry
870 rename dscary288 ds_carry
871 rename hpshop288 hp_shop
872 rename dsshop288 ds_shop
873 rename hpclen288 hp_clean
874 rename dsclen288 ds_clean
875 rename hpmeal288 hp_meal
876 rename dsmeal288 ds_meal
877 rename hpmeds288 hp_meds
878 rename dsmeds288 ds_meds
879 rename hpbill288 hp_bills
880 rename dsbill288 ds_bills
881 rename iadlhp288 iadl_dep
882 rename iadlds288 iadl_dis
883 rename hp_mob288 hp_mobil
884 rename ds_mob288 ds_mobil
885 rename bloc_c288 bloc_cat
886 rename blocks288 blocksb1
887 rename adl24_288 adl24_b1
888 rename badl14288 badl14b1
889 rename iadl10288 iadl10b1
890 rename mobdis288 mobdisb1
891 rename mpase9bw mpase9cw
892 rename mpase288 mpaseb1
893 rename smoker_288 smoker_b1
894 rename * raw_*
895 * Generate a new variable called interval and assign the number 17 to each observation
896 gen interval = 17
897 * Rename raw_studyID variable to ensure consistency with master dataset
898 rename raw_studyID STUDYID
899 * Change working directory
900 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
901 * Save 288-month follow-up dataset with a new name
902 save other_vrb17.dta
903 clear
904

```

```

905 * Import 306-month follow-up data
906 import sas using "P:\projects\PEP\Master\fu306face.sas7bdat"
907 * Keep relevant variables
908 keep studyID fc5_pase306 age_306 female white edu_cont agecat306 BMI_306 highbp mi chf stroke Cancer
diabetes hipfx arthrit lung_d hbp_306 mi_306 chf_306 strok_306 cancr_306 diab_306 hipfx_306 lungd_306
arth_306 ccsum306a ccsum306b ccsum306c d_4adl306 d_7adl306 d_3adl306 hp3adl306 walkd_306 hpwalk306
dswalk306 hpstir306 dsstir306 hpcary306 dscary306 hpshop306 dsshop306 hpclen306 dsclen306 hpmeal306
dsmeal306 hpmeds306 dsmeds306 hpbill306 dsbill306 iadlhp306 iadlds306 hp_mob306 ds_mob306 bloc_c306
blocks306 adl24_306 adl12ms badl14306 iadl10306 mobdis306 intdate306 mpase1w mpase2w mpase3w mpase4w
mpase5w mpase6w mpase7w mpase8aw mpase8bw mpase8cw mpase8dw mpase9bw mpase306 smoker_306

909 * Rename variables to ensure consistency across intervals
910 rename fc5_pase306 fc5_paseb1
911 rename BMI_306 BMI_b1
912 rename agecat306 age5cat
913 rename age_306 age_b
914 rename ccsum306a ccsumb11
915 rename ccsum306c ccsumb12
916 rename Cancer cancer
917 rename d_4adl306 diff4adl
918 rename d_7adl306 diff7adl
919 rename d_3adl306 diff3adl
920 rename hp3adl306 dep3adl
921 rename walkd_306 walk_dev
922 rename hpwalk306 hp_walk
923 rename dswalk306 ds_walk
924 rename hpstir306 hp_stair
925 rename dsstir306 ds_stair
926 rename hpcary306 hp_carry
927 rename dscary306 ds_carry
928 rename hpshop306 hp_shop
929 rename dsshop306 ds_shop
930 rename hpclen306 hp_clean
931 rename dsclen306 ds_clean
932 rename hpmeal306 hp_meal
933 rename dsmeal306 ds_meal
934 rename hpmeds306 hp_meds
935 rename dsmeds306 ds_meds
936 rename hpbill306 hp_bills
937 rename dsbill306 ds_bills
938 rename iadlhp306 iadl_dep
939 rename iadlds306 iadl_dis
940 rename hp_mob306 hp_mobil
941 rename ds_mob306 ds_mobil
942 rename bloc_c306 bloc_cat
943 rename blocks306 blocksb1
944 rename adl24_306 adl24_b1
945 rename badl14306 badl14b1
946 rename iadl10306 iadl10b1
947 rename mobdis306 mobdisb1
948 rename mpase9bw mpase9cw
949 rename mpase306 mpaseb1
950 rename smoker_306 smoker_b1
951 rename * raw_*
952 * Generate a new variable called interval and assign the number 18 to each observation
953 gen interval = 18
954 * Rename raw_studyID variable to ensure consistency with master dataset
955 rename raw_studyID STUDYID
956 * Change working directory
957 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
958 * Save 306-month follow-up dataset with a new name
959 save other_vrb18.dta
960 clear
961

```

```

962 * Append follow-up comprehensive assessment datasets to baseline dataset
963 use other_vrb1.dta
964 append using other_vrb2.dta
965 append using other_vrb3.dta
966 append using other_vrb4.dta
967 append using other_vrb5.dta
968 append using other_vrb6.dta
969 append using other_vrb7.dta
970 append using other_vrb9.dta
971 append using other_vrb10.dta
972 append using other_vrb11.dta
973 append using other_vrb12.dta
974 append using other_vrb13.dta
975 append using other_vrb14.dta
976 append using other_vrb15.dta
977 append using other_vrb16.dta
978 append using other_vrb17.dta
979 append using other_vrb18.dta
980 * Sort by participant ID and interval (lowest to highest)
981 sort STUDYID interval
982 * Save dataset with a new name
983 save raw_other.dta
984 * Count total number of participants and observations
985 unique STUDYID
986 * 754 individuals, 5525 observations
987 clear
988
989 * Use master dataset
990 use master.dta
991 * Rename StudyID variable to ensure consistency with combined comprehensive assessment dataset
992 rename StudyID STUDYID
993 * Overwrite dataset, by replacing the previously saved file
994 save master.dta, replace
995 * Merge master dataset with combined comprehensive assessment dataset
996 merge 1:1 STUDYID interval using raw_other.dta, generate(merge_rawother)
997 * Keep comprehensive assessments that were not refused
998 keep if inlist(merge_rawother,1,3)
999 * Sort by participant ID and interval (lowest to highest)
1000 sort STUDYID interval
1001 * Save updated master dataset with a new name
1002 save f2f.dta
1003 * Count total number of participants and observations
1004 unique STUDYID
1005 * 754 individuals, 5168 observations
1006 clear
1007
1008 * Import raw baseline data
1009 import sas using "P:\projects\PEP\Master\pepbase.sas7bdat"
1010 * Keep relevant variables
1011 keep STUDYID GDI1 GDI2 GDI3 GDI8 HT1 HT2 WT2A ALC1 ALC2 ALC3 ALC4 ALC5 SM01 MPASE1 MPASE1A MPASE2
MPASE2A MPASE3 MPASE3A MPASE4 MPASE4A MPASE5 MPASE5A MPASE6 MPASE7 MPASE8A MPASE8B MPASE8C MPASE8D
MPASE9A MPASE9B MPASE9C INC GDI9 PF1 PF2 PF3 PF4 PF5A PF5B PF5C PF5D PF6A PF6B PF7A PF7B PF8A PF8B
PF9A PF9B PF10A PF10B PF11A PF11B PF12A PF12B PF13A PF13B PF14A PF14B PF15A PF15B PF16A PF16B PF17
CC1A CC1B CC2 CC3 CC4A CC4B1 CC4B2 CC4B3 CC4B4 CC5 CC6A CC6B CC6C CC7 CC8A CC8B1 CC8B2 CC8B3 CC8B4
CC9A CC9B1 CC9B2 CC9B3 CC9B4 CC9B5 CC10 CC11A CC11B CC11C CC12 CC13A CC13B CC14
1012 * Rename variables to ensure consistency across intervals
1013 rename GDI1 gender_p
1014 rename GDI2 race_p
1015 rename GDI3 age_p
1016 rename GDI8 education_p
1017 rename WT2A wt1a
1018 rename ALC1 alc1
1019 rename ALC2 alc2

```

```
1020  rename ALC3 alc3
1021  rename ALC4 alc4
1022  rename ALC5 alc5
1023  rename SMO1 smo1
1024  rename MPASE1 mpase1
1025  rename MPASE1A mpase1a
1026  rename MPASE2 mpase2
1027  rename MPASE2A mpase2a
1028  rename MPASE3 mpase3
1029  rename MPASE3A mpase3a
1030  rename MPASE4 mpase4
1031  rename MPASE4A mpase4a
1032  rename MPASE5 mpase5
1033  rename MPASE5A mpase5a
1034  rename MPASE6 mpase6
1035  rename MPASE7 mpase7
1036  rename MPASE8A mpase8a
1037  rename MPASE8B mpase8b
1038  rename MPASE8C mpase8c
1039  rename MPASE8D mpase8d
1040  rename MPASE9A mpase9a
1041  rename MPASE9B mpase9b
1042  rename MPASE9C mpase9c
1043  rename GDI9 pf18
1044  rename PF1 pf1b
1045  rename PF2 pf2b
1046  rename PF3 pf3b
1047  rename PF4 pf4b
1048  rename PF5A pf5a
1049  rename PF5B pf5b
1050  rename PF5C pf5c
1051  rename PF5D pf5d
1052  rename PF6A pf6a
1053  rename PF6B pf6b
1054  rename PF7A pf7a
1055  rename PF7B pf7b
1056  rename PF8A pf8a
1057  rename PF8B pf8b
1058  rename PF9A pf9a
1059  rename PF9B pf9b
1060  rename PF10A pf10a
1061  rename PF10B pf10b
1062  rename PF11A pf11a
1063  rename PF11B pf11b
1064  rename PF12A pf12a
1065  rename PF12B pf12b
1066  rename PF13A pf13a
1067  rename PF13B pf13b
1068  rename PF14A pf14a
1069  rename PF14B pf14b
1070  rename PF15A pf15a
1071  rename PF15B pf15b
1072  rename PF16A pf16a
1073  rename PF16B pf16b
1074  rename PF17 pf17
1075  rename CC1A cc1a
1076  rename CC1B cc1b
1077  rename CC2 cc2
1078  rename CC3 cc3
1079  rename CC4A cc4a
1080  rename CC4B1 cc4b1
1081  rename CC4B2 cc4b2
1082  rename CC4B3 cc4b3
```

```

1083 rename CC4B4 cc4b4
1084 rename CC5 cc5
1085 rename CC6A cc6a
1086 rename CC6B cc6b
1087 rename CC6C cc6c
1088 rename CC7 cc7
1089 rename CC9A cc8a
1090 rename CC9B1 cc8b1
1091 rename CC9B2 cc8b2
1092 rename CC9B3 cc8b3
1093 rename CC9B4 cc8b4
1094 rename CC9B5 cc8b5
1095 rename CC13A cc9a
1096 rename CC13B cc9b
1097 rename * mfu_*
1098 * Generate a new variable called interval and assign the number 1 to each observation
1099 gen interval = 1
1100 * Rename mfu_STUDYID variable to ensure consistency with updated master dataset
1101 rename mfu_STUDYID STUDYID
1102 * Change working directory
1103 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1104 * Save raw baseline dataset with a new name
1105 save other_mfu1.dta
1106 clear
1107
1108 * Import raw 18-month follow-up data
1109 import sas using "P:\projects\PEP\sasdata\face18m.sas7bdat"
1110 * Keep relevant variables
1111 keep StudyID wt1a alc1 alc2 alc3 alc4 alc5 smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4
mpase4a mpase5 mpase5a mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d
pf1a pf1b pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b
pf10a pf10b pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b pf17 pf18 cc1a
cc1b cc2 cc3 cc4a cc4b1 cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5
cc9a cc9b cc10a cc10b cc11
1112 * Rename variables to ensure consistency across intervals
1113 rename * mfu_*
1114 * Generate a new variable called interval and assign the number 2 to each observation
1115 gen interval = 2
1116 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1117 rename mfu_StudyID STUDYID
1118 * Change working directory
1119 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1120 * Save raw 18-month follow-up dataset with a new name
1121 save other_mfu2.dta
1122 clear
1123
1124 * Import raw 36-month follow-up data
1125 import sas using "P:\projects\PEP\sasdata\face36mfu.sas7bdat"
1126 * Keep relevant variables
1127 keep StudyID wt1a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1 bath2a bath3 pf2a
pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b pf11a
pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b pf17 cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b cc10a cc10b cc11
1128 * Rename variables to ensure consistency across intervals
1129 rename bath3 bath3g
1130 rename bath2a bath3
1131 rename bath1 bath2a
1132 rename * mfu_*
1133 * Generate a new variable called interval and assign the number 3 to each observation
1134 gen interval = 3
1135 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1136 rename mfu_StudyID STUDYID

```

```

1137 * Change working directory
1138 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1139 * Save raw 36-month follow-up dataset with a new name
1140 save other_mfu3.dta
1141 clear
1142
1143 * Import raw 54-month follow-up data
1144 import sas using "P:\projects\PEP\sasdata\face54mfu.sas7bdat"
1145 * Keep relevant variables
1146 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf2e pf3a pf3b pf3e pf4a pf4b pf4e pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a
pf9b pf10a pf10b pf11a pf11b pf12a pf12b pf12e pf13a pf13b pf13e pf14a pf14b pf14e pf15a pf15b pf15e
pf16a pf16b pf16c pf16d pf16e cc1a cc1b cc2 cc3 cc4a cc4b1 cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7
cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1147 * Rename variables to ensure consistency across intervals
1148 rename wt2a wt1a
1149 rename bath2a bath3
1150 rename bath1a bath2a
1151 rename bath3c bath3g
1152 rename pf2b pf2help3m
1153 rename pf3b pf3help3m
1154 rename pf4b pf4help3m
1155 rename pf2e pf2e4
1156 rename pf3e pf3e4
1157 rename pf4e pf4e4
1158 rename pf12e pf12e4
1159 rename pf13e pf13e4
1160 rename pf14e pf14e4
1161 rename pf15e pf15e4
1162 rename pf16e pf16e4
1163 rename * mfu_*
1164 * Generate a new variable called interval and assign the number 4 to each observation
1165 gen interval = 4
1166 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1167 rename mfu_StudyID STUDYID
1168 * Change working directory
1169 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1170 * Save raw 54-month follow-up dataset with a new name
1171 save other_mfu4.dta
1172 clear
1173
1174 * Import raw 72-month follow-up data
1175 import sas using "P:\projects\PEP\sasdata\face72mfu.sas7bdat"
1176 * Keep relevant variables
1177 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3g
pf2a pf2b pf2f pf3a pf3b pf3f pf4a pf4b pf4f pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a
pf9b pf10a pf10b pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2
cc3 cc4a cc4b1 cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1178 * Rename variables to ensure consistency across intervals
1179 rename wt2a wt1a
1180 rename bath2a bath3
1181 rename bath1a bath2a
1182 rename pf2b pf2help1m
1183 rename pf3b pf3help1m
1184 rename pf4b pf4help1m
1185 rename * mfu_*
1186 * Generate a new variable called interval and assign the number 5 to each observation
1187 gen interval = 5
1188 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1189 rename mfu_StudyID STUDYID
1190 * Change working directory

```

```

1191 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1192 * Save raw 72-month follow-up dataset with a new name
1193 save other_mfu5.dta
1194 clear
1195
1196 * Import raw 90-month follow-up data
1197 import sas using "P:\projects\PEP\sasdata\face90mfu.sas7bdat"
1198 * Keep relevant variables
1199 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
    mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3g
    pf2a pf2b pf2f pf3a pf3b pf3f pf4a pf4b pf4f pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a
    pf9b pf10a pf10b pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2
    cc3 cc4a cc4b1 cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1200 * Rename variables to ensure consistency across intervals
1201 rename wt2a wt1a
1202 rename bath2a bath3
1203 rename bath1a bath2a
1204 rename pf2b pf2help12m
1205 rename pf3b pf3help12m
1206 rename pf4b pf4help12m
1207 rename * mfu_*
1208 * Generate a new variable called interval and assign the number 6 to each observation
1209 gen interval = 6
1210 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1211 rename mfu_StudyID STUDYID
1212 * Change working directory
1213 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1214 * Save raw 90-month follow-up dataset with a new name
1215 save other_mfu6.dta
1216 clear
1217
1218 * Import raw 108-month follow-up data
1219 import sas using "P:\projects\PEP\sasdata\face108mfu.sas7bdat"
1220 * Keep relevant variables
1221 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
    mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
    pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
    pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
    cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1222 * Rename variables to ensure consistency across intervals
1223 rename wt2a wt1a
1224 rename bath2a bath3
1225 rename bath1a bath2a
1226 rename bath3c bath3g
1227 rename * mfu_*
1228 * Generate a new variable called interval and assign the number 7 to each observation
1229 gen interval = 7
1230 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1231 rename mfu_StudyID STUDYID
1232 * Change working directory
1233 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1234 * Save raw 108-month follow-up dataset with a new name
1235 save other_mfu7.dta
1236 clear
1237
1238 * Import raw 144-month follow-up data
1239 import sas using "P:\projects\PEP\sasdata\face144mfu.sas7bdat"
1240 * Keep relevant variables
1241 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
    mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
    pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
    pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
    cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b

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1242 * Rename variables to ensure consistency across intervals
1243 rename wt2a wt1a
1244 rename bath2a bath3
1245 rename bath1a bath2a
1246 rename bath3c bath3g
1247 rename * mfu_*
1248 * Generate a new variable called interval and assign the number 9 to each observation
1249 gen interval = 9
1250 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1251 rename mfu_StudyID STUDYID
1252 * Change working directory
1253 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1254 * Save raw 144-month follow-up dataset with a new name
1255 save other_mfu9.dta
1256 clear
1257
1258 * Import raw 162-month follow-up data
1259 import sas using "P:\projects\PEP\sasdata\face162mfu.sas7bdat"
1260 * Keep relevant variables
1261 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1262 * Rename variables to ensure consistency across intervals
1263 rename wt2a wt1a
1264 rename bath2a bath3
1265 rename bath1a bath2a
1266 rename bath3c bath3g
1267 rename * mfu_*
1268 * Generate a new variable called interval and assign the number 10 to each observation
1269 gen interval = 10
1270 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1271 rename mfu_StudyID STUDYID
1272 * Change working directory
1273 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1274 * Save raw 162-month follow-up dataset with a new name
1275 save other_mfu10.dta
1276 clear
1277
1278 * Import raw 180-month follow-up data
1279 import sas using "P:\projects\PEP\sasdata\face180mfu.sas7bdat"
1280 * Keep relevant variables
1281 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1282 * Rename variables to ensure consistency across intervals
1283 rename wt2a wt1a
1284 rename bath2a bath3
1285 rename bath1a bath2a
1286 rename bath3c bath3g
1287 rename * mfu_*
1288 * Generate a new variable called interval and assign the number 11 to each observation
1289 gen interval = 11
1290 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1291 rename mfu_StudyID STUDYID
1292 * Change working directory
1293 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1294 * Save raw 180-month follow-up dataset with a new name
1295 save other_mfu11.dta
1296 clear

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1297
1298 * Import raw 198-month follow-up data
1299 import sas using "P:\projects\PEP\sasdata\face198mfu.sas7bdat"
1300 * Keep relevant variables
1301 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1302 * Rename variables to ensure consistency across intervals
1303 rename wt2a wt1a
1304 rename bath2a bath3
1305 rename bath1a bath2a
1306 rename bath3c bath3g
1307 rename * mfu_*
1308 * Generate a new variable called interval and assign the number 12 to each observation
1309 gen interval = 12
1310 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1311 rename mfu_StudyID STUDYID
1312 * Change working directory
1313 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1314 * Save raw 198-month follow-up dataset with a new name
1315 save other_mfu12.dta
1316 clear
1317
1318 * Import raw 216-month follow-up data
1319 import sas using "P:\projects\PEP\sasdata\face216mfu.sas7bdat"
1320 * Keep relevant variables
1321 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1322 * Rename variables to ensure consistency across intervals
1323 rename wt2a wt1a
1324 rename bath2a bath3
1325 rename bath1a bath2a
1326 rename bath3c bath3g
1327 rename * mfu_*
1328 * Generate a new variable called interval and assign the number 13 to each observation
1329 gen interval = 13
1330 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1331 rename mfu_StudyID STUDYID
1332 * Change working directory
1333 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1334 * Save raw 216-month follow-up dataset with a new name
1335 save other_mfu13.dta
1336 clear
1337
1338 * Import raw 234-month follow-up data
1339 import sas using "P:\projects\PEP\sasdata\face234mfu.sas7bdat"
1340 * Keep relevant variables
1341 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1342 * Rename variables to ensure consistency across intervals
1343 rename wt2a wt1a
1344 rename bath2a bath3
1345 rename bath1a bath2a
1346 rename bath3c bath3g
1347 rename * mfu_*

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1348 * Generate a new variable called interval and assign the number 14 to each observation
1349 gen interval = 14
1350 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1351 rename mfu_StudyID STUDYID
1352 * Change working directory
1353 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1354 * Save raw 234-month follow-up dataset with a new name
1355 save other_mfu14.dta
1356 clear
1357
1358 * Import raw 252-month follow-up data
1359 import sas using "P:\projects\PEP\sasdata\face252mfu.sas7bdat"
1360 * Keep relevant variables
1361 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1362 * Rename variables to ensure consistency across intervals
1363 rename wt2a wt1a
1364 rename bath2a bath3
1365 rename bath1a bath2a
1366 rename bath3c bath3g
1367 rename * mfu_*
1368 * Generate a new variable called interval and assign the number 15 to each observation
1369 gen interval = 15
1370 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1371 rename mfu_StudyID STUDYID
1372 * Change working directory
1373 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1374 * Save raw 252-month follow-up dataset with a new name
1375 save other_mfu15.dta
1376 clear
1377
1378 * Import raw 270-month follow-up data
1379 import sas using "P:\projects\PEP\sasdata\face270mfu_amend_wrc.sas7bdat"
1380 * Keep relevant variables
1381 keep StudyID wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1382 * Rename variables to ensure consistency across intervals
1383 rename wt2a wt1a
1384 rename bath2a bath3
1385 rename bath1a bath2a
1386 rename bath3c bath3g
1387 rename * mfu_*
1388 * Generate a new variable called interval and assign the number 16 to each observation
1389 gen interval = 16
1390 * Rename mfu_StudyID variable to ensure consistency with updated master dataset
1391 rename mfu_StudyID STUDYID
1392 * Change working directory
1393 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1394 * Save raw 270-month follow-up dataset with a new name
1395 save other_mfu16.dta
1396 clear
1397
1398 * Import raw 288-month follow-up data
1399 import sas using "P:\projects\PEP\sasdata\face288mfu.sas7bdat"
1400 * Keep relevant variables
1401 keep studyid wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c

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pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1402 * Rename variables to ensure consistency across intervals
1403 rename wt2a wt1a
1404 rename bath2a bath3
1405 rename bath1a bath2a
1406 rename bath3c bath3g
1407 rename * mfu_*
1408 * Generate a new variable called interval and assign the number 17 to each observation
1409 gen interval = 17
1410 * Rename mfu_studyid variable to ensure consistency with updated master dataset
1411 rename mfu_studyid STUDYID
1412 * Change working directory
1413 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1414 * Save raw 288-month follow-up dataset with a new name
1415 save other_mfu17.dta
1416 clear
1417
1418 * Import raw 306-month follow-up data
1419 import sas using "P:\projects\PEP\sasdata\face306mfu.sas7bdat"
1420 * Keep relevant variables
1421 keep studyid wt2a smo1 mpase1 mpase1a mpase2 mpase2a mpase3 mpase3a mpase4 mpase4a mpase5 mpase5a
mpase6 mpase7 mpase8a mpase8b mpase8c mpase8d mpase9a mpase9b mpase9c mpase9d bath1a bath2a bath3c
pf2a pf2b pf3a pf3b pf4a pf4b pf5a pf5b pf5c pf5d pf6a pf6b pf7a pf7b pf8a pf8b pf9a pf9b pf10a pf10b
pf11a pf11b pf12a pf12b pf13a pf13b pf14a pf14b pf15a pf15b pf16a pf16b cc1a cc1b cc2 cc3 cc4a cc4b1
cc4b2 cc4b3 cc4b4 cc5 cc6a cc6b cc6c cc7 cc8a cc8b1 cc8b2 cc8b3 cc8b4 cc8b5 cc9a cc9b
1422 * Rename variables to ensure consistency across intervals
1423 rename wt2a wt1a
1424 rename bath2a bath3
1425 rename bath1a bath2a
1426 rename bath3c bath3g
1427 rename * mfu_*
1428 * Generate a new variable called interval and assign the number 18 to each observation
1429 gen interval = 18
1430 * Rename mfu_studyid variable to ensure consistency with updated master dataset
1431 rename mfu_studyid STUDYID
1432 * Change working directory
1433 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1434 * Save raw 306-month follow-up dataset with a new name
1435 save other_mfu18.dta
1436 clear
1437
1438 * Append raw follow-up comprehensive assessment datasets to raw baseline dataset
1439 use other_mfu1.dta
1440 append using other_mfu2.dta
1441 append using other_mfu3.dta
1442 append using other_mfu4.dta
1443 append using other_mfu5.dta
1444 append using other_mfu6.dta
1445 append using other_mfu7.dta
1446 append using other_mfu9.dta
1447 append using other_mfu10.dta
1448 append using other_mfu11.dta
1449 append using other_mfu12.dta
1450 append using other_mfu13.dta
1451 append using other_mfu14.dta
1452 append using other_mfu15.dta
1453 append using other_mfu16.dta
1454 append using other_mfu17.dta
1455 append using other_mfu18.dta
1456 * Sort by participant ID and interval (lowest to highest)
1457 sort STUDYID interval

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1458 * Save dataset with a new name
1459 save mfu_other.dta
1460 * Count total number of participants and observations
1461 unique STUDYID
1462 * 848 individuals, 5201 observations
1463 clear
1464
1465 * Use updated master dataset
1466 use f2f.dta
1467 * Merge updated master dataset with combined raw comprehensive assessment dataset
1468 merge 1:1 STUDYID interval using mfu_other.dta, generate(merge_mfuother)
1469 * Keep comprehensive assessments that were not refused
1470 keep if inlist(merge_mfuother,1,3)
1471 * Sort by participant ID and interval (lowest to highest)
1472 sort STUDYID interval
1473 * Overwrite dataset, by replacing the previously saved file
1474 save f2f.dta, replace
1475 * Count total number of participants and observations
1476 unique STUDYID
1477 * 754 individuals, 5168 observations
1478 clear
1479
1480 * Import raw baseline data
1481 import sas using "P:\projects\PEP\Master\pepbase.sas7bdat"
1482 * Keep relevant variables
1483 keep STUDYID GDI4MO GDI4DY GDI4YR
1484 * Generate a new variable called interval and assign the number 1 to each observation
1485 gen interval = 1
1486 * Change working directory
1487 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1488 * Save raw baseline dataset with a new name
1489 save birthmfu.dta
1490 clear
1491
1492 * Use updated master dataset
1493 use f2f.dta
1494 * Merge updated master dataset with raw baseline dataset
1495 merge 1:1 STUDYID interval using birthmfu.dta, generate(merge_mfubdate)
1496 * Keep comprehensive assessments that were not refused
1497 keep if inlist(merge_mfubdate,1,3)
1498 * Sort by participant ID and interval (lowest to highest)
1499 sort STUDYID interval
1500 * Overwrite dataset, by replacing the previously saved file
1501 save f2f.dta, replace
1502 * Count total number of participants and observations
1503 unique STUDYID
1504 * 754 individuals, 5168 observations
1505 clear
1506
1507 * Import corrected date of birth data
1508 import sas using "P:\projects\PEP\sasdata\subjects_080521_dobfix.sas7bdat"
1509 * Keep relevant variables
1510 keep StudyID BDate DDate bdate_ss
1511 * Rename variables
1512 rename StudyID STUDYID
1513 rename BDate BDates
1514 rename DDate DDates
1515 rename bdate_ss bdates_ss
1516 * Count total number of participants
1517 unique STUDYID
1518 * 754 individuals
1519 * Change working directory
1520 cd "P:\projects\PEP\OMalkowski\Active life expectancy"

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1521 * Save corrected date of birth dataset with a new name
1522 save dobfix.dta
1523 clear
1524
1525 * Use updated master dataset
1526 use f2f.dta
1527 * Merge updated master dataset with corrected date of birth dataset
1528 merge m:1 STUDYID using dobfix.dta, generate(merge_dobfix)
1529 * Keep if matched
1530 keep if merge_dobfix==3
1531 * Sort by participant ID and interval (lowest to highest)
1532 sort STUDYID interval
1533 * Save dataset with a new name
1534 save comprehensive.dta
1535 * Count total number of participants and observations
1536 unique STUDYID
1537 * 754 individuals, 5168 observations
1538 clear
1539
1540 * Import baseline Area Deprivation Index (ADI) data
1541 import sas using "P:\projects\PEP\ADI_Life_expectancy\New_ADI_update\pep_adi_atbaseline.sas7bdat"
1542 * Keep relevant variables
1543 keep studyID interval StateDecile ADId910
1544 * Rename variables
1545 rename studyID STUDYID
1546 rename StateDecile StateDecileb
1547 rename ADId910 ADId910b
1548 * Count total number of participants
1549 unique STUDYID
1550 * 754 individuals
1551 * Change working directory
1552 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1553 * Save ADI dataset with a new name
1554 save ADI_baseline.dta
1555 clear
1556
1557 * Use updated master dataset
1558 use comprehensive.dta
1559 * Merge updated master dataset with ADI dataset
1560 merge 1:1 STUDYID interval using ADI_baseline.dta, generate(merge_ADIB)
1561 * Keep comprehensive assessments that were not refused
1562 keep if inlist(merge_ADIB,1,3)
1563 * Sort by participant ID and interval (lowest to highest)
1564 sort STUDYID interval
1565 * Overwrite dataset, by replacing the previously saved file
1566 save comprehensive.dta, replace
1567 * Count total number of participants and observations
1568 unique STUDYID
1569 * 754 individuals, 5168 observations
1570 clear
1571
1572 * Import imputed income data
1573 import sas using "P:\projects\PEP\ADI_Life_expectancy\New_ADI_update\incblmiforeg.sas7bdat"
1574 * Keep relevant variables
1575 keep StudyID INC_MImdn
1576 * Rename StudyID variable to ensure consistency with updated master dataset
1577 rename StudyID STUDYID
1578 * Count total number of participants
1579 unique STUDYID
1580 * 754 individuals
1581 * Change working directory
1582 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
1583 * Save imputed income dataset with a new name

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1584 save inc_MI.dta
1585 clear
1586
1587 * Use updated master dataset
1588 use comprehensive.dta
1589 * Merge updated master dataset with imputed income dataset
1590 merge m:1 STUDYID using inc_MI.dta, generate(merge_inc)
1591 * Keep if matched
1592 keep if merge_inc==3
1593 * Sort by participant ID and interval (lowest to highest)
1594 sort STUDYID interval
1595 * Overwrite dataset, by replacing the previously saved file
1596 save comprehensive.dta, replace
1597 * Count total number of participants and observations
1598 unique STUDYID
1599 * 754 individuals, 5168 observations
1600
1601 * Individual- and area-level indicators of socio-economic status (SES)
1602 ** Years of education (time-constant)
1603 * Generate a new variable duplicating the education variable at baseline
1604 gen edu_cons = mfu_education_p if interval==1
1605 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
1606 tsset STUDYID interval
1607 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1608 bysort STUDYID: carryforward edu_cons, replace
1609 * Code "lower education (less than 12 years)" as 0
1610 gen edu_cons_bi = 0 if edu_cons<12
1611 * Code "higher education (12+ years)" as 1
1612 replace edu_cons_bi = 1 if edu_cons>=12
1613
1614 ** Income (time-constant)
1615 * Generate a new version of the imputed income variable ranging from 0 to 12 (instead of 1 to 13)
1616 gen INC_MImdn_new = INC_MImdn-1
1617 * Generate a new variable duplicating the imputed income variable at baseline
1618 gen inc_consmi = INC_MImdn_new if interval==1
1619 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
1620 tsset STUDYID interval
1621 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1622 bysort STUDYID: carryforward inc_consmi, replace
1623 * Code "lower income (< $15000)" as 0
1624 gen inc_bimi = 0 if inlist(inc_consmi,0,1,2,3,4)
1625 * Code "higher income (≥ $15000)" as 1
1626 replace inc_bimi = 1 if inlist(inc_consmi,5,6,7,8,9,10,11,12)
1627
1628 ** Neighbourhood disadvantage (time-constant)
1629 * Generate a new variable duplicating the ADI variable at baseline
1630 gen adi_cons = ADId910b if interval==1
1631 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
1632 tsset STUDYID interval
1633 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1634 bysort STUDYID: carryforward adi_cons, replace
1635 * Reverse the variable such that disadvantaged neighborhoods (81 to 100 percentile) are coded as 0
and nondisadvantaged neighborhoods (1 to 80 percentile) are coded as 1
1636 replace adi_cons = 2 if adi_cons==0
1637 replace adi_cons = 0 if adi_cons==1
1638 replace adi_cons = 1 if adi_cons==2
1639
1640 * Descriptive variables
1641 ** Age (years)
1642 gen age_fix = floor((intdateF2F-BDates) / 365.25)

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1643 gen age_fix_dec = ((intdateF2F-BDates) / 365.25)
1644
1645 ** Biological sex (time-constant)
1646 * Generate a new variable duplicating the biological sex variable at baseline
1647 gen sex_cons = mfu_gender_p if interval==1
1648 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
1649 tsset STUDYID interval
1650 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1651 bysort STUDYID: carryforward sex_cons, replace
1652 * Code "male" as 0
1653 replace sex_cons = 0 if sex_cons==1
1654 * Code "female" as 1
1655 replace sex_cons = 1 if sex_cons==2
1656
1657 * Covariates
1658 ** Race or ethnicity (time-constant)
1659 * Generate a new variable duplicating the race or ethnicity variable at baseline
1660 gen eth_cons = mfu_race_p if interval==1
1661 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
1662 tsset STUDYID interval
1663 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1664 bysort STUDYID: carryforward eth_cons, replace
1665 * Code "White" as 0
1666 gen eth_bi = 0 if eth_cons==1
1667 * Code "non-White" as 1
1668 replace eth_bi = 1 if inlist(eth_cons,2,3,5)
1669
1670 ** Number of chronic conditions
1671 *** Hypertension
1672 * Code as 1 if the participant has been told by a doctor that they have high blood pressure or
hypertension and the participant is currently taking any medicine for high blood pressure
1673 gen highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1
1674 * Code as 0 otherwise
1675 replace highbl = 0 if highbl!=1
1676 * Code as 0 if the participant responds "Don't know" to the question about having high blood pressure
or the question about taking medicine for high blood pressure
1677 replace highbl = 0 if mfu_cc1a==8 | mfu_cc1b==8
1678 * Code as missing if the participant refused to answer the question (or it was left blank) about
having high blood pressure or the question about taking medicine for high blood pressure
1679 replace highbl = . if inlist(mfu_cc1a,.,7) | mfu_cc1b==7
1680 * Code as missing if the comprehensive assessment is not at baseline or 18-month follow-up (i.e.,
intervals 1-2)
1681 replace highbl = . if ! inlist(interval,1,2)
1682
1683 * Carryforward observations (high blood pressure and on medication at the current comprehensive
assessment or any prior comprehensive assessment) with respect to the time variable "interval" (i.e.,
to all subsequent comprehensive assessments) by participant ID
1684 gen hbp18_cons = raw_hbp_18 if interval==2
1685 tsset STUDYID interval
1686 bysort STUDYID: carryforward hbp18_cons, replace
1687 gen hbp36_cons = raw_hbp_36 if interval==3
1688 tsset STUDYID interval
1689 bysort STUDYID: carryforward hbp36_cons, replace
1690 gen hbp54_cons = raw_hbp_54 if interval==4
1691 tsset STUDYID interval
1692 bysort STUDYID: carryforward hbp54_cons, replace
1693 gen hbp72_cons = raw_hbp_72 if interval==5
1694 tsset STUDYID interval
1695 bysort STUDYID: carryforward hbp72_cons, replace
1696 gen hbp90_cons = raw_hbp_90 if interval==6
1697 tsset STUDYID interval

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1698 bysort STUDYID: carryforward hbp90_cons, replace
1699 gen hbp108_cons = raw_hbp_108 if interval==7
1700 tsset STUDYID interval
1701 bysort STUDYID: carryforward hbp108_cons, replace
1702 gen hbp144_cons = raw_hbp_144 if interval==9
1703 tsset STUDYID interval
1704 bysort STUDYID: carryforward hbp144_cons, replace
1705 gen hbp162_cons = raw_hbp_162 if interval==10
1706 tsset STUDYID interval
1707 bysort STUDYID: carryforward hbp162_cons, replace
1708 gen hbp180_cons = raw_hbp_180 if interval==11
1709 tsset STUDYID interval
1710 bysort STUDYID: carryforward hbp180_cons, replace
1711 gen hbp198_cons = raw_hbp_198 if interval==12
1712 tsset STUDYID interval
1713 bysort STUDYID: carryforward hbp198_cons, replace
1714 gen hbp216_cons = raw_hbp_216 if interval==13
1715 tsset STUDYID interval
1716 bysort STUDYID: carryforward hbp216_cons, replace
1717 gen hbp234_cons = raw_hbp_234 if interval==14
1718 tsset STUDYID interval
1719 bysort STUDYID: carryforward hbp234_cons, replace
1720 gen hbp252_cons = raw_hbp_252 if interval==15
1721 tsset STUDYID interval
1722 bysort STUDYID: carryforward hbp252_cons, replace
1723 gen hbp270_cons = raw_hbp_270 if interval==16
1724 tsset STUDYID interval
1725 bysort STUDYID: carryforward hbp270_cons, replace
1726 gen hbp288_cons = raw_hbp_288 if interval==17
1727 tsset STUDYID interval
1728 bysort STUDYID: carryforward hbp288_cons, replace
1729
1730 * For intervals 3-18, code as 1 if the participant has been told by a doctor that they have high
blood pressure or hypertension and the participant is currently taking any medicine for high blood
pressure, and the participant was coded as NOT having hypertension at all prior comprehensive
assessments (i.e., new condition)
1731 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp18_cons==0 & interval==3
1732 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp36_cons==0 & interval==4
1733 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp54_cons==0 & interval==5
1734 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp72_cons==0 & interval==6
1735 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp90_cons==0 & interval==7
1736 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp108_cons==0 & interval==9
1737 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp144_cons==0 & interval==10
1738 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp162_cons==0 & interval==11
1739 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp180_cons==0 & interval==12
1740 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp198_cons==0 & interval==13
1741 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp216_cons==0 & interval==14
1742 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp234_cons==0 & interval==15
1743 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp252_cons==0 & interval==16
1744 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp270_cons==0 & interval==17
1745 replace highbl = 1 if mfu_cc1a==1 & mfu_cc1b==1 & hbp288_cons==0 & interval==18
1746
1747 * Code as 0 otherwise
1748 replace highbl = 0 if highbl!=1
1749 * Code as 0 if the participant responds "Don't know" to the question about having high blood pressure
or the question about taking medicine for high blood pressure
1750 replace highbl = 0 if mfu_cc1a==8 | mfu_cc1b==8
1751 * Code as missing if the participant refused to answer the question (or it was left blank) about
having high blood pressure or the question about taking medicine for high blood pressure
1752 replace highbl = . if inlist(mfu_cc1a,.,7) | mfu_cc1b==7
1753 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
1754 replace highbl = . if STUDYID==2007 & interval==2

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1755 * Compare values with variable from combined comprehensive assessment dataset
1756 gen bplminus = highbl - raw_highbpl
1757
1758 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1759 gen hbp_cons = raw_highbpl if interval==1
1760 tsset STUDYID interval
1761 bysort STUDYID: carryforward hbp_cons, replace
1762 * For each comprehensive assessment, code as 1 if the participant was newly coded as having
hypertension at the corresponding comprehensive assessment or the participant was coded as having
hypertension at any prior comprehensive assessment(s)
1763 * Code as 0 otherwise
1764 gen hbp18 = 1 if (highbl==1 | hbp_cons==1) & interval==2
1765 replace hbp18 = 0 if hbp18!=1 & interval==2
1766 gen hbp36 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1) & interval==3
1767 replace hbp36 = 0 if hbp36!=1 & interval==3
1768 gen hbp54 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1) & interval==4
1769 replace hbp54 = 0 if hbp54!=1 & interval==4
1770 gen hbp72 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1) & interval
==5
1771 replace hbp72 = 0 if hbp72!=1 & interval==5
1772 gen hbp90 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 | hbp72_cons
==1) & interval==6
1773 replace hbp90 = 0 if hbp90!=1 & interval==6
1774 gen hbp108 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1) & interval==7
1775 replace hbp108 = 0 if hbp108!=1 & interval==7
1776 gen hbp144 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1) & interval==9
1777 replace hbp144 = 0 if hbp144!=1 & interval==9
1778 gen hbp162 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1) & interval==10
1779 replace hbp162 = 0 if hbp162!=1 & interval==10
1780 gen hbp180 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1) & interval==11
1781 replace hbp180 = 0 if hbp180!=1 & interval==11
1782 gen hbp198 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1) &
interval==12
1783 replace hbp198 = 0 if hbp198!=1 & interval==12
1784 gen hbp216 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1) & interval==13
1785 replace hbp216 = 0 if hbp216!=1 & interval==13
1786 gen hbp234 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1 | hbp216_cons==1) & interval==14
1787 replace hbp234 = 0 if hbp234!=1 & interval==14
1788 gen hbp252 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1 | hbp216_cons==1 | hbp234_cons==1) & interval==15
1789 replace hbp252 = 0 if hbp252!=1 & interval==15
1790 gen hbp270 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1 | hbp216_cons==1 | hbp234_cons==1 | hbp252_cons==1) & interval==16
1791 replace hbp270 = 0 if hbp270!=1 & interval==16
1792 gen hbp288 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1 | hbp216_cons==1 | hbp234_cons==1 | hbp252_cons==1 | hbp270_cons==1) & interval==17
1793 replace hbp288 = 0 if hbp288!=1 & interval==17
1794 gen hbp306 = 1 if (highbl==1 | hbp_cons==1 | hbp18_cons==1 | hbp36_cons==1 | hbp54_cons==1 |
hbp72_cons==1 | hbp90_cons==1 | hbp108_cons==1 | hbp144_cons==1 | hbp162_cons==1 | hbp180_cons==1 |
hbp198_cons==1 | hbp216_cons==1 | hbp234_cons==1 | hbp252_cons==1 | hbp270_cons==1 | hbp288_cons==1) &

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        interval==18
1795 replace hbp306 = 0 if hbp306!=1 & interval==18
1796
1797 *** Myocardial infarction
1798 * Code as 1 if the participant has been told by a doctor that they had a heart attack, or coronary,
or myocardial infarction, and had to be hospitalized
1799 gen heart = 1 if mfu_cc2==1
1800 * Code as 0 if the participant responds "Suspect or possible" or "No"
1801 replace heart = 0 if inlist(mfu_cc2,2,3)
1802 * Code as 0 if the participant responds "Don't know"
1803 replace heart = 0 if mfu_cc2==8
1804 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had a myocardial infarction
1805 replace heart = . if inlist(mfu_cc2,.,7)
1806 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
1807 replace heart = . if STUDYID==2007 & interval==2
1808 * Compare values with variable from combined comprehensive assessment dataset
1809 gen mminus = heart - raw_mi
1810
1811 * Carryforward observations (myocardial infarction and hospitalization at the current comprehensive
assessment or any prior comprehensive assessment) with respect to the time variable "interval" (i.e.,
to all subsequent comprehensive assessments) by participant ID
1812 gen mi_cons = raw_mi if interval==1
1813 tsset STUDYID interval
1814 bysort STUDYID: carryforward mi_cons, replace
1815 gen mi18_cons = raw_mi_18 if interval==2
1816 tsset STUDYID interval
1817 bysort STUDYID: carryforward mi18_cons, replace
1818 gen mi36_cons = raw_mi_36 if interval==3
1819 tsset STUDYID interval
1820 bysort STUDYID: carryforward mi36_cons, replace
1821 gen mi54_cons = raw_mi_54 if interval==4
1822 tsset STUDYID interval
1823 bysort STUDYID: carryforward mi54_cons, replace
1824 gen mi72_cons = raw_mi_72 if interval==5
1825 tsset STUDYID interval
1826 bysort STUDYID: carryforward mi72_cons, replace
1827 gen mi90_cons = raw_mi_90 if interval==6
1828 tsset STUDYID interval
1829 bysort STUDYID: carryforward mi90_cons, replace
1830 gen mi108_cons = raw_mi_108 if interval==7
1831 tsset STUDYID interval
1832 bysort STUDYID: carryforward mi108_cons, replace
1833 gen mi144_cons = raw_mi_144 if interval==9
1834 tsset STUDYID interval
1835 bysort STUDYID: carryforward mi144_cons, replace
1836 gen mi162_cons = raw_mi_162 if interval==10
1837 tsset STUDYID interval
1838 bysort STUDYID: carryforward mi162_cons, replace
1839 gen mi180_cons = raw_mi_180 if interval==11
1840 tsset STUDYID interval
1841 bysort STUDYID: carryforward mi180_cons, replace
1842 gen mi198_cons = raw_mi_198 if interval==12
1843 tsset STUDYID interval
1844 bysort STUDYID: carryforward mi198_cons, replace
1845 gen mi216_cons = raw_mi_216 if interval==13
1846 tsset STUDYID interval
1847 bysort STUDYID: carryforward mi216_cons, replace
1848 gen mi234_cons = raw_mi_234 if interval==14
1849 tsset STUDYID interval
1850 bysort STUDYID: carryforward mi234_cons, replace
1851 gen mi252_cons = raw_mi_252 if interval==15

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1852 tsset STUDYID interval
1853 bysort STUDYID: carryforward mi252_cons, replace
1854 gen mi270_cons = raw_mi_270 if interval==16
1855 tsset STUDYID interval
1856 bysort STUDYID: carryforward mi270_cons, replace
1857 gen mi288_cons = raw_mi_288 if interval==17
1858 tsset STUDYID interval
1859 bysort STUDYID: carryforward mi288_cons, replace
1860
1861 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had a
myocardial infarction and having been hospitalized at the corresponding comprehensive assessment or
the participant was coded as having had a myocardial infarction at any prior comprehensive
assessment(s)
1862 * Code as 0 otherwise
1863 gen mi18 = 1 if (heart==1 | mi_cons==1) & interval==2
1864 replace mi18 = 0 if mi18!=1 & interval==2
1865 gen mi36 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1) & interval==3
1866 replace mi36 = 0 if mi36!=1 & interval==3
1867 gen mi54 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1) & interval==4
1868 replace mi54 = 0 if mi54!=1 & interval==4
1869 gen mi72 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1) & interval==5
1870 replace mi72 = 0 if mi72!=1 & interval==5
1871 gen mi90 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1) &
interval==6
1872 replace mi90 = 0 if mi90!=1 & interval==6
1873 gen mi108 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1) & interval==7
1874 replace mi108 = 0 if mi108!=1 & interval==7
1875 gen mi144 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1) & interval==9
1876 replace mi144 = 0 if mi144!=1 & interval==9
1877 gen mi162 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1) & interval==10
1878 replace mi162 = 0 if mi162!=1 & interval==10
1879 gen mi180 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1) & interval==11
1880 replace mi180 = 0 if mi180!=1 & interval==11
1881 gen mi198 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1) & interval==12
1882 replace mi198 = 0 if mi198!=1 & interval==12
1883 gen mi216 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1) &
interval==13
1884 replace mi216 = 0 if mi216!=1 & interval==13
1885 gen mi234 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1 |
mi216_cons==1) & interval==14
1886 replace mi234 = 0 if mi234!=1 & interval==14
1887 gen mi252 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1 |
mi216_cons==1 | mi234_cons==1) & interval==15
1888 replace mi252 = 0 if mi252!=1 & interval==15
1889 gen mi270 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1 |
mi216_cons==1 | mi234_cons==1 | mi252_cons==1) & interval==16
1890 replace mi270 = 0 if mi270!=1 & interval==16
1891 gen mi288 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1 |
mi216_cons==1 | mi234_cons==1 | mi252_cons==1 | mi270_cons==1) & interval==17
1892 replace mi288 = 0 if mi288!=1 & interval==17
1893 gen mi306 = 1 if (heart==1 | mi_cons==1 | mi18_cons==1 | mi36_cons==1 | mi54_cons==1 | mi72_cons==1 |
mi90_cons==1 | mi108_cons==1 | mi144_cons==1 | mi162_cons==1 | mi180_cons==1 | mi198_cons==1 |
mi216_cons==1 | mi234_cons==1 | mi252_cons==1 | mi270_cons==1 | mi288_cons==1) & interval==18

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1894  replace mi306 = 0 if mi306!=1 & interval==18
1895
1896  *** Congestive heart failure
1897  * Code as 1 if the participant has been told by a doctor that they had heart failure or congestive
    heart failure
1898  gen chf = 1 if mfu_cc3==1
1899  * Code as 0 otherwise
1900  replace chf = 0 if chf!=1
1901  * Code as missing if the participant refused to answer the question (or it was left blank) about
    having had congestive heart failure
1902  replace chf = . if inlist(mfu_cc3,7,.)
1903  * Code as missing if the comprehensive assessment is not at baseline or 18-month follow-up (i.e.,
    intervals 1-2)
1904  replace chf = . if ! inlist(interval,1,2)
1905
1906  * Carryforward observations (congestive heart failure at the current comprehensive assessment or any
    prior comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
    comprehensive assessments) by participant ID
1907  gen chf18_cons = raw_chf_18 if interval==2
1908  tsset STUDYID interval
1909  bysort STUDYID: carryforward chf18_cons, replace
1910  gen chf36_cons = raw_chf_36 if interval==3
1911  tsset STUDYID interval
1912  bysort STUDYID: carryforward chf36_cons, replace
1913  gen chf54_cons = raw_chf_54 if interval==4
1914  tsset STUDYID interval
1915  bysort STUDYID: carryforward chf54_cons, replace
1916  gen chf72_cons = raw_chf_72 if interval==5
1917  tsset STUDYID interval
1918  bysort STUDYID: carryforward chf72_cons, replace
1919  gen chf90_cons = raw_chf_90 if interval==6
1920  tsset STUDYID interval
1921  bysort STUDYID: carryforward chf90_cons, replace
1922  gen chf108_cons = raw_chf_108 if interval==7
1923  tsset STUDYID interval
1924  bysort STUDYID: carryforward chf108_cons, replace
1925  gen chf144_cons = raw_chf_144 if interval==9
1926  tsset STUDYID interval
1927  bysort STUDYID: carryforward chf144_cons, replace
1928  gen chf162_cons = raw_chf_162 if interval==10
1929  tsset STUDYID interval
1930  bysort STUDYID: carryforward chf162_cons, replace
1931  gen chf180_cons = raw_chf_180 if interval==11
1932  tsset STUDYID interval
1933  bysort STUDYID: carryforward chf180_cons, replace
1934  gen chf198_cons = raw_chf_198 if interval==12
1935  tsset STUDYID interval
1936  bysort STUDYID: carryforward chf198_cons, replace
1937  gen chf216_cons = raw_chf_216 if interval==13
1938  tsset STUDYID interval
1939  bysort STUDYID: carryforward chf216_cons, replace
1940  gen chf234_cons = raw_chf_234 if interval==14
1941  tsset STUDYID interval
1942  bysort STUDYID: carryforward chf234_cons, replace
1943  gen chf252_cons = raw_chf_252 if interval==15
1944  tsset STUDYID interval
1945  bysort STUDYID: carryforward chf252_cons, replace
1946  gen chf270_cons = raw_chf_270 if interval==16
1947  tsset STUDYID interval
1948  bysort STUDYID: carryforward chf270_cons, replace
1949  gen chf288_cons = raw_chf_288 if interval==17
1950  tsset STUDYID interval
1951  bysort STUDYID: carryforward chf288_cons, replace

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1952
1953 * For intervals 3-18, code as 1 if the participant has been told by a doctor that they had heart
failure or congestive heart failure, and the participant was coded as NOT having had congestive heart
failure at all prior comprehensive assessments (i.e., new condition)
1954 replace chf = 1 if mfu_cc3==1 & chf18_cons==0 & interval==3
1955 replace chf = 1 if mfu_cc3==1 & chf36_cons==0 & interval==4
1956 replace chf = 1 if mfu_cc3==1 & chf54_cons==0 & interval==5
1957 replace chf = 1 if mfu_cc3==1 & chf72_cons==0 & interval==6
1958 replace chf = 1 if mfu_cc3==1 & chf90_cons==0 & interval==7
1959 replace chf = 1 if mfu_cc3==1 & chf108_cons==0 & interval==9
1960 replace chf = 1 if mfu_cc3==1 & chf144_cons==0 & interval==10
1961 replace chf = 1 if mfu_cc3==1 & chf162_cons==0 & interval==11
1962 replace chf = 1 if mfu_cc3==1 & chf180_cons==0 & interval==12
1963 replace chf = 1 if mfu_cc3==1 & chf198_cons==0 & interval==13
1964 replace chf = 1 if mfu_cc3==1 & chf216_cons==0 & interval==14
1965 replace chf = 1 if mfu_cc3==1 & chf234_cons==0 & interval==15
1966 replace chf = 1 if mfu_cc3==1 & chf252_cons==0 & interval==16
1967 replace chf = 1 if mfu_cc3==1 & chf270_cons==0 & interval==17
1968 replace chf = 1 if mfu_cc3==1 & chf288_cons==0 & interval==18
1969
1970 * Code as 0 otherwise
1971 replace chf = 0 if chf!=1
1972 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had congestive heart failure
1973 replace chf = . if inlist(mfu_cc3,7,.)
1974 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
1975 replace chf = . if STUDYID==2007 & interval==1
1976 * Compare values with variable from combined comprehensive assessment dataset
1977 gen chf_mis = raw_chf
1978
1979 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
1980 gen chf_cons = raw_chf_cons
1981 tssort STUDYID interval
1982 bys STUDYID (sort) interval, replace
1983 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had
congestive heart failure at the corresponding comprehensive assessment or the participant was coded
as having had congestive heart failure at any prior comprehensive assessment(s)
1984 * Code as 0 otherwise
1985 gen chf18 = 1 if (chf==1 | chf_cons==1) & interval==3
1986 replace chf18 = 0 if chf18!=1 & interval==3
1987 gen chf36 = 1 if (chf==1 | chf_cons==1 | chf18==1) & interval==4
1988 replace chf36 = 0 if chf36!=1 & interval==4
1989 gen chf54 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1) & interval==5
1990 replace chf54 = 0 if chf54!=1 & interval==5
1991 gen chf72 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1) & interval==6
1992 replace chf72 = 0 if chf72!=1 & interval==6
1993 gen chf90 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1) & interval==7
1994 replace chf90 = 0 if chf90!=1 & interval==7
1995 gen chf108 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1 | chf90==1) & interval==9
1996 replace chf108 = 0 if chf108!=1 & interval==9
1997 gen chf144 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1 | chf90==1 | chf108==1) & interval==10
1998 replace chf144 = 0 if chf144!=1 & interval==10
1999 gen chf162 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1 | chf90==1 | chf108==1 |
chf144==1) & interval==11
2000 replace chf162 = 0 if chf162!=1 & interval==11
2001 gen chf180 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1 | chf90==1 | chf108==1 |
chf144==1 | chf162==1) & interval==12
2002 replace chf180 = 0 if chf180!=1 & interval==12
2003 gen chf198 = 1 if (chf==1 | chf_cons==1 | chf18==1 | chf36==1 | chf54==1 | chf72==1 | chf90==1 | chf108==1 |
chf144==1 | chf162==1 | chf180==1) & interval==13
2004 replace chf198 = 0 if chf198!=1 & interval==13

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2005  gen chf216 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      ) & interval==13
2006  replace chf216 = 0 if chf216!=1 & interval==13
2007  gen chf234 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      | chf216_cons==1) & interval==14
2008  replace chf234 = 0 if chf234!=1 & interval==14
2009  gen chf252 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      | chf216_cons==1 | chf234_cons==1) & interval==15
2010  replace chf252 = 0 if chf252!=1 & interval==15
2011  gen chf270 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      | chf216_cons==1 | chf234_cons==1 | chf252_cons==1) & interval==16
2012  replace chf270 = 0 if chf270!=1 & interval==16
2013  gen chf288 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      | chf216_cons==1 | chf234_cons==1 | chf252_cons==1 | chf270_cons==1) & interval==17
2014  replace chf288 = 0 if chf288!=1 & interval==17
2015  gen chf306 = 1 if (chf==1 | chf_cons==1 | chf18_cons==1 | chf36_cons==1 | chf54_cons==1 | chf72_cons==
      1 | chf90_cons==1 | chf108_cons==1 | chf144_cons==1 | chf162_cons==1 | chf180_cons==1 | chf198_cons==1
      | chf216_cons==1 | chf234_cons==1 | chf252_cons==1 | chf270_cons==1 | chf288_cons==1) & interval==18
2016  replace chf306 = 0 if chf306!=1 & interval==18
2017
2018  *** Stroke
2019  * Code as 1 if the participant has been told by a doctor that they had a stroke or brain hemorrhage
      and had to be hospitalized
2020  gen stroke = 1 if mfu_cc4a==1
2021  * Code as 0 if the participant responds "Suspect or possible" or "No"
2022  replace stroke = 0 if inlist(mfu_cc4a,2,3)
2023  * Code as 0 if the participant responds "Don't know"
2024  replace stroke = 0 if mfu_cc4a==8
2025  * Code as missing if the participant refused to answer the question (or it was left blank) about
      having had a stroke
2026  replace stroke = . if inlist(mfu_cc4a,.,7)
2027  * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
      mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2028  replace stroke = . if STUDYID==2007 & interval==2
2029  * Compare values with variable from combined comprehensive assessment dataset
2030  gen stminus = stroke - raw_stroke
2031
2032  * Carryforward observations (stroke and hospitalization at the current comprehensive assessment or
      any prior comprehensive assessment) with respect to the time variable "interval" (i.e., to all
      subsequent comprehensive assessments) by participant ID
2033  gen stroke_cons = raw_stroke if interval==1
2034  tsset STUDYID interval
2035  bysort STUDYID: carryforward stroke_cons, replace
2036  gen stroke18_cons = raw_strok_18 if interval==2
2037  tsset STUDYID interval
2038  bysort STUDYID: carryforward stroke18_cons, replace
2039  gen stroke36_cons = raw_strok_36 if interval==3
2040  tsset STUDYID interval
2041  bysort STUDYID: carryforward stroke36_cons, replace
2042  gen stroke54_cons = raw_strok_54 if interval==4
2043  tsset STUDYID interval
2044  bysort STUDYID: carryforward stroke54_cons, replace
2045  gen stroke72_cons = raw_strok_72 if interval==5
2046  tsset STUDYID interval
2047  bysort STUDYID: carryforward stroke72_cons, replace
2048  gen stroke90_cons = raw_strok_90 if interval==6
2049  tsset STUDYID interval
2050  bysort STUDYID: carryforward stroke90_cons, replace

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2051 gen stroke108_cons = raw_strok_108 if interval==7
2052 tsset STUDYID interval
2053 bysort STUDYID: carryforward stroke108_cons, replace
2054 gen stroke144_cons = raw_strok_144 if interval==9
2055 tsset STUDYID interval
2056 bysort STUDYID: carryforward stroke144_cons, replace
2057 gen stroke162_cons = raw_strok_162 if interval==10
2058 tsset STUDYID interval
2059 bysort STUDYID: carryforward stroke162_cons, replace
2060 gen stroke180_cons = raw_strok_180 if interval==11
2061 tsset STUDYID interval
2062 bysort STUDYID: carryforward stroke180_cons, replace
2063 gen stroke198_cons = raw_strok_198 if interval==12
2064 tsset STUDYID interval
2065 bysort STUDYID: carryforward stroke198_cons, replace
2066 gen stroke216_cons = raw_strok_216 if interval==13
2067 tsset STUDYID interval
2068 bysort STUDYID: carryforward stroke216_cons, replace
2069 gen stroke234_cons = raw_strok_234 if interval==14
2070 tsset STUDYID interval
2071 bysort STUDYID: carryforward stroke234_cons, replace
2072 gen stroke252_cons = raw_strok_252 if interval==15
2073 tsset STUDYID interval
2074 bysort STUDYID: carryforward stroke252_cons, replace
2075 gen stroke270_cons = raw_strok_270 if interval==16
2076 tsset STUDYID interval
2077 bysort STUDYID: carryforward stroke270_cons, replace
2078 gen stroke288_cons = raw_strok_288 if interval==17
2079 tsset STUDYID interval
2080 bysort STUDYID: carryforward stroke288_cons, replace
2081
2082 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had a
stroke and having been hospitalized at the corresponding comprehensive assessment or the participant
was coded as having had a stroke at any prior comprehensive assessment(s)
2083 * Code as 0 otherwise
2084 gen stroke18 = 1 if (stroke==1 | stroke_cons==1) & interval==2
2085 replace stroke18 = 0 if stroke18!=1 & interval==2
2086 gen stroke36 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1) & interval==3
2087 replace stroke36 = 0 if stroke36!=1 & interval==3
2088 gen stroke54 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1) & interval==4
2089 replace stroke54 = 0 if stroke54!=1 & interval==4
2090 gen stroke72 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1) & interval==5
2091 replace stroke72 = 0 if stroke72!=1 & interval==5
2092 gen stroke90 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1) & interval==6
2093 replace stroke90 = 0 if stroke90!=1 & interval==6
2094 gen stroke108 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1) & interval==7
2095 replace stroke108 = 0 if stroke108!=1 & interval==7
2096 gen stroke144 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1) & interval==9
2097 replace stroke144 = 0 if stroke144!=1 & interval==9
2098 gen stroke162 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1) & interval==10
2099 replace stroke162 = 0 if stroke162!=1 & interval==10
2100 gen stroke180 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1)
& interval==11
2101 replace stroke180 = 0 if stroke180!=1 & interval==11
2102 gen stroke198 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1) & interval==12

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2103 replace stroke198 = 0 if stroke198!=1 & interval==12
2104 gen stroke216 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1) & interval==13
2105 replace stroke216 = 0 if stroke216!=1 & interval==13
2106 gen stroke234 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1 | stroke216_cons==1) & interval==14
2107 replace stroke234 = 0 if stroke234!=1 & interval==14
2108 gen stroke252 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1 | stroke216_cons==1 | stroke234_cons==1) & interval==15
2109 replace stroke252 = 0 if stroke252!=1 & interval==15
2110 gen stroke270 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1 | stroke216_cons==1 | stroke234_cons==1 | stroke252_cons==1) &
interval==16
2111 replace stroke270 = 0 if stroke270!=1 & interval==16
2112 gen stroke288 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1 | stroke216_cons==1 | stroke234_cons==1 | stroke252_cons==1 |
stroke270_cons==1) & interval==17
2113 replace stroke288 = 0 if stroke288!=1 & interval==17
2114 gen stroke306 = 1 if (stroke==1 | stroke_cons==1 | stroke18_cons==1 | stroke36_cons==1 | stroke54_cons
==1 | stroke72_cons==1 | stroke90_cons==1 | stroke108_cons==1 | stroke144_cons==1 | stroke162_cons==1
| stroke180_cons==1 | stroke198_cons==1 | stroke216_cons==1 | stroke234_cons==1 | stroke252_cons==1 |
stroke270_cons==1 | stroke288_cons==1) & interval==18
2115 replace stroke306 = 0 if stroke306!=1 & interval==18
2116
2117 *** Cancer
2118 * Code as 1 if the participant has been told by a doctor that they had cancer or a malignant tumor,
excluding minor skin cancers
2119 gen cancer = 1 if mfu_cc5==1
2120 * Code as 0 if the participant responds "Suspect or possible" or "No"
2121 replace cancer = 0 if inlist(mfu_cc5,2,3)
2122 * Code as 0 if the participant responds "Don't know"
2123 replace cancer = 0 if mfu_cc5==8
2124 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had cancer
2125 replace cancer = . if inlist(mfu_cc5,.,7)
2126 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2127 replace cancer = . if STUDYID==2007 & interval==2
2128 * Compare values with variable from combined comprehensive assessment dataset
2129 gen caminus = cancer - raw_cancer
2130
2131 * Carryforward observations (cancer at the current comprehensive assessment or any prior
comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
comprehensive assessments) by participant ID
2132 gen cancer_cons = raw_cancer if interval==1
2133 tsset STUDYID interval
2134 bysort STUDYID: carryforward cancer_cons, replace
2135 gen cancer18_cons = raw_cancr_18 if interval==2
2136 tsset STUDYID interval
2137 bysort STUDYID: carryforward cancer18_cons, replace
2138 gen cancer36_cons = raw_cancr_36 if interval==3
2139 tsset STUDYID interval
2140 bysort STUDYID: carryforward cancer36_cons, replace
2141 gen cancer54_cons = raw_cancr_54 if interval==4
2142 tsset STUDYID interval
2143 bysort STUDYID: carryforward cancer54_cons, replace
2144 gen cancer72_cons = raw_cancr_72 if interval==5
2145 tsset STUDYID interval

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2146 bysort STUDYID: carryforward cancer72_cons, replace
2147 gen cancer90_cons = raw_cancr_90 if interval==6
2148 tsset STUDYID interval
2149 bysort STUDYID: carryforward cancer90_cons, replace
2150 gen cancer108_cons = raw_cancr_108 if interval==7
2151 tsset STUDYID interval
2152 bysort STUDYID: carryforward cancer108_cons, replace
2153 gen cancer144_cons = raw_cancr_144 if interval==9
2154 tsset STUDYID interval
2155 bysort STUDYID: carryforward cancer144_cons, replace
2156 gen cancer162_cons = raw_cancr_162 if interval==10
2157 tsset STUDYID interval
2158 bysort STUDYID: carryforward cancer162_cons, replace
2159 gen cancer180_cons = raw_cancr_180 if interval==11
2160 tsset STUDYID interval
2161 bysort STUDYID: carryforward cancer180_cons, replace
2162 gen cancer198_cons = raw_cancr_198 if interval==12
2163 tsset STUDYID interval
2164 bysort STUDYID: carryforward cancer198_cons, replace
2165 gen cancer216_cons = raw_cancr_216 if interval==13
2166 tsset STUDYID interval
2167 bysort STUDYID: carryforward cancer216_cons, replace
2168 gen cancer234_cons = raw_cancr_234 if interval==14
2169 tsset STUDYID interval
2170 bysort STUDYID: carryforward cancer234_cons, replace
2171 gen cancer252_cons = raw_cancr_252 if interval==15
2172 tsset STUDYID interval
2173 bysort STUDYID: carryforward cancer252_cons, replace
2174 gen cancer270_cons = raw_cancr_270 if interval==16
2175 tsset STUDYID interval
2176 bysort STUDYID: carryforward cancer270_cons, replace
2177 gen cancer288_cons = raw_cancr_288 if interval==17
2178 tsset STUDYID interval
2179 bysort STUDYID: carryforward cancer288_cons, replace
2180
2181 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had
2182 cancer at the corresponding comprehensive assessment or the participant was coded as having had
2183 cancer at any prior comprehensive assessment(s)
2184 * Code as 0 otherwise
2185 gen cancer18 = 1 if (cancer==1 | cancer_cons==1) & interval==2
2186 replace cancer18 = 0 if cancer18!=1 & interval==2
2187 gen cancer36 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1) & interval==3
2188 replace cancer36 = 0 if cancer36!=1 & interval==3
2189 gen cancer54 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1) & interval==4
2190 replace cancer54 = 0 if cancer54!=1 & interval==4
2191 gen cancer72 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2192 ==1) & interval==5
2193 replace cancer72 = 0 if cancer72!=1 & interval==5
2194 gen cancer90 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2195 ==1 | cancer72_cons==1) & interval==6
2196 replace cancer90 = 0 if cancer90!=1 & interval==6
2197 gen cancer108 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2198 ==1 | cancer72_cons==1 | cancer90_cons==1) & interval==7
2199 replace cancer108 = 0 if cancer108!=1 & interval==7
2200 gen cancer144 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2201 ==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1) & interval==9
2202 replace cancer144 = 0 if cancer144!=1 & interval==9
2203 gen cancer162 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2204 ==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1) & interval==10
2205 replace cancer162 = 0 if cancer162!=1 & interval==10
2206 gen cancer180 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
2207 ==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1)
2208 & interval==11

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2200 replace cancer180 = 0 if cancer180!=1 & interval==11
2201 gen cancer198 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1) & interval==12
2202 replace cancer198 = 0 if cancer198!=1 & interval==12
2203 gen cancer216 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1) & interval==13
2204 replace cancer216 = 0 if cancer216!=1 & interval==13
2205 gen cancer234 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1 | cancer216_cons==1) & interval==14
2206 replace cancer234 = 0 if cancer234!=1 & interval==14
2207 gen cancer252 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1 | cancer216_cons==1 | cancer234_cons==1) & interval==15
2208 replace cancer252 = 0 if cancer252!=1 & interval==15
2209 gen cancer270 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1 | cancer216_cons==1 | cancer234_cons==1 | cancer252_cons==1) &
interval==16
2210 replace cancer270 = 0 if cancer270!=1 & interval==16
2211 gen cancer288 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1 | cancer216_cons==1 | cancer234_cons==1 | cancer252_cons==1 |
cancer270_cons==1) & interval==17
2212 replace cancer288 = 0 if cancer288!=1 & interval==17
2213 gen cancer306 = 1 if (cancer==1 | cancer_cons==1 | cancer18_cons==1 | cancer36_cons==1 | cancer54_cons
==1 | cancer72_cons==1 | cancer90_cons==1 | cancer108_cons==1 | cancer144_cons==1 | cancer162_cons==1
| cancer180_cons==1 | cancer198_cons==1 | cancer216_cons==1 | cancer234_cons==1 | cancer252_cons==1 |
cancer270_cons==1 | cancer288_cons==1) & interval==18
2214 replace cancer306 = 0 if cancer306!=1 & interval==18
2215
2216 *** Diabetes mellitus
2217 * Code as 1 if the participant has been told by a doctor that they had diabetes, sugar in their
urine, or high blood sugar ("Yes" or "Suspect or possible"), and the participant is now using
medication that they swallow to treat or control their diabetes or the participant is now using
insulin injections
2218 gen diabetes_bl = 1 if inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)
2219 * Code as 0 otherwise
2220 replace diabetes_bl = 0 if diabetes_bl!=1
2221 * Code as missing if the participant responds "Refused" or "Don't know" to the question about having
had diabetes
2222 replace diabetes_bl = . if inlist(mfu_cc6a,7,8)
2223 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
2224 gen diab_cons = diabetes_bl if interval==1
2225 tsset STUDYID interval
2226 bysort STUDYID: carryforward diab_cons, replace
2227 * Generate a new variable replicating the "diabetes_bl" variable for all baseline comprehensive
assessments
2228 gen diabetes = diabetes_bl if interval==1
2229 * Code as 1 if the participant has been told by a doctor that they had diabetes, sugar in their
urine, or high blood sugar ("Yes" or "Suspect or possible"), and the participant is now using
medication that they swallow to treat or control their diabetes or the participant is now using
insulin injections, and the participant was coded as NOT having diabetes at all prior comprehensive
assessments (i.e., new condition)
2230 replace diabetes = 1 if diab_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1))
2231 * Code as 0 otherwise
2232 replace diabetes = 0 if diabetes!=1
2233 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had diabetes
2234 replace diabetes = . if inlist(mfu_cc6a,.,7)

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2235 * Code as missing if the comprehensive assessment is not at baseline or 18-month follow-up (i.e.,
2236 intervals 1-2)
2237 replace diabetes = . if ! inlist(interval,1,2)
2238
2239 * Carryforward observations (diabetes at the current comprehensive assessment or any prior
2240 comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
2241 comprehensive assessments) by participant ID
2239 gen diab18_cons = raw_diab_18 if interval==2
2240 tsset STUDYID interval
2241 bysort STUDYID: carryforward diab18_cons, replace
2242 gen diab36_cons = raw_diab_36 if interval==3
2243 tsset STUDYID interval
2244 bysort STUDYID: carryforward diab36_cons, replace
2245 gen diab54_cons = raw_diab_54 if interval==4
2246 tsset STUDYID interval
2247 bysort STUDYID: carryforward diab54_cons, replace
2248 gen diab72_cons = raw_diab_72 if interval==5
2249 tsset STUDYID interval
2250 bysort STUDYID: carryforward diab72_cons, replace
2251 gen diab90_cons = raw_diab_90 if interval==6
2252 tsset STUDYID interval
2253 bysort STUDYID: carryforward diab90_cons, replace
2254 gen diab108_cons = raw_diab_108 if interval==7
2255 tsset STUDYID interval
2256 bysort STUDYID: carryforward diab108_cons, replace
2257 gen diab144_cons = raw_diab_144 if interval==9
2258 tsset STUDYID interval
2259 bysort STUDYID: carryforward diab144_cons, replace
2260 gen diab162_cons = raw_diab_162 if interval==10
2261 tsset STUDYID interval
2262 bysort STUDYID: carryforward diab162_cons, replace
2263 gen diab180_cons = raw_diab_180 if interval==11
2264 tsset STUDYID interval
2265 bysort STUDYID: carryforward diab180_cons, replace
2266 gen diab198_cons = raw_diab_198 if interval==12
2267 tsset STUDYID interval
2268 bysort STUDYID: carryforward diab198_cons, replace
2269 gen diab216_cons = raw_diab_216 if interval==13
2270 tsset STUDYID interval
2271 bysort STUDYID: carryforward diab216_cons, replace
2272 gen diab234_cons = raw_diab_234 if interval==14
2273 tsset STUDYID interval
2274 bysort STUDYID: carryforward diab234_cons, replace
2275 gen diab252_cons = raw_diab_252 if interval==15
2276 tsset STUDYID interval
2277 bysort STUDYID: carryforward diab252_cons, replace
2278 gen diab270_cons = raw_diab_270 if interval==16
2279 tsset STUDYID interval
2280 bysort STUDYID: carryforward diab270_cons, replace
2281 gen diab288_cons = raw_diab_288 if interval==17
2282 tsset STUDYID interval
2283 bysort STUDYID: carryforward diab288_cons, replace
2284
2285 * For intervals 3-18, code as 1 if the participant has been told by a doctor that they had diabetes,
2286 sugar in their urine, or high blood sugar ("Yes" or "Suspect or possible"), and the participant is
2287 now using medication that they swallow to treat or control their diabetes or the participant is now
2288 using insulin injections, and the participant was coded as NOT having diabetes at all prior
2289 comprehensive assessments (i.e., new condition)
2286 replace diabetes = 1 if diab18_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
2287 interval==3
2287 replace diabetes = 1 if diab36_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
2288 interval==4
2288 replace diabetes = 1 if diab54_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &

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interval==5
2289 * Note that in the original syntax ("P:\projects\PEP\sas\FACE Master pgm\fu90face_3.sas" - not
written by the authors of the current article) for the 90-month follow-up comprehensive assessment
dataset compilation, "diab54_cons" was used in the following line of code rather than "diab72_cons" -
this has been corrected here (the mistake in the comprehensive assessment dataset does not affect the
raw_ccsumbl2 variable)
2290 replace diabetes = 1 if diab72_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==6
2291 replace diabetes = 1 if diab90_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==7
2292 replace diabetes = 1 if diab108_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==9
2293 replace diabetes = 1 if diab144_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==10
2294 replace diabetes = 1 if diab162_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==11
2295 replace diabetes = 1 if diab180_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==12
2296 replace diabetes = 1 if diab198_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==13
2297 replace diabetes = 1 if diab216_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==14
2298 replace diabetes = 1 if diab234_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==15
2299 replace diabetes = 1 if diab252_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==16
2300 replace diabetes = 1 if diab270_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==17
2301 replace diabetes = 1 if diab288_cons==0 & (inlist(mfu_cc6a,1,3) & (mfu_cc6b==1 | mfu_cc6c==1)) &
interval==18
2302
2303 * Code as 0 otherwise
2304 replace diabetes = 0 if diabetes!=1
2305 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had diabetes
2306 replace diabetes = . if inlist(mfu_cc6a,.,7)
2307 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2308 replace diabetes = . if STUDYID==2007 & interval==2
2309 * Compare values with variable from combined comprehensive assessment dataset
2310 gen diabminus = diabetes - raw_diabetes
2311
2312 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had
diabetes at the corresponding comprehensive assessment or the participant was coded as having had
diabetes at any prior comprehensive assessment(s)
2313 * Code as 0 otherwise
2314 gen diab18 = 1 if (diabetes==1 | diab_cons==1) & interval==2
2315 replace diab18 = 0 if diab18!=1 & interval==2
2316 gen diab36 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1) & interval==3
2317 replace diab36 = 0 if diab36!=1 & interval==3
2318 gen diab54 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1) & interval==4
2319 replace diab54 = 0 if diab54!=1 & interval==4
2320 gen diab72 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1) &
interval==5
2321 replace diab72 = 0 if diab72!=1 & interval==5
2322 gen diab90 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1) & interval==6
2323 replace diab90 = 0 if diab90!=1 & interval==6
2324 gen diab108 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1) & interval==7
2325 replace diab108 = 0 if diab108!=1 & interval==7
2326 gen diab144 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1) & interval==9

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2327 replace diab144 = 0 if diab144!=1 & interval==9
2328 gen diab162 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1) & interval==10
2329 replace diab162 = 0 if diab162!=1 & interval==10
2330 gen diab180 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1) & interval==11
2331 replace diab180 = 0 if diab180!=1 & interval==11
2332 gen diab198 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1) & interval==12
2333 replace diab198 = 0 if diab198!=1 & interval==12
2334 gen diab216 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1) & interval==13
2335 replace diab216 = 0 if diab216!=1 & interval==13
2336 gen diab234 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1 | diab216_cons==1) & interval==14
2337 replace diab234 = 0 if diab234!=1 & interval==14
2338 gen diab252 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1 | diab216_cons==1 | diab234_cons==1) & interval==15
2339 replace diab252 = 0 if diab252!=1 & interval==15
2340 gen diab270 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1 | diab216_cons==1 | diab234_cons==1 | diab252_cons==1) & interval==16
2341 replace diab270 = 0 if diab270!=1 & interval==16
2342 gen diab288 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1 | diab216_cons==1 | diab234_cons==1 | diab252_cons==1 | diab270_cons==1) &
interval==17
2343 replace diab288 = 0 if diab288!=1 & interval==17
2344 gen diab306 = 1 if (diabetes==1 | diab_cons==1 | diab18_cons==1 | diab36_cons==1 | diab54_cons==1 |
diab72_cons==1 | diab90_cons==1 | diab108_cons==1 | diab144_cons==1 | diab162_cons==1 | diab180_cons==
1 | diab198_cons==1 | diab216_cons==1 | diab234_cons==1 | diab252_cons==1 | diab270_cons==1 |
diab288_cons==1) & interval==18
2345 replace diab306 = 0 if diab306!=1 & interval==18
2346
2347 *** Hip fracture
2348 * Code as 1 if the participant has been told by a doctor that they had a broken or fractured hip and
had to be hospitalized
2349 gen hipf = 1 if mfu_cc7==1
2350 * Code as 0 if the participant responds "Suspect or possible" or "No"
2351 replace hipf = 0 if inlist(mfu_cc7,2,3)
2352 * Code as 0 if the participant responds "Don't know"
2353 replace hipf = 0 if mfu_cc7==8
2354 * Code as missing if the participant refused to answer the question (or it was left blank) about
having had a hip fracture
2355 replace hipf = . if inlist(mfu_cc7,.,7)
2356 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2357 replace hipf = . if STUDYID==2007 & interval==2
2358 * Compare values with variable from combined comprehensive assessment dataset
2359 gen hfminus = hipf - raw_hipfx
2360
2361 * Carryforward observations (hip fracture at the current comprehensive assessment or any prior
comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
comprehensive assessments) by participant ID
2362 gen hipfx_cons = raw_hipfx if interval==1
2363 tsset STUDYID interval
2364 bysort STUDYID: carryforward hipfx_cons, replace
2365 gen hipfx18_cons = raw_hipfx_18 if interval==2
2366 tsset STUDYID interval

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2367 bysort STUDYID: carryforward hipfx18_cons, replace
2368 gen hipfx36_cons = raw_hipfx_36 if interval==3
2369 tsset STUDYID interval
2370 bysort STUDYID: carryforward hipfx36_cons, replace
2371 gen hipfx54_cons = raw_hipfx_54 if interval==4
2372 tsset STUDYID interval
2373 bysort STUDYID: carryforward hipfx54_cons, replace
2374 gen hipfx72_cons = raw_hipfx_72 if interval==5
2375 tsset STUDYID interval
2376 bysort STUDYID: carryforward hipfx72_cons, replace
2377 gen hipfx90_cons = raw_hipfx_90 if interval==6
2378 tsset STUDYID interval
2379 bysort STUDYID: carryforward hipfx90_cons, replace
2380 gen hipfx108_cons = raw_hipfx_108 if interval==7
2381 tsset STUDYID interval
2382 bysort STUDYID: carryforward hipfx108_cons, replace
2383 gen hipfx144_cons = raw_hipfx_144 if interval==9
2384 tsset STUDYID interval
2385 bysort STUDYID: carryforward hipfx144_cons, replace
2386 gen hipfx162_cons = raw_hipfx_162 if interval==10
2387 tsset STUDYID interval
2388 bysort STUDYID: carryforward hipfx162_cons, replace
2389 gen hipfx180_cons = raw_hipfx_180 if interval==11
2390 tsset STUDYID interval
2391 bysort STUDYID: carryforward hipfx180_cons, replace
2392 gen hipfx198_cons = raw_hipfx_198 if interval==12
2393 tsset STUDYID interval
2394 bysort STUDYID: carryforward hipfx198_cons, replace
2395 gen hipfx216_cons = raw_hipfx_216 if interval==13
2396 tsset STUDYID interval
2397 bysort STUDYID: carryforward hipfx216_cons, replace
2398 gen hipfx234_cons = raw_hipfx_234 if interval==14
2399 tsset STUDYID interval
2400 bysort STUDYID: carryforward hipfx234_cons, replace
2401 gen hipfx252_cons = raw_hipfx_252 if interval==15
2402 tsset STUDYID interval
2403 bysort STUDYID: carryforward hipfx252_cons, replace
2404 gen hipfx270_cons = raw_hipfx_270 if interval==16
2405 tsset STUDYID interval
2406 bysort STUDYID: carryforward hipfx270_cons, replace
2407 gen hipfx288_cons = raw_hipfx_288 if interval==17
2408 tsset STUDYID interval
2409 bysort STUDYID: carryforward hipfx288_cons, replace
2410
2411 * For each comprehensive assessment, code as 1 if the participant was newly coded as having had a hip
2412 * fracture at the corresponding comprehensive assessment or the participant was coded as having had a
2413 * hip fracture at any prior comprehensive assessment(s)
2414 * Code as 0 otherwise
2415 gen hipfx18 = 1 if (hipf==1 | hipfx_cons==1) & interval==2
2416 replace hipfx18 = 0 if hipfx18!=1 & interval==2
2417 gen hipfx36 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1) & interval==3
2418 replace hipfx36 = 0 if hipfx36!=1 & interval==3
2419 gen hipfx54 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1) & interval==4
2420 replace hipfx54 = 0 if hipfx54!=1 & interval==4
2421 gen hipfx72 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1) &
2422 interval==5
2423 replace hipfx72 = 0 if hipfx72!=1 & interval==5
2424 gen hipfx90 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
2425 hipfx72_cons==1) & interval==6
2426 replace hipfx90 = 0 if hipfx90!=1 & interval==6
2427 gen hipfx108 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
2428 hipfx72_cons==1 | hipfx90_cons==1) & interval==7
2429 replace hipfx108 = 0 if hipfx108!=1 & interval==7

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2425 gen hipfx144 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1) & interval==9
2426 replace hipfx144 = 0 if hipfx144!=1 & interval==9
2427 gen hipfx162 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1) & interval==10
2428 replace hipfx162 = 0 if hipfx162!=1 & interval==10
2429 gen hipfx180 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1) & interval
==11
2430 replace hipfx180 = 0 if hipfx180!=1 & interval==11
2431 gen hipfx198 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1) & interval==12
2432 replace hipfx198 = 0 if hipfx198!=1 & interval==12
2433 gen hipfx216 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1) & interval==13
2434 replace hipfx216 = 0 if hipfx216!=1 & interval==13
2435 gen hipfx234 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1 | hipfx216_cons==1) & interval==14
2436 replace hipfx234 = 0 if hipfx234!=1 & interval==14
2437 gen hipfx252 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1 | hipfx216_cons==1 | hipfx234_cons==1) & interval==15
2438 replace hipfx252 = 0 if hipfx252!=1 & interval==15
2439 gen hipfx270 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1 | hipfx216_cons==1 | hipfx234_cons==1 | hipfx252_cons==1) &
interval==16
2440 replace hipfx270 = 0 if hipfx270!=1 & interval==16
2441 gen hipfx288 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1 | hipfx216_cons==1 | hipfx234_cons==1 | hipfx252_cons==1 |
hipfx270_cons==1) & interval==17
2442 replace hipfx288 = 0 if hipfx288!=1 & interval==17
2443 gen hipfx306 = 1 if (hipf==1 | hipfx_cons==1 | hipfx18_cons==1 | hipfx36_cons==1 | hipfx54_cons==1 |
hipfx72_cons==1 | hipfx90_cons==1 | hipfx108_cons==1 | hipfx144_cons==1 | hipfx162_cons==1 |
hipfx180_cons==1 | hipfx198_cons==1 | hipfx216_cons==1 | hipfx234_cons==1 | hipfx252_cons==1 |
hipfx270_cons==1 | hipfx288_cons==1) & interval==18
2444 replace hipfx306 = 0 if hipfx306!=1 & interval==18
2445
2446 *** Arthritis
2447 * For all baseline comprehensive assessments, code as 1 if the participant has seen a doctor
specifically for arthritis or rheumatism and they had pain and/or stiffness in their hands/fingers,
shoulders, knees, hips, and/or back/spine
2448 gen arth = 1 if inlist(mfu_cc8a,1,3) & (mfu_cc8b1==1 | mfu_cc8b2==1 | mfu_cc8b3==1 | mfu_cc8b4==1 |
mfu_cc8b5==1) & interval==1
2449 * Code as 0 if the participant has not seen a doctor specifically for arthritis or rheumatism, or if
the participant has seen a doctor specifically for arthritis or rheumatism but they did not have pain
and/or stiffness in their joints
2450 replace arth = 0 if (mfu_cc8a==2 | (inlist(mfu_cc8a,1,3) & (mfu_cc8b1!=1 & mfu_cc8b2!=1 & mfu_cc8b3!=1
& mfu_cc8b4!=1 & mfu_cc8b5!=1))) & interval==1
2451 * For all follow-up comprehensive assessments, code as 1 if the participant has seen a doctor
specifically for arthritis or rheumatism and they had pain and/or stiffness in their hands/fingers,
shoulders, knees, hips, and/or back/spine
2452 replace arth = 1 if inlist(mfu_cc8a,1,3) & (mfu_cc8b1==1 | mfu_cc8b2==1 | mfu_cc8b3==1 | mfu_cc8b4==1
| mfu_cc8b5==1) & interval!=1
2453 * Code as 0 if the participant has not seen a doctor specifically for arthritis or rheumatism, or if
the participant has seen a doctor specifically for arthritis or rheumatism but they did not have pain
and/or stiffness in their joints
2454 replace arth = 0 if (mfu_cc8a==2 | (inlist(mfu_cc8a,1,3) & (mfu_cc8b1!=1 & mfu_cc8b2!=1 & mfu_cc8b3!=1
& mfu_cc8b4!=1 & mfu_cc8b5!=1))) & interval!=1

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2455 * Code as 0 if the participant responds "Don't know"
2456 replace arth = 0 if mfu_cc8a==8 & interval!=1
2457 * Code as missing if the participant refused to answer the question (or it was left blank) about
having seen a doctor specifically for arthritis or rheumatism
2458 replace arth = . if inlist(mfu_cc8a,.,7) & interval!=1
2459 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2460 replace arth = . if STUDYID==2007 & interval==2
2461 * Compare values with variable from combined comprehensive assessment dataset
2462 gen arminus = arth - raw_arthrit
2463
2464 * Carryforward observations (arthritis at the current comprehensive assessment or any prior
comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
comprehensive assessments) by participant ID
2465 gen arth_cons = raw_arthrit if interval==1
2466 tsset STUDYID interval
2467 bysort STUDYID: carryforward arth_cons, replace
2468 gen arth18_cons = raw_arth_18 if interval==2
2469 tsset STUDYID interval
2470 bysort STUDYID: carryforward arth18_cons, replace
2471 gen arth36_cons = raw_arth_36 if interval==3
2472 tsset STUDYID interval
2473 bysort STUDYID: carryforward arth36_cons, replace
2474 gen arth54_cons = raw_arth_54 if interval==4
2475 tsset STUDYID interval
2476 bysort STUDYID: carryforward arth54_cons, replace
2477 gen arth72_cons = raw_arth_72 if interval==5
2478 tsset STUDYID interval
2479 bysort STUDYID: carryforward arth72_cons, replace
2480 gen arth90_cons = raw_arth_90 if interval==6
2481 tsset STUDYID interval
2482 bysort STUDYID: carryforward arth90_cons, replace
2483 gen arth108_cons = raw_arth_108 if interval==7
2484 tsset STUDYID interval
2485 bysort STUDYID: carryforward arth108_cons, replace
2486 gen arth144_cons = raw_arth_144 if interval==9
2487 tsset STUDYID interval
2488 bysort STUDYID: carryforward arth144_cons, replace
2489 gen arth162_cons = raw_arth_162 if interval==10
2490 tsset STUDYID interval
2491 bysort STUDYID: carryforward arth162_cons, replace
2492 gen arth180_cons = raw_arth_180 if interval==11
2493 tsset STUDYID interval
2494 bysort STUDYID: carryforward arth180_cons, replace
2495 gen arth198_cons = raw_arth_198 if interval==12
2496 tsset STUDYID interval
2497 bysort STUDYID: carryforward arth198_cons, replace
2498 gen arth216_cons = raw_arth_216 if interval==13
2499 tsset STUDYID interval
2500 bysort STUDYID: carryforward arth216_cons, replace
2501 gen arth234_cons = raw_arth_234 if interval==14
2502 tsset STUDYID interval
2503 bysort STUDYID: carryforward arth234_cons, replace
2504 gen arth252_cons = raw_arth_252 if interval==15
2505 tsset STUDYID interval
2506 bysort STUDYID: carryforward arth252_cons, replace
2507 gen arth270_cons = raw_arth_270 if interval==16
2508 tsset STUDYID interval
2509 bysort STUDYID: carryforward arth270_cons, replace
2510 gen arth288_cons = raw_arth_288 if interval==17
2511 tsset STUDYID interval
2512 bysort STUDYID: carryforward arth288_cons, replace
2513

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2514 * For each comprehensive assessment, code as 1 if the participant was newly coded as having arthritis
      at the corresponding comprehensive assessment or the participant was coded as having arthritis at any
      prior comprehensive assessment(s)
2515 * Code as 0 otherwise
2516 gen arth18 = 1 if (arth==1 | arth_cons==1) & interval==2
2517 replace arth18 = 0 if arth18!=1 & interval==2
2518 gen arth36 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1) & interval==3
2519 replace arth36 = 0 if arth36!=1 & interval==3
2520 gen arth54 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1) & interval==4
2521 replace arth54 = 0 if arth54!=1 & interval==4
2522 gen arth72 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1) &
      interval==5
2523 replace arth72 = 0 if arth72!=1 & interval==5
2524 gen arth90 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1) & interval==6
2525 replace arth90 = 0 if arth90!=1 & interval==6
2526 gen arth108 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1) & interval==7
2527 replace arth108 = 0 if arth108!=1 & interval==7
2528 gen arth144 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1) & interval==9
2529 replace arth144 = 0 if arth144!=1 & interval==9
2530 gen arth162 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1) & interval==10
2531 replace arth162 = 0 if arth162!=1 & interval==10
2532 gen arth180 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1) & interval==11
2533 replace arth180 = 0 if arth180!=1 & interval==11
2534 gen arth198 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1) & interval==12
2535 replace arth198 = 0 if arth198!=1 & interval==12
2536 gen arth216 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1) & interval==13
2537 replace arth216 = 0 if arth216!=1 & interval==13
2538 gen arth234 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1 | arth216_cons==1) & interval==14
2539 replace arth234 = 0 if arth234!=1 & interval==14
2540 gen arth252 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1 | arth216_cons==1 | arth234_cons==1) & interval==15
2541 replace arth252 = 0 if arth252!=1 & interval==15
2542 gen arth270 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1 | arth216_cons==1 | arth234_cons==1 | arth252_cons==1) & interval==16
2543 replace arth270 = 0 if arth270!=1 & interval==16
2544 gen arth288 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1 | arth216_cons==1 | arth234_cons==1 | arth252_cons==1 | arth270_cons==1) &
      interval==17
2545 replace arth288 = 0 if arth288!=1 & interval==17
2546 gen arth306 = 1 if (arth==1 | arth_cons==1 | arth18_cons==1 | arth36_cons==1 | arth54_cons==1 |
      arth72_cons==1 | arth90_cons==1 | arth108_cons==1 | arth144_cons==1 | arth162_cons==1 | arth180_cons==
      1 | arth198_cons==1 | arth216_cons==1 | arth234_cons==1 | arth252_cons==1 | arth270_cons==1 |
      arth288_cons==1) & interval==18
2547 replace arth306 = 0 if arth306!=1 & interval==18
2548
2549 *** Chronic lung disease
2550 * For all baseline comprehensive assessments, code as 1 if the participant has been told by a doctor
      that they have chronic lung disease such as chronic bronchitis, COPD, asthma, or emphysema ("Yes"),
      or the participant has been told by a doctor that they possibly have chronic lung disease ("Suspect

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or possible") and the participant's lung disease limits their usual activities such as household
chores
2551 gen copd = 1 if (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1)) & interval==1
2552 * Code as 0 otherwise
2553 replace copd = 0 if copd!=1 & interval==1
2554 * Code as missing if either question was left blank
2555 replace copd = . if mfu_cc9a==. & mfu_cc9b==. & interval==1
2556 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
2557 gen copd_cons = copd if interval==1
2558 tsset STUDYID interval
2559 bysort STUDYID: carryforward copd_cons, replace
2560 * For all 18-month follow-up comprehensive assessments, code as 1 if the participant has been told by
a doctor that they have chronic lung disease such as chronic bronchitis, COPD, asthma, or emphysema
("Yes"), or the participant has been told by a doctor that they possibly have chronic lung disease
("Suspect or possible") and the participant's lung disease limits their usual activities such as
household chores, and the participant was coded as NOT having chronic lung disease at the baseline
comprehensive assessment (i.e., new condition)
2561 replace copd = 1 if (copd_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==2
2562 * Code as 0 otherwise
2563 replace copd = 0 if copd!=1 & interval==2
2564 * Code as missing if the participant refused to answer the question (or it was left blank) about
having chronic lung disease
2565 replace copd = . if inlist(mfu_cc9a,.,7)
2566 * Code as missing if the comprehensive assessment is not at baseline or 18-month follow-up (i.e.,
intervals 1-2)
2567 replace copd = . if ! inlist(interval,1,2)
2568
2569 * Carryforward observations (chronic lung disease at the current comprehensive assessment or any
prior comprehensive assessment) with respect to the time variable "interval" (i.e., to all subsequent
comprehensive assessments) by participant ID
2570 gen lungd18_cons = raw_lungd_18 if interval==2
2571 tsset STUDYID interval
2572 bysort STUDYID: carryforward lungd18_cons, replace
2573 gen lungd36_cons = raw_lungd_36 if interval==3
2574 tsset STUDYID interval
2575 bysort STUDYID: carryforward lungd36_cons, replace
2576 gen lungd54_cons = raw_lungd_54 if interval==4
2577 tsset STUDYID interval
2578 bysort STUDYID: carryforward lungd54_cons, replace
2579 gen lungd72_cons = raw_lungd_72 if interval==5
2580 tsset STUDYID interval
2581 bysort STUDYID: carryforward lungd72_cons, replace
2582 gen lungd90_cons = raw_lungd_90 if interval==6
2583 tsset STUDYID interval
2584 bysort STUDYID: carryforward lungd90_cons, replace
2585 gen lungd108_cons = raw_lungd_108 if interval==7
2586 tsset STUDYID interval
2587 bysort STUDYID: carryforward lungd108_cons, replace
2588 gen lungd144_cons = raw_lungd_144 if interval==9
2589 tsset STUDYID interval
2590 bysort STUDYID: carryforward lungd144_cons, replace
2591 gen lungd162_cons = raw_lungd_162 if interval==10
2592 tsset STUDYID interval
2593 bysort STUDYID: carryforward lungd162_cons, replace
2594 gen lungd180_cons = raw_lungd_180 if interval==11
2595 tsset STUDYID interval
2596 bysort STUDYID: carryforward lungd180_cons, replace
2597 gen lungd198_cons = raw_lungd_198 if interval==12
2598 tsset STUDYID interval
2599 bysort STUDYID: carryforward lungd198_cons, replace
2600 gen lungd216_cons = raw_lungd_216 if interval==13
2601 tsset STUDYID interval

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2602 bysort STUDYID: carryforward lungd216_cons, replace
2603 gen lungd234_cons = raw_lungd_234 if interval==14
2604 tsset STUDYID interval
2605 bysort STUDYID: carryforward lungd234_cons, replace
2606 gen lungd252_cons = raw_lungd_252 if interval==15
2607 tsset STUDYID interval
2608 bysort STUDYID: carryforward lungd252_cons, replace
2609 gen lungd270_cons = raw_lungd_270 if interval==16
2610 tsset STUDYID interval
2611 bysort STUDYID: carryforward lungd270_cons, replace
2612 gen lungd288_cons = raw_lungd_288 if interval==17
2613 tsset STUDYID interval
2614 bysort STUDYID: carryforward lungd288_cons, replace
2615
2616 * For intervals 3-18, code as 1 if the participant has been told by a doctor that they have chronic
lung disease such as chronic bronchitis, COPD, asthma, or emphysema ("Yes"), or the participant has
been told by a doctor that they possibly have chronic lung disease ("Suspect or possible") and the
participant's lung disease limits their usual activities such as household chores, and the
participant was coded as NOT having chronic lung disease at all prior comprehensive assessments
(i.e., new condition)
2617 replace copd = 1 if (lungd18_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==3
2618 replace copd = 1 if (lungd36_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==4
2619 replace copd = 1 if (lungd54_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==5
2620 replace copd = 1 if (lungd72_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==6
2621 replace copd = 1 if (lungd90_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==7
2622 replace copd = 1 if (lungd108_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==9
2623 replace copd = 1 if (lungd144_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==10
2624 replace copd = 1 if (lungd162_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==11
2625 replace copd = 1 if (lungd180_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==12
2626 replace copd = 1 if (lungd198_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==13
2627 replace copd = 1 if (lungd216_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==14
2628 replace copd = 1 if (lungd234_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==15
2629 replace copd = 1 if (lungd252_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==16
2630 replace copd = 1 if (lungd270_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==17
2631 replace copd = 1 if (lungd288_cons==0 & (mfu_cc9a==1 | (mfu_cc9a==3 & mfu_cc9b==1))) & interval==18
2632
2633 * Code as 0 otherwise
2634 replace copd = 0 if copd!=1
2635 * Code as missing if the participant refused to answer the question (or it was left blank) about
having chronic lung disease
2636 replace copd = . if inlist(mfu_cc9a,.,7)
2637 * Generate a new variable replicating the "copd" variable
2638 gen lung_d = copd
2639 * Code as 0 if the participant responds "Don't know"
2640 replace lung_d = 0 if mfu_cc9a==8
2641 * If mfu_cc1a or mfu_cc1b; mfu_cc2; mfu_cc3; mfu_cc4a; mfu_cc5; mfu_cc6a or mfu_cc6b or mfu_cc6c;
mfu_cc7; mfu_cc8a; and mfu_cc9a are coded as "Don't know", treat observation as a missing case
2642 replace lung_d = . if STUDYID==2007 & interval==2
2643 * Compare values with variable from combined comprehensive assessment dataset
2644 gen lungminus = lung_d - raw_lung_d
2645 * If mfu_cc9a is coded as "Suspect or possible" and mfu_cc9b is coded as "Yes", but chronic lung
disease at the previous comprehensive assessment was left blank, code as 1
2646 replace lung_d = 1 if STUDYID==510 & interval==3
2647 * Compare values with variable from combined comprehensive assessment dataset
2648 gen lungminus2 = lung_d - raw_lung_d
2649
2650 * For each comprehensive assessment, code as 1 if the participant was newly coded as having chronic
lung disease at the corresponding comprehensive assessment or the participant was coded as having
chronic lung disease at any prior comprehensive assessment(s)
2651 * Code as 0 otherwise
2652 gen lungd18 = 1 if (lung_d==1 | copd_cons==1) & interval==2
2653 replace lungd18 = 0 if lungd18!=1 & interval==2
2654 gen lungd36 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1) & interval==3

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2655 replace lungd36 = 0 if lungd36!=1 & interval==3
2656 gen lungd54 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1) & interval==4
2657 replace lungd54 = 0 if lungd54!=1 & interval==4
2658 gen lungd72 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1) &
interval==5
2659 replace lungd72 = 0 if lungd72!=1 & interval==5
2660 gen lungd90 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1) & interval==6
2661 replace lungd90 = 0 if lungd90!=1 & interval==6
2662 gen lungd108 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1) & interval==7
2663 replace lungd108 = 0 if lungd108!=1 & interval==7
2664 gen lungd144 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1) & interval==9
2665 replace lungd144 = 0 if lungd144!=1 & interval==9
2666 gen lungd162 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1) & interval==10
2667 replace lungd162 = 0 if lungd162!=1 & interval==10
2668 gen lungd180 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1) & interval
==11
2669 replace lungd180 = 0 if lungd180!=1 & interval==11
2670 gen lungd198 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1) & interval==12
2671 replace lungd198 = 0 if lungd198!=1 & interval==12
2672 gen lungd216 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1) & interval==13
2673 replace lungd216 = 0 if lungd216!=1 & interval==13
2674 gen lungd234 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1 | lungd216_cons==1) & interval==14
2675 replace lungd234 = 0 if lungd234!=1 & interval==14
2676 gen lungd252 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1 | lungd216_cons==1 | lungd234_cons==1) & interval==15
2677 replace lungd252 = 0 if lungd252!=1 & interval==15
2678 gen lungd270 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1 | lungd216_cons==1 | lungd234_cons==1 | lungd252_cons==1) &
interval==16
2679 replace lungd270 = 0 if lungd270!=1 & interval==16
2680 gen lungd288 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1 | lungd216_cons==1 | lungd234_cons==1 | lungd252_cons==1 |
lungd270_cons==1) & interval==17
2681 replace lungd288 = 0 if lungd288!=1 & interval==17
2682 gen lungd306 = 1 if (lung_d==1 | copd_cons==1 | lungd18_cons==1 | lungd36_cons==1 | lungd54_cons==1 |
lungd72_cons==1 | lungd90_cons==1 | lungd108_cons==1 | lungd144_cons==1 | lungd162_cons==1 |
lungd180_cons==1 | lungd198_cons==1 | lungd216_cons==1 | lungd234_cons==1 | lungd252_cons==1 |
lungd270_cons==1 | lungd288_cons==1) & interval==18
2683 replace lungd306 = 0 if lungd306!=1 & interval==18
2684
2685 *** Sum
2686 tab raw_ccsumbl2
2687 * At baseline and each follow-up comprehensive assessment, generate a count of the number of chronic
conditions
2688 gen chronic3 = (hbp18 + mi18 + chf18 + stroke18 + cancer18 + diab18 + hipfx18 + lungd18 + arth18) if
interval==2
2689 replace chronic3 = (hbp36 + mi36 + chf36 + stroke36 + cancer36 + diab36 + hipfx36 + lungd36 + arth36)
if interval==3
2690 replace chronic3 = (hbp54 + mi54 + chf54 + stroke54 + cancer54 + diab54 + hipfx54 + lungd54 + arth54)

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if interval==4
2691 replace chronic3 = (hbp72 + mi72 + chf72 + stroke72 + cancer72 + diab72 + hipfx72 + lungd72 + arth72)
if interval==5
2692 replace chronic3 = (hbp90 + mi90 + chf90 + stroke90 + cancer90 + diab90 + hipfx90 + lungd90 + arth90)
if interval==6
2693 replace chronic3 = (hbp108 + mi108 + chf108 + stroke108 + cancer108 + diab108 + hipfx108 + lungd108 +
arth108) if interval==7
2694 replace chronic3 = (hbp144 + mi144 + chf144 + stroke144 + cancer144 + diab144 + hipfx144 + lungd144 +
arth144) if interval==9
2695 replace chronic3 = (hbp162 + mi162 + chf162 + stroke162 + cancer162 + diab162 + hipfx162 + lungd162 +
arth162) if interval==10
2696 replace chronic3 = (hbp180 + mi180 + chf180 + stroke180 + cancer180 + diab180 + hipfx180 + lungd180 +
arth180) if interval==11
2697 replace chronic3 = (hbp198 + mi198 + chf198 + stroke198 + cancer198 + diab198 + hipfx198 + lungd198 +
arth198) if interval==12
2698 replace chronic3 = (hbp216 + mi216 + chf216 + stroke216 + cancer216 + diab216 + hipfx216 + lungd216 +
arth216) if interval==13
2699 replace chronic3 = (hbp234 + mi234 + chf234 + stroke234 + cancer234 + diab234 + hipfx234 + lungd234 +
arth234) if interval==14
2700 replace chronic3 = (hbp252 + mi252 + chf252 + stroke252 + cancer252 + diab252 + hipfx252 + lungd252 +
arth252) if interval==15
2701 replace chronic3 = (hbp270 + mi270 + chf270 + stroke270 + cancer270 + diab270 + hipfx270 + lungd270 +
arth270) if interval==16
2702 replace chronic3 = (hbp288 + mi288 + chf288 + stroke288 + cancer288 + diab288 + hipfx288 + lungd288 +
arth288) if interval==17
2703 replace chronic3 = (hbp306 + mi306 + chf306 + stroke306 + cancer306 + diab306 + hipfx306 + lungd306 +
arth306) if interval==18
2704 replace chronic3 = (highb1 + heart + chf + stroke + cancer + diabetes + hipf + arth + lung_d) if
interval==1
2705 * Compare values with variable from combined comprehensive assessment dataset
2706 gen ch3minus = chronic3 - raw_ccsumbl2
2707
2708 * Lifestyle risk factors
2709 ** Alcohol consumption (time-constant)
2710 * Code items 2 to 5 as missing if the participant responded "Refused" or "Don't know"
2711 * Code items 2 to 5 as 0 if the question was "Not applicable"
2712 gen alc2_n = mfu_alc2
2713 replace alc2_n = . if inlist(alc2_n,7,8)
2714 replace alc2_n = 0 if inlist(alc2_n,9)
2715 gen alc3_n = mfu_alc3
2716 replace alc3_n = . if inlist(alc3_n,7,8)
2717 replace alc3_n = 0 if inlist(alc3_n,9)
2718 gen alc4_n = mfu_alc4
2719 replace alc4_n = . if inlist(alc4_n,7,8)
2720 replace alc4_n = 0 if inlist(alc4_n,9)
2721 gen alc5_n = mfu_alc5
2722 replace alc5_n = . if inlist(alc5_n,7,8)
2723 replace alc5_n = 0 if inlist(alc5_n,9)
2724 * Sum items
2725 gen alc_sum = mfu_alc1+alc2_n+alc3_n+alc4_n+alc5_n
2726 * Code as 0 (lower risk) if the participant had a score of 4 or less
2727 gen alc_bi = 0 if inlist(alc_sum,0,1,2,3,4)
2728 * Code as 1 (higher risk) if the participant had a score of 5 or more
2729 replace alc_bi = 1 if alc_sum >=5 & alc_sum!=.
2730 * Compare values with variable from combined comprehensive assessment dataset
2731 gen alcminus = raw_alc_b1 - alc_bi
2732 * Generate a new variable duplicating the alcohol consumption variable at baseline
2733 gen alc_cons = alc_bi if interval==1
2734 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
2735 tsset STUDYID interval
2736 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID
2737 bysort STUDYID: carryforward alc_cons, replace

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2738
2739 ** Smoking (time-varying)
2740 * Code as 0 (lower risk) if the participant reported having never smoked
2741 gen smoking = 0 if mfu_smo1==1
2742 * Code as 1 (higher risk) if the participant reported being a former or current smoker
2743 replace smoking = 1 if inlist(mfu_smo1,2,3)
2744 * Hard codes documented in "P:\projects\PEP\sas\FACE Master pgm\fu18face_3.sas",
    "P:\projects\PEP\sas\FACE Master pgm\fu36face_3.sas", and "P:\projects\PEP\sas\FACE Master
    pgm\fu54face_3.sas"
2745 replace smoking = 1 if inlist(STUDYID,1402,2570) & interval==2
2746 replace smoking = 0 if STUDYID==1888 & interval==3
2747 replace smoking = 1 if inlist(STUDYID,1402,1646,2946,3170) & interval==3
2748 replace smoking = 0 if inlist(STUDYID,291,331,1223,1888,2076) & interval==4
2749 replace smoking = 1 if inlist(STUDYID,1252,1259,2853,2946,3027,3102) & interval==4
2750 * Compare values with variable from combined comprehensive assessment dataset
2751 gen smokminus = smoking - raw_smoker_b1
2752
2753 ** Low physical activity (time-varying)
2754 * Score by following the instructions for the Physical Activity Scale for the Elderly (Washburn RA,
    Smith KW, Jette AM, Janney CA. The physical activity scale for the elderly (PASE): development and
    evaluation. J Clin Epidemiol. 1993;46:153-62.)
2755 gen mpase1w = 0 if mfu_mpase1==1
2756 replace mpase1w = 0.11 if mfu_mpase1==2 & mfu_mpase1a==1
2757 replace mpase1w = 0.32 if mfu_mpase1==2 & mfu_mpase1a==2
2758 replace mpase1w = 0.64 if mfu_mpase1==2 & mfu_mpase1a==3
2759 replace mpase1w = 1.07 if mfu_mpase1==2 & mfu_mpase1a==4
2760 replace mpase1w = 0.25 if mfu_mpase1==3 & mfu_mpase1a==1
2761 replace mpase1w = 0.75 if mfu_mpase1==3 & mfu_mpase1a==2
2762 replace mpase1w = 1.50 if mfu_mpase1==3 & mfu_mpase1a==3
2763 replace mpase1w = 2.50 if mfu_mpase1==3 & mfu_mpase1a==4
2764 replace mpase1w = 0.43 if mfu_mpase1==4 & mfu_mpase1a==1
2765 replace mpase1w = 1.29 if mfu_mpase1==4 & mfu_mpase1a==2
2766 replace mpase1w = 2.57 if mfu_mpase1==4 & mfu_mpase1a==3
2767 replace mpase1w = 4.29 if mfu_mpase1==4 & mfu_mpase1a==4
2768 replace mpase1w = mpase1w*20
2769 * Hard code documented in "P:\projects\PEP\sas\FACE Master pgm\baseface_3.sas"
2770 replace mfu_mpase1a = 1 if STUDYID==1884 & interval==1
2771 replace mpase1w = 0.11*20 if STUDYID==1884 & interval==1
2772 * Compare values with variable from combined comprehensive assessment dataset
2773 gen mpa1minus = mpase1w - raw_mpase1w
2774
2775 gen mpase2w = 0 if mfu_mpase2==1
2776 replace mpase2w = 0.11 if mfu_mpase2==2 & mfu_mpase2a==1
2777 replace mpase2w = 0.32 if mfu_mpase2==2 & mfu_mpase2a==2
2778 replace mpase2w = 0.64 if mfu_mpase2==2 & mfu_mpase2a==3
2779 replace mpase2w = 1.07 if mfu_mpase2==2 & mfu_mpase2a==4
2780 replace mpase2w = 0.25 if mfu_mpase2==3 & mfu_mpase2a==1
2781 replace mpase2w = 0.75 if mfu_mpase2==3 & mfu_mpase2a==2
2782 replace mpase2w = 1.50 if mfu_mpase2==3 & mfu_mpase2a==3
2783 replace mpase2w = 2.50 if mfu_mpase2==3 & mfu_mpase2a==4
2784 replace mpase2w = 0.43 if mfu_mpase2==4 & mfu_mpase2a==1
2785 replace mpase2w = 1.29 if mfu_mpase2==4 & mfu_mpase2a==2
2786 replace mpase2w = 2.57 if mfu_mpase2==4 & mfu_mpase2a==3
2787 replace mpase2w = 4.29 if mfu_mpase2==4 & mfu_mpase2a==4
2788 replace mpase2w = mpase2w*21
2789 * Compare values with variable from combined comprehensive assessment dataset
2790 gen mpa2minus = mpase2w - raw_mpase2w
2791
2792 gen mpase3w = 0 if mfu_mpase3==1
2793 replace mpase3w = 0.11 if mfu_mpase3==2 & mfu_mpase3a==1
2794 replace mpase3w = 0.32 if mfu_mpase3==2 & mfu_mpase3a==2
2795 replace mpase3w = 0.64 if mfu_mpase3==2 & mfu_mpase3a==3
2796 replace mpase3w = 1.07 if mfu_mpase3==2 & mfu_mpase3a==4

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2797 replace mpase3w = 0.25 if mfu_mpase3==3 & mfu_mpase3a==1
2798 replace mpase3w = 0.75 if mfu_mpase3==3 & mfu_mpase3a==2
2799 replace mpase3w = 1.50 if mfu_mpase3==3 & mfu_mpase3a==3
2800 replace mpase3w = 2.50 if mfu_mpase3==3 & mfu_mpase3a==4
2801 replace mpase3w = 0.43 if mfu_mpase3==4 & mfu_mpase3a==1
2802 replace mpase3w = 1.29 if mfu_mpase3==4 & mfu_mpase3a==2
2803 replace mpase3w = 2.57 if mfu_mpase3==4 & mfu_mpase3a==3
2804 replace mpase3w = 4.29 if mfu_mpase3==4 & mfu_mpase3a==4
2805 replace mpase3w = mpase3w*23
2806 * Compare values with variable from combined comprehensive assessment dataset
2807 gen mpa3minus = mpase3w - raw_mpase3w
2808
2809 gen mpase4w = 0 if mfu_mpase4==1
2810 replace mpase4w = 0.11 if mfu_mpase4==2 & mfu_mpase4a==1
2811 replace mpase4w = 0.32 if mfu_mpase4==2 & mfu_mpase4a==2
2812 replace mpase4w = 0.64 if mfu_mpase4==2 & mfu_mpase4a==3
2813 replace mpase4w = 1.07 if mfu_mpase4==2 & mfu_mpase4a==4
2814 replace mpase4w = 0.25 if mfu_mpase4==3 & mfu_mpase4a==1
2815 replace mpase4w = 0.75 if mfu_mpase4==3 & mfu_mpase4a==2
2816 replace mpase4w = 1.50 if mfu_mpase4==3 & mfu_mpase4a==3
2817 replace mpase4w = 2.50 if mfu_mpase4==3 & mfu_mpase4a==4
2818 replace mpase4w = 0.43 if mfu_mpase4==4 & mfu_mpase4a==1
2819 replace mpase4w = 1.29 if mfu_mpase4==4 & mfu_mpase4a==2
2820 replace mpase4w = 2.57 if mfu_mpase4==4 & mfu_mpase4a==3
2821 replace mpase4w = 4.29 if mfu_mpase4==4 & mfu_mpase4a==4
2822 replace mpase4w = mpase4w*23
2823 * Compare values with variable from combined comprehensive assessment dataset
2824 gen mpa4minus = mpase4w - raw_mpase4w
2825
2826 gen mpase5w = 0 if mfu_mpase5==1
2827 replace mpase5w = 0.11 if mfu_mpase5==2 & mfu_mpase5a==1
2828 replace mpase5w = 0.32 if mfu_mpase5==2 & mfu_mpase5a==2
2829 replace mpase5w = 0.64 if mfu_mpase5==2 & mfu_mpase5a==3
2830 replace mpase5w = 1.07 if mfu_mpase5==2 & mfu_mpase5a==4
2831 replace mpase5w = 0.25 if mfu_mpase5==3 & mfu_mpase5a==1
2832 replace mpase5w = 0.75 if mfu_mpase5==3 & mfu_mpase5a==2
2833 replace mpase5w = 1.50 if mfu_mpase5==3 & mfu_mpase5a==3
2834 replace mpase5w = 2.50 if mfu_mpase5==3 & mfu_mpase5a==4
2835 replace mpase5w = 0.43 if mfu_mpase5==4 & mfu_mpase5a==1
2836 replace mpase5w = 1.29 if mfu_mpase5==4 & mfu_mpase5a==2
2837 replace mpase5w = 2.57 if mfu_mpase5==4 & mfu_mpase5a==3
2838 replace mpase5w = 4.29 if mfu_mpase5==4 & mfu_mpase5a==4
2839 replace mpase5w = mpase5w*30
2840 * Compare values with variable from combined comprehensive assessment dataset
2841 gen mpa5minus = mpase5w - raw_mpase5w
2842
2843 * Recode as 0 if the participant responded "No", "Refused", or "Don't know"
2844 recode mfu_mpase6 mfu_mpase7 mfu_mpase8a mfu_mpase8b mfu_mpase8c mfu_mpase8d (2=0)(7=0)(8=0), prefix(
new)
2845 gen mpase6w = newmfu_mpase6*25
2846 * Compare values with variable from combined comprehensive assessment dataset
2847 gen mpa6minus = mpase6w - raw_mpase6w
2848 gen mpase7w = newmfu_mpase7*25
2849 * Compare values with variable from combined comprehensive assessment dataset
2850 gen mpa7minus = mpase7w - raw_mpase7w
2851 gen mpase8aw = newmfu_mpase8a*30
2852 * Compare values with variable from combined comprehensive assessment dataset
2853 gen mpa8aminus = mpase8aw - raw_mpase8aw
2854 gen mpase8bw = newmfu_mpase8b*36
2855 * Compare values with variable from combined comprehensive assessment dataset
2856 gen mpa8bminus = mpase8bw - raw_mpase8bw
2857 gen mpase8cw = newmfu_mpase8c*20
2858 * Compare values with variable from combined comprehensive assessment dataset

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2859 gen mpa8cminus = mpase8cw - raw_mpase8cw
2860 gen mpase8dw = newmfu_mpase8d*35
2861 * Compare values with variable from combined comprehensive assessment dataset
2862 gen mpa8dminus = mpase8dw - raw_mpase8dw
2863
2864 * Generate a new variable duplicating the variable "How many hours per week did you work for pay
and/or as a volunteer?"
2865 gen newmpase9b = mfu_mpase9b
2866 * For intervals 2 to 16, code as 0 if the participant responded "No", "Refused", or "Don't know" to
the question "During the past 7 days, did you work for pay or as a volunteer?" and responded
"Refused", "Don't know", or "Not applicable" to the question "How many hours per week did you work
for pay and/or as a volunteer?"
2867 replace newmpase9b = 0 if inlist(mfu_mpase9a,2,7,8) & inlist(newmpase9b,99,97,98) & inlist(interval,2,
3,4,5,6,7,9,10,11,12,13,14,15,16)
2868 * Hard code documented in "P:\projects\PEP\Evelyne\Analysis\fu270face_3.sas"
2869 replace newmpase9b = 0 if STUDYID==129 & interval==16
2870 * For intervals 17 and 18, code as 0 if the participant responded "No", "Refused", or "Don't know" to
the question "During the past 7 days, did you work for pay or as a volunteer?" and responded
"Refused", "Don't know", or "Not applicable" to the question "How many hours per week did you work
for pay and/or as a volunteer?" (or left it blank)
2871 replace newmpase9b = 0 if inlist(mfu_mpase9a,2,7,8) & inlist(newmpase9b,.,99,97,98) & inlist(interval,
17,18)
2872 * For interval 1, code as missing
2873 replace newmpase9b = . if interval==1
2874 * Generate a new variable duplicating the variable "How many hours per week did you work for pay
and/or as a volunteer?" at baseline
2875 gen newmpase9c = mfu_mpase9c if interval==1
2876 * Code as 0 if the participant responded "No", "Refused", or "Don't know" to the question "During the
past 7 days, did you work for pay or as a volunteer?" and responded "Refused", "Don't know", or "Not
applicable" to the question "How many hours per week did you work for pay and/or as a volunteer?"
2877 replace newmpase9c = 0 if inlist(mfu_mpase9a,2,7,8) & inlist(newmpase9c,99,97,98)
2878 gen mpase9w = (newmpase9b/7)*21 if interval!=1
2879 replace mpase9w = (newmpase9c/7)*21 if interval==1
2880 * Compare values with variable from combined comprehensive assessment dataset
2881 gen mpa9minus = mpase9w - raw_mpase9cw
2882
2883 * Sum items
2884 gen mpase_new = mpase1w + mpase2w + mpase3w + mpase4w + mpase5w + mpase6w + mpase7w + mpase8aw +
mpase8bw + mpase8cw + mpase8dw + mpase9w
2885 * Compare values (in terms of precision) with variable from combined comprehensive assessment dataset
2886 gen mpminus = mpase_new - raw_mpaseb1
2887 gen mpase_final = raw_mpaseb1
2888 * Code as 0 (lower risk) if the participant scored 64 or more and was a man
2889 gen low_pa = 0 if mpase_final>=64 & mpase_final!=. & sex_cons==0
2890 * Code as 0 (lower risk) if the participant scored 52 or more and was a woman
2891 replace low_pa = 0 if mpase_final>=52 & mpase_final!=. & sex_cons==1
2892 * Code as 1 (higher risk) if the participant scored less than 64 and was a man
2893 replace low_pa = 1 if mpase_final<64 & sex_cons==0
2894 * Code as 1 (higher risk) if the participant scored less than 52 and was a woman
2895 replace low_pa = 1 if mpase_final<52 & sex_cons==1
2896
2897 ** Obesity (time-varying)
2898 * Convert height in feet to inches
2899 gen ht_inch = mfu_HT1*12
2900 replace ht_inch = ht_inch+mfu_HT2
2901 * Convert height in inches to meters
2902 gen meters = ht_inch/39.37
2903 * Generate a new variable duplicating the height (in meters) variable at baseline
2904 gen hei_cons = meters if interval==1
2905 * Declare a panel dataset with participant ID "STUDYID" and time variable "interval"
2906 tsset STUDYID interval
2907 * Carryforward observations with respect to the time variable "interval" (i.e., from baseline to the
follow-up comprehensive assessments) by participant ID

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2908 bysort STUDYID: carryforward hei_cons, replace
2909 * Convert weight in pounds to kilograms
2910 gen weight = mfu_wt1a/2.205
2911 * Observations coded as "Refused" or "Don't know" were treated as missing cases
2912 replace weight = . if inlist(mfu_wt1a,997,998)
2913 * Calculate body mass index (BMI) using the standard formula
2914 gen BMI_raw = weight/(hei_cons*hei_cons)
2915 * Compare values with variable from combined comprehensive assessment dataset
2916 gen bmiminus = BMI_raw - raw_BMI_b1
2917 * Code as 0 (lower risk) if the participant had a BMI below 30 kg/m2
2918 gen obesity = 0 if BMI_raw <30.00
2919 * Code as 1 (higher risk) if the participant had a BMI above or equal to 30 kg/m2
2920 replace obesity = 1 if BMI_raw >=30.00 & BMI_raw!=.
2921 * Compare values with variable from combined comprehensive assessment dataset
2922 gen obminus = BMI_ge30 - obesity
2923
2924 * Disability at the comprehensive assessments
2925 replace mfu_pf1a = 1 if interval==1
2926 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to bathe (wash and dry your whole body)?"
2927 gen bathing = 0 if mfu_pf1a==1
2928 * Code as 1 if the participant responds "Help" or "Unable to do"
2929 replace bathing = 1 if inlist(mfu_pf1a,2,3)
2930 * Code as 0 if the participant responds "No help"
2931 replace bathing = 0 if mfu_bath3==1 & interval>=3
2932 * Code as 1 if the participant responds "Help" or "Unable to do"
2933 replace bathing = 1 if inlist(mfu_bath3,2,3) & interval>=3
2934
2935 replace mfu_pf2a = 1 if interval==1
2936 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to dress (like putting on a shirt or shoes, buttoning and zippering)?"
2937 gen dressing = 0 if mfu_pf2a==1
2938 * Code as 1 if the participant responds "Help" or "Unable to do"
2939 replace dressing = 1 if inlist(mfu_pf2a,2,3)
2940
2941 replace mfu_pf3a = 1 if interval==1
2942 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to get in and out of a chair?"
2943 gen transferring = 0 if mfu_pf3a==1
2944 * Code as 1 if the participant responds "Help" or "Unable to do"
2945 replace transferring = 1 if inlist(mfu_pf3a,2,3)
2946
2947 replace mfu_pf4a = 1 if interval==1
2948 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to walk around your house or apartment?"
2949 gen waround = 0 if mfu_pf4a==1
2950 * Code as 1 if the participant responds "Help" or "Unable to do"
2951 replace waround = 1 if inlist(mfu_pf4a,2,3)
2952
2953 ** Sum
2954 * Generate a count of the number of essential activities of daily living (ADL) requiring personal
assistance
2955 gen adldep = bathing + dressing + transferring + waround
2956 * Code as 0 if the participant is fully independent
2957 gen adldep_bi = 0 if adldep==0
2958 * Code as 1 if the participant is dependent in one or more ADL
2959 replace adldep_bi = 1 if inlist(adldep,1,2,3,4)
2960
2961 * Overwrite dataset, by replacing the previously saved file
2962 save comprehensive_analyses.dta, replace
2963 clear
2964
2965 * Import monthly data

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2966 import sas using "P:\projects\PEP\Master\pepmoonth_le123124\pepmoonth_le123124.sas7bdat"
2967 * Keep data pertaining only to decedents
2968 keep if Died==1
2969 * Keep data from the first monthly interview
2970 keep if intbloc==1
2971 * Rename StudyID variable to ensure consistency with updated master dataset
2972 rename StudyID STUDYID
2973 * Keep relevant variables
2974 keep STUDYID DDate
2975 * Rename DDate variable to differentiate it from the less recent DDate variable in the updated master
dataset
2976 rename DDate DDate2
2977 * Change working directory
2978 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
2979 * Save corrected date of death dataset with a new name
2980 save deaths.dta
2981 clear
2982
2983 * Use updated master dataset
2984 use comprehensive_analyses.dta
2985 * Merge updated master dataset with corrected date of death dataset
2986 merge m:1 STUDYID using deaths.dta, generate(merge_death)
2987 * Count total number of participants and observations
2988 unique STUDYID
2989 * 754 individuals, 5168 observations
2990 * Overwrite dataset, by replacing the previously saved file
2991 save comprehensive_analyses.dta, replace
2992
2993 * Drop interval 8 (no comprehensive assessments took place at the 126-month follow-up owing to lack
of funds)
2994 drop if interval==8
2995 * Count total number of participants and observations
2996 unique STUDYID
2997 * 754 individuals, 4867 observations
2998 * Amount of missing data
2999 sum age_fix
3000 tab edu_cons_bi if interval==1
3001 tab inc_bimi if interval==1
3002 tab adi_cons if interval==1
3003 tab alc_cons if interval==1
3004 tab smoking
3005 tab low_pa
3006 tab obesity
3007 tab sex_cons if interval==1
3008 tab eth_bi if interval==1
3009 sum raw_ccsumbl2
3010 clear
3011
3012 * Import monthly data
3013 import sas using "P:\projects\PEP\Master\pepmoonth_le123124\pepmoonth_le123124.sas7bdat"
3014 * Rename StudyID variable to ensure consistency with updated master dataset
3015 rename StudyID STUDYID
3016 * Generate a variable that assigns the observation number (i.e., 1 for first data collection
timepoint, 2 for second data collection timepoint) to each row by participant ID
3017 bysort STUDYID(intbloc): gen obsnr_m = _n
3018 * Generate a variable that assigns the number of total observations to each row of data for a given
participant
3019 bysort STUDYID: gen obscount_m = _N
3020
3021 ** Disability during the monthly interviews
3022 gen act1n = ACT1
3023 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to bathe (wash and dry your whole body)?"

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3024 replace act1n = 0 if act1n==1
3025 * Code as 1 if the participant responds "Help" or "Unable to do"
3026 replace act1n = 1 if inlist(act1n,2,3)
3027 * Observations coded as "Refused" or "Don't know" were treated as missing cases
3028 replace act1n = . if inlist(act1n,7,8)
3029 gen act2n = ACT2
3030 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to walk around your home or apartment?"
3031 replace act2n = 0 if act2n==1
3032 * Code as 1 if the participant responds "Help" or "Unable to do"
3033 replace act2n = 1 if inlist(act2n,2,3)
3034 * Observations coded as "Refused" or "Don't know" were treated as missing cases
3035 replace act2n = . if inlist(act2n,7,8)
3036 gen act3n = ACT3
3037 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to dress (like putting on a shirt or shoes, buttoning and zipping)?"
3038 replace act3n = 0 if act3n==1
3039 * Code as 1 if the participant responds "Help" or "Unable to do"
3040 replace act3n = 1 if inlist(act3n,2,3)
3041 * Observations coded as "Refused" or "Don't know" were treated as missing cases
3042 replace act3n = . if inlist(act3n,7,8)
3043 gen act4n = ACT4
3044 * Code as 0 if the participant responds "No help" to the question "At the present time, do you need
help from another person to get in and out of a chair?"
3045 replace act4n = 0 if act4n==1
3046 * Code as 1 if the participant responds "Help" or "Unable to do"
3047 replace act4n = 1 if inlist(act4n,2,3)
3048 * Observations coded as "Refused" or "Don't know" were treated as missing cases
3049 replace act4n = . if inlist(act4n,7,8)
3050
3051 *** Sum
3052 * Generate a count of the number of ADL requiring personal assistance
3053 gen essential = act1n + act2n + act3n + act4n
3054 * Create a binary variable (fully independent versus dependent in one or more ADL)
3055 gen essential_bi = essential
3056 replace essential_bi = 1 if inlist(essential_bi,2,3,4)
3057 * Count total number of participants and observations
3058 unique STUDYID
3059 * 754 individuals, 87704 observations
3060 * Save dataset with a new name
3061 save monthly.dta
3062 clear
3063
3064 * Use updated master dataset
3065 use comprehensive_analyses.dta
3066 * Keep interview dates of comprehensive assessments and other relevant variables
3067 keep STUDYID interval intdateF2F
3068 * Reshape data into wide format for observations identified by participant ID and add interval as an
identifying time period
3069 reshape wide intdateF2F, i(STUDYID) j(interval)
3070 * Generate a new variable called interval and assign the number 1 to each observation
3071 generate interval = 1
3072 * Save interview dates dataset with a new name
3073 save wide_intdateF2F.dta
3074 clear
3075
3076 * Use monthly dataset
3077 use monthly.dta
3078 * Merge monthly dataset with interview dates dataset
3079 merge m:1 STUDYID using wide_intdateF2F.dta, generate(merge_date)
3080 * Keep if matched
3081 keep if merge_date==3
3082 * Count total number of participants and observations

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3083 unique STUDYID
3084 * 754 individuals, 87704 observations
3085 * Overwrite dataset, by replacing the previously saved file
3086 save monthly.dta, replace
3087 clear
3088
3089 * Import temporary interview dates
3090 import sas using "P:\projects\PEP\sas\OTHER Source pgm\f2fdates_306.sas7bdat"
3091 * Change working directory
3092 cd "P:\projects\PEP\OMalkowski\Active life expectancy"
3093 * Rename StudyID variable to ensure consistency with monthly dataset
3094 rename StudyID STUDYID
3095 * Save temporary interview dates dataset with a new name
3096 save wide_tmpdates.dta
3097 clear
3098
3099 * Use monthly dataset
3100 use monthly.dta
3101 * Merge monthly dataset with temporary interview dates dataset
3102 merge m:1 STUDYID using wide_tmpdates.dta, generate(merge_date2)
3103 * Keep if matched
3104 keep if merge_date2==3
3105 * Count total number of participants and observations
3106 unique STUDYID
3107 * 754 individuals, 87704 observations
3108 * Overwrite dataset, by replacing the previously saved file
3109 save monthly.dta, replace
3110
3111 * Generate a variable that allows data from the comprehensive assessments to be merged with and
3112   carried forward to all monthly observations between consecutive comprehensive assessments
3113 gen interval4 = 1 if intdate>=tmpdate0 & intdate<tmpdate18
3114 replace interval4 = 2 if intdate>=tmpdate18 & intdate<tmpdate36
3115 replace interval4 = 3 if intdate>=tmpdate36 & intdate<tmpdate54
3116 replace interval4 = 4 if intdate>=tmpdate54 & intdate<tmpdate72
3117 replace interval4 = 5 if intdate>=tmpdate72 & intdate<tmpdate90
3118 replace interval4 = 6 if intdate>=tmpdate90 & intdate<tmpdate108
3119 replace interval4 = 7 if intdate>=tmpdate108 & intdate<tmpdate126
3120 replace interval4 = 8 if intdate>=tmpdate126 & intdate<tmpdate144
3121 replace interval4 = 9 if intdate>=tmpdate144 & intdate<tmpdate162
3122 replace interval4 = 10 if intdate>=tmpdate162 & intdate<tmpdate180
3123 replace interval4 = 11 if intdate>=tmpdate180 & intdate<tmpdate198
3124 replace interval4 = 12 if intdate>=tmpdate198 & intdate<tmpdate216
3125 replace interval4 = 13 if intdate>=tmpdate216 & intdate<tmpdate234
3126 replace interval4 = 14 if intdate>=tmpdate234 & intdate<tmpdate252
3127 replace interval4 = 15 if intdate>=tmpdate252 & intdate<tmpdate270
3128 replace interval4 = 16 if intdate>=tmpdate270 & intdate<tmpdate288
3129 replace interval4 = 17 if intdate>=tmpdate288 & intdate<tmpdate306
3130 replace interval4 = 18 if intdate>=tmpdate306 & intdate<.
3131 * Two participants died before their first monthly interview, but took part in the baseline
3132   comprehensive assessment
3133 replace interval4 = 1 if STUDYID==1523 & intbloc==1
3134 replace interval4 = 1 if STUDYID==2409 & intbloc==1
3135 * Overwrite dataset, by replacing the previously saved file
3136 save monthly.dta, replace
3137 clear
3138
3139 * Use updated master dataset
3140 use comprehensive_analyses.dta
3141 * Carryforward observations with respect to the time variable "interval" (i.e., from interval 7 to
3142   the follow-up comprehensive assessments) by participant ID, and replace values at interval 8 with
3143   values from interval 7
3144 gen raw_ccsumbl2_cons = raw_ccsumbl2 if interval==7
3145 tsset STUDYID interval

```

```

3142 bysort STUDYID: carryforward raw_ccsumbl2_cons, replace
3143 replace raw_ccsumbl2 = raw_ccsumbl2_cons if interval==8
3144 gen smoking_cons = smoking if interval==7
3145 tsset STUDYID interval
3146 bysort STUDYID: carryforward smoking_cons, replace
3147 replace smoking = smoking_cons if interval==8
3148 gen low_pa_cons = low_pa if interval==7
3149 tsset STUDYID interval
3150 bysort STUDYID: carryforward low_pa_cons, replace
3151 replace low_pa = low_pa_cons if interval==8
3152 gen obesity_cons = obesity if interval==7
3153 tsset STUDYID interval
3154 bysort STUDYID: carryforward obesity_cons, replace
3155 replace obesity = obesity_cons if interval==8
3156 gen adldep_bi_cons = adldep_bi if interval==7
3157 tsset STUDYID interval
3158 bysort STUDYID: carryforward adldep_bi_cons, replace
3159 replace adldep_bi = adldep_bi_cons if interval==8
3160 * Overwrite dataset, by replacing the previously saved file
3161 save comprehensive_analyses.dta, replace
3162 * Keep necessary variables from the comprehensive assessments
3163 keep STUDYID interval edu_cons_bi inc_bimi adi_cons age_fix age_fix_dec sex_cons eth_bi raw_ccsumbl2
alc_cons smoking low_pa obesity adldep_bi DDate2 BDates
3164 * Rename variable to ensure consistency with the monthly dataset
3165 rename interval interval4
3166 * Save comprehensive assessment variables dataset with a new name
3167 save comprehensive_merge.dta
3168 clear
3169
3170 * Use monthly dataset
3171 use monthly.dta
3172 * Merge monthly dataset with comprehensive assessment variables dataset
3173 merge m:1 STUDYID interval4 using comprehensive_merge.dta, generate(merge_f2f)
3174 * Keep all observations from the monthly dataset
3175 keep if inlist(merge_f2f,1,3)
3176 * Count total number of participants and observations
3177 unique STUDYID
3178 * 754 individuals, 87704 observations
3179 * Save dataset with a new name
3180 save monthly_analysis.dta
3181
3182 * Time to death from monthly interview (i.e., date of death minus date of monthly interview rounded
to the nearest integer) in months and years
3183 gen timetodeath = (DDate-intdate)/(365/12)
3184 replace timetodeath = round(timetodeath,1)
3185 gen timetodeath_yr = timetodeath/12
3186
3187 * Time to death from baseline (i.e., date of death minus date of baseline comprehensive assessment)
in months and years
3188 gen timetodeath_bl2 = (DDate-intdateF2F1)/(365/12)
3189 gen timetodeath_bl2_yr = timetodeath_bl2/12
3190 * Calculate the median time to death from baseline
3191 sum timetodeath_bl2_yr if intbloc==1,d
3192 * Perform a Shapiro-Wilk test for normality
3193 swilk timetodeath_bl2_yr if intbloc==1
3194 * Calculate the interquartile range
3195 tabstat timetodeath_bl2_yr if intbloc==1, statistics(iqr)
3196
3197 * Time to refusal from baseline (i.e., date of refusal minus date of baseline comprehensive
assessment) in months
3198 gen timebaserefusal = (daterefuse-intdateF2F1)/(365/12)
3199 * Keep one observation per participant and limit observations to permanent refusers
3200 replace timebaserefusal = .if intbloc!=1

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```

3201 replace timebaserefusal = . if permrefuser==0
3202 * Calculate the median time to refusal from baseline
3203 sum timebaserefusal,d
3204 * Perform a Shapiro-Wilk test for normality
3205 swilk timebaserefusal
3206 * Calculate the interquartile range
3207 tabstat timebaserefusal, statistics(iqr)
3208 * Overwrite dataset, by replacing the previously saved file
3209 save monthly_analysis.dta, replace
3210
3211 * Keep necessary variables for discrete-time multistate modeling analyses
3212 keep STUDYID interval edu_cons_bi inc_bimi adi_cons age_fix age_fix_dec sex_cons eth_bi raw_ccsumbl2
alc_cons smoking low_pa obesity adldep_bi DDate2 BDates intdate intbloc interval4 DDate Died
essential_bi timetodeath timetodeath_yr timetodeath_bl2 timetodeath_bl2_yr timebaserefusal permrefuser
daterefuse
3213 * Assigns a number in ascending order to each row of observations
3214 gen ascnr = _n
3215 * Generate a variable that assigns the observation number (i.e., 1 for first data collection
timepoint, 2 for second data collection timepoint) to each row by participant ID
3216 bysort STUDYID(intdate): gen obsnr = _n
3217 * Generate a variable that assigns the number of total observations to each row of data for a given
participant
3218 bysort STUDYID: gen obscount = _N
3219 * Count total number of participants and observations
3220 unique STUDYID
3221 * 754 individuals, 87704 observations
3222 * Save analysis dataset with a new name
3223 save multistate.dta
3224 clear
3225
3226 * Use updated master dataset
3227 use comprehensive_analyses.dta
3228 * Keep baseline data only
3229 keep if interval==1
3230 * Keep relevant variables
3231 keep STUDYID interval edu_cons_bi inc_bimi adi_cons age_fix age_fix_dec sex_cons eth_bi raw_ccsumbl2
alc_cons smoking low_pa obesity adldep_bi DDate2 BDates intdateF2F
3232 * Generate a new variable called intbloc and assign the number 0 to each observation (as the baseline
comprehensive assessment precedes the first block of monthly interviews)
3233 gen intbloc=0
3234 * Generate a new variable with the same name as the binary ADL variable in the monthly dataset and
assign the number 0 to each observation (as all participants were nondisabled at baseline)
3235 gen essential_bi=0
3236 * Save baseline ADL dataset with a new name
3237 save base_adl.dta
3238 clear
3239
3240 * Use analysis dataset
3241 use multistate.dta
3242 * Merge monthly dataset with baseline ADL dataset
3243 merge 1:1 STUDYID intbloc using base_adl.dta, generate(merge_adl)
3244 * Count total number of participants and observations
3245 unique STUDYID
3246 * 754 individuals, 88458 observations
3247 * Sort by participant ID and intbloc (lowest to highest)
3248 sort STUDYID intbloc
3249 * Declare a panel dataset with participant ID "STUDYID" and time variable "intbloc"
3250 tsset STUDYID intbloc
3251 * Generate a completely balanced dataset (i.e., all participants have a row for each intbloc)
3252 tsfill, full
3253 * Count total number of participants and observations
3254 unique STUDYID
3255 * 754 individuals, 240526 observations

```

```

3256 * Sort by participant ID and intbloc (lowest to highest)
3257 sort STUDYID intbloc
3258
3259 * Generate a variable duplicating the binary ADL variable
3260 gen state = essential_bi
3261 * Generate a variable duplicating any variable assessed during the comprehensive assessments with no
missing data (in this case biological sex)
3262 gen constant = sex_cons
3263 * Generate a variable duplicating the binary, time-constant alcohol consumption variable to provide a
reference for which participants were missing data on alcohol consumption at baseline
3264 gen constant2 = alc_cons
3265
3266 * Create lagged versions of the following variables by participant ID
3267 by STUDYID: gen obsnr_l = L.obsnr
3268 by STUDYID: gen obscount_l = L.obscount
3269 by STUDYID: gen Died_l = L.Died
3270 by STUDYID: gen timetodeath_l = L.timetodeath
3271 by STUDYID: gen permrefuser_l = L.permrefuser
3272
3273 * Code as dead if (1) the current state variable (fully independent versus dependent in one or more
ADL) is blank, (2) the previous observation was the participant's final monthly interview, (3) the
participant has since died, (4) and the participant's date of death is within two months of their
final monthly interview (or the baseline comprehensive assessment for those who died before their
first monthly interview)
3274 replace state = 2 if state==. & obsnr_l==obscount_l & Died_l==1 & inlist(timetodeath_l,0,1)
3275 tab permrefuser_l if state==. & obsnr_l==obscount_l & Died_l==1
3276 tab timetodeath_l if state==. & obsnr_l==obscount_l & Died_l==1 & permrefuser_l==1
3277 tab timetodeath_l if state==. & obsnr_l==obscount_l & Died_l==1 & permrefuser_l==0
3278 replace state = 2 if state==. & obsnr_l==obscount_l & Died_l==1 & inlist(timetodeath_l,2)
3279 tab STUDYID if state==. & obsnr_l==obscount_l & Died_l==1
3280 tab STUDYID if state==. & obsnr_l==obscount_l & Died_l==1 & permrefuser_l==0
3281 tab STUDYID if state==1 & obsnr_l==obscount_l & Died_l==1
3282 replace state = 2 if state==. & obsnr_l==186 & obscount_l==187 & L.Died==1 & inlist(timetodeath_l,0,1)
3283 replace state = 2 if intbloc==1 & STUDYID==2409
3284 replace state = 2 if intbloc==1 & STUDYID==1523
3285
3286 * Generate a new variable duplicating the date of birth variable at the time of each participant's
first monthly interview
3287 gen BDates_cons = BDates if obsnr==1
3288 * Declare a panel dataset with participant ID "STUDYID" and time variable "intbloc"
3289 tsset STUDYID intbloc
3290 * Carryforward observations with respect to the time variable "intbloc" (i.e., from the first monthly
interview to the follow-up monthly interviews) by participant ID
3291 bysort STUDYID: carryforward BDates_cons, replace
3292 * Replace any missing values with the participant's date of birth
3293 replace BDates_cons=BDates if BDates_cons==.
3294
3295 * Generate a new variable duplicating the date of death variable at the time of each participant's
first monthly interview
3296 gen DDates_cons = DDate2 if obsnr==1
3297 * Declare a panel dataset with participant ID "STUDYID" and time variable "intbloc"
3298 tsset STUDYID intbloc
3299 * Carryforward observations with respect to the time variable "intbloc" (i.e., from the first monthly
interview to the follow-up monthly interviews) by participant ID
3300 bysort STUDYID: carryforward DDates_cons, replace
3301 * Replace any missing values with the participant's date of death
3302 replace DDates_cons=DDate2 if DDates_cons==.
3303
3304 * Age (months)
3305 format BDates_cons %td
3306 format intdate %td
3307 gen age_months_dec = datediff_frac(BDates_cons,intdate,"month")
3308 gen age_months = floor(age_months_dec)

```

```

3309 replace age_fix_dec = age_fix_dec*12
3310 replace age_fix_dec = floor(age_fix_dec)
3311 replace age_months = age_fix_dec if intbloc==0
3312 format DDates_cons %td
3313 gen age_months_death = datediff_frac(BDates_cons,DDates_cons,"month")
3314 gen age_months_death_f1 = floor(age_months_death)
3315 replace age_months = age_months_death_f1 if state==2
3316
3317 * Sort by participant ID and intbloc (lowest to highest)
3318 sort STUDYID intbloc
3319 * Generate a new age in months variable (i.e., calculated at baseline and incremented by one month at
each subsequent telephone interview) to be used in the discrete-time multistate modeling analyses, as
we assume that longitudinal data are observed in regular intervals
3320 by STUDYID: gen newage = age_months[1] + _n-1
3321
3322 * Create a new variable replacing the numeric state values with character strings
3323 gen state2 = "Nondisabled" if state==0
3324 replace state2 = "Disabled" if state==1
3325 replace state2 = "Dead" if state==2
3326
3327 * For the following time-constant variables, replace any missing values with the corresponding
(time-constant) value
3328 bysort STUDYID(intbloc): replace edu_cons_bi = edu_cons_bi[_n-1] if missing(edu_cons_bi)
3329 bysort STUDYID(intbloc): replace inc_bimi = inc_bimi[_n-1] if missing(inc_bimi)
3330 bysort STUDYID(intbloc): replace adi_cons = adi_cons[_n-1] if missing(adi_cons)
3331 bysort STUDYID(intbloc): replace sex_cons = sex_cons[_n-1] if missing(sex_cons)
3332 bysort STUDYID(intbloc): replace eth_bi = eth_bi[_n-1] if missing(eth_bi)
3333 bysort STUDYID(intbloc): replace alc_cons = alc_cons[_n-1] if missing(alc_cons)
3334
3335 * Sort by participant ID and intbloc (lowest to highest)
3336 sort STUDYID intbloc
3337 * Generate a new variable duplicating the smoking variable
3338 by STUDYID: gen last_nm_smok = smoking
3339 * For a given participant, replace with the value from the previous row of data (from the same
participant) if smoking data were missing for the current row
3340 by STUDYID: replace last_nm_smok = last_nm_smok[_n-1] if missing(last_nm_smok)
3341 * Replace with a missing value if smoking data were missing at the corresponding comprehensive
assessment (i.e., at the comprehensive assessment preceding the respective monthly interview)
3342 replace last_nm_smok = . if constant!=. & smoking==.
3343
3344 * Sort by participant ID and intbloc (lowest to highest)
3345 sort STUDYID intbloc
3346 * Generate a new variable duplicating the low physical activity variable
3347 by STUDYID: gen last_nm_pa = low_pa
3348 * For a given participant, replace with the value from the previous row of data (from the same
participant) if low physical activity data were missing for the current row
3349 by STUDYID: replace last_nm_pa = last_nm_pa[_n-1] if missing(last_nm_pa)
3350 * Replace with a missing value if low physical activity data were missing at the corresponding
comprehensive assessment (i.e., at the comprehensive assessment preceding the respective monthly
interview)
3351 replace last_nm_pa = . if constant!=. & low_pa==.
3352
3353 * Sort by participant ID and intbloc (lowest to highest)
3354 sort STUDYID intbloc
3355 * Generate a new variable duplicating the obesity variable
3356 by STUDYID: gen last_nm_ob = obesity
3357 * For a given participant, replace with the value from the previous row of data (from the same
participant) if obesity data were missing for the current row
3358 by STUDYID: replace last_nm_ob = last_nm_ob[_n-1] if missing(last_nm_ob)
3359 * Replace with a missing value if obesity data were missing at the corresponding comprehensive
assessment (i.e., at the comprehensive assessment preceding the respective monthly interview)
3360 replace last_nm_ob = . if constant!=. & obesity==.
3361

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```

3362 * Sort by participant ID and intbloc (lowest to highest)
3363 sort STUDYID intbloc
3364 * Generate a new variable duplicating the number of chronic conditions variable
3365 by STUDYID: gen last_nm_cc = raw_ccsumbl2
3366 * For a given participant, replace with the value from the previous row of data (from the same
participant) if data on the number of chronic conditions were missing for the current row
3367 by STUDYID: replace last_nm_cc = last_nm_cc[_n-1] if missing(last_nm_cc)
3368 * Replace with a missing value if data on the number of chronic conditions were missing at the
corresponding comprehensive assessment (i.e., at the comprehensive assessment preceding the
respective monthly interview)
3369 replace last_nm_cc = . if constant!=. & raw_ccsumbl2==.
3370
3371 * Sort by participant ID and intbloc (lowest to highest)
3372 sort STUDYID intbloc
3373 * Generate a new variable duplicating the age in years variable
3374 by STUDYID: gen last_nm_age = age_fix
3375 * For a given participant, replace with the value from the previous row of data (from the same
participant) if age in years data were missing for the current row
3376 by STUDYID: replace last_nm_age = last_nm_age[_n-1] if missing(last_nm_age)
3377 * Replace with a missing value if age in years data were missing at the corresponding comprehensive
assessment (i.e., at the comprehensive assessment preceding the respective monthly interview)
3378 replace last_nm_age = . if constant!=. & age_fix==.
3379
3380 * Compare the age in months variables
3381 gen ageminus = newage-age_months
3382 tab ageminus if state!=.
3383 tab ageminus if state!=. & L.state!=.
3384
3385 * Calculate the median time (and interquartile range) between monthly interviews in days
3386 gen interview = intdate
3387 replace interview = intdateF2F if intbloc==0
3388 replace interview = DDate if inlist(STUDYID,1523,2409) & intbloc==1
3389 bysort STUDYID(intbloc): gen days_diff = interview-interview[_n-1]
3390 sum days_diff,d
3391 tabstat days_diff, statistics(iqr)
3392 * Save dataset with a new name
3393 save intermediate.dta
3394
3395 * Keep relevant variables
3396 keep STUDYID intbloc age_months newage state last_nm_age last_nm_cc last_nm_ob last_nm_pa last_nm_smok
alc_cons eth_bi sex_cons adi_cons inc_bimi edu_cons_bi state2 permrefuser
3397 * Sort by participant ID and intbloc (lowest to highest)
3398 sort STUDYID intbloc
3399 * Increment the age in months variable by one month for each subsequent row containing missing data
3400 local missing_count = 1
3401 while `missing_count' > 0 {
3402     replace age_months = age_months[_n-1] + 1 if missing(age_months)
3403     count if missing(age_months)
3404     local missing_count = r(N)
3405 }
3406 * Generate time-constant versions of the following time-varying variables for use in the
baseline-only discrete-time multistate modeling analyses
3407 gen cc_cons = last_nm_cc if intbloc==0
3408 tsset STUDYID intbloc
3409 bysort STUDYID: carryforward cc_cons, replace
3410 gen sm_cons = last_nm_smok if intbloc==0
3411 tsset STUDYID intbloc
3412 bysort STUDYID: carryforward sm_cons, replace
3413 gen pa_cons = last_nm_pa if intbloc==0
3414 tsset STUDYID intbloc
3415 bysort STUDYID: carryforward pa_cons, replace
3416 gen ob_cons = last_nm_ob if intbloc==0
3417 tsset STUDYID intbloc

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3418 bysort STUDYID: carryforward ob_cons, replace
3419 * Save dataset with a new name
3420 save intermediate2.dta
3421
3422 * Amount of missing data
3423 gen has_val_alc = !missing(alc_cons)
3424 bysort STUDYID: egen ever_had_alc = max(has_val_alc)
3425 list STUDYID if ever_had_alc==0
3426 gen has_val_smo = !missing(last_nm_smo)
3427 bysort STUDYID: egen ever_had_smo = max(has_val_smo)
3428 list STUDYID if ever_had_smo==0
3429 gen has_val_pa = !missing(last_nm_pa)
3430 bysort STUDYID: egen ever_had_pa = max(has_val_pa)
3431 list STUDYID if ever_had_pa==0
3432 gen has_val_ob = !missing(last_nm_ob)
3433 bysort STUDYID: egen ever_had_ob = max(has_val_ob)
3434 list STUDYID if ever_had_ob==0
3435
3436 * Total sample size after transitions starting or ending in a missing value are dropped
3437 unique STUDYID if L.state!=. & state!=.
3438 * Total sample size after transitions with incomplete data on any of the lifestyle risk factors (for
the adjusted analyses) are dropped
3439 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons!=. & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.edu_cons_bi!=.
3440 * Sample size for adjusted discrete-time multistate modeling analyses among men
3441 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.edu_cons_bi!=.
3442 * 266 individuals, 27943 observations
3443 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.inc_bimi!=.
3444 * 266 individuals, 27943 observations
3445 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.adi_cons!=.
3446 * 266 individuals, 27943 observations
3447
3448 * Sample size for adjusted discrete-time multistate modeling analyses among women
3449 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.edu_cons_bi!=.
3450 * 486 individuals, 58307 observations
3451 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.inc_bimi!=.
3452 * 486 individuals, 58307 observations
3453 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.last_nm_smo!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.adi_cons!=.
3454 * 486 individuals, 58307 observations
3455
3456 * Sample size for baseline-only discrete-time multistate modeling analyses among men
3457 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.edu_cons_bi!=.
3458 * 266 individuals, 28092 observations
3459 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.inc_bimi!=.
3460 * 266 individuals, 28092 observations
3461 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==0 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.adi_cons!=.
3462 * 266 individuals, 28092 observations
3463
3464 * Sample size for baseline-only discrete-time multistate modeling analyses among women
3465 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.edu_cons_bi!=.
3466 * 486 individuals, 59221 observations
3467 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.inc_bimi!=.

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```

3468 * 486 individuals, 59221 observations
3469 unique STUDYID if L.state!=. & state!=. & L.newage!=. & L.sex_cons==1 & L.eth_bi!=. & L.alc_cons!=. &
L.sm_cons!=. & L.pa_cons!=. & L.ob_cons!=. & L.cc_cons!=. & L.adi_cons!=.
3470 * 486 individuals, 59221 observations
3471
3472 * Compare the age in months variables
3473 gen ageminus = age_months-newage
3474 tab ageminus if L.state!=. & state!=. & L.newage!=. & L.sex_cons!=. & L.eth_bi!=. & L.alc_cons!=. & L.
last_nm_smok!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.adi_cons!=.
3475 sort STUDYID intbloc
3476 gen gap = D.age_months
3477 tab gap if L.state!=. & state!=. & L.newage!=. & L.sex_cons!=. & L.eth_bi!=. & L.alc_cons!=. & L.
last_nm_smok!=. & L.last_nm_pa!=. & L.last_nm_ob!=. & L.last_nm_cc!=. & L.adi_cons!=.
3478 * Overwrite dataset, by replacing the previously saved file
3479 save intermediate2.dta, replace
3480
3481 * Baseline characteristics of participants by sex and/or indicator of SES (Table 1)
3482 keep if intbloc==0
3483 bysort sex_cons: tab eth_bi
3484 bysort sex_cons: tab alc_cons
3485 bysort sex_cons: tab last_nm_smok
3486 bysort sex_cons: tab last_nm_pa
3487 bysort sex_cons: tab last_nm_ob
3488 bysort sex_cons: sum last_nm_cc
3489 bysort sex_cons: sum last_nm_age
3490
3491 bysort sex_cons: tab edu_cons_bi sex_cons
3492 bysort sex_cons: tab edu_cons_bi eth_bi
3493 bysort sex_cons: tab edu_cons_bi alc_cons
3494 bysort sex_cons: tab edu_cons_bi last_nm_smok
3495 bysort sex_cons: tab edu_cons_bi last_nm_pa
3496 bysort sex_cons: tab edu_cons_bi last_nm_ob
3497
3498 bysort sex_cons: tab inc_bimi sex_cons
3499 bysort sex_cons: tab inc_bimi eth_bi
3500 bysort sex_cons: tab inc_bimi alc_cons
3501 bysort sex_cons: tab inc_bimi last_nm_smok
3502 bysort sex_cons: tab inc_bimi last_nm_pa
3503 bysort sex_cons: tab inc_bimi last_nm_ob
3504
3505 bysort sex_cons: tab adi_cons sex_cons
3506 bysort sex_cons: tab adi_cons eth_bi
3507 bysort sex_cons: tab adi_cons alc_cons
3508 bysort sex_cons: tab adi_cons last_nm_smok
3509 bysort sex_cons: tab adi_cons last_nm_pa
3510 bysort sex_cons: tab adi_cons last_nm_ob
3511
3512 bysort sex_cons: sum last_nm_cc if edu_cons_bi==0
3513 bysort sex_cons: sum last_nm_cc if edu_cons_bi==1
3514 bysort sex_cons: sum last_nm_cc if inc_bimi==0
3515 bysort sex_cons: sum last_nm_cc if inc_bimi==1
3516 bysort sex_cons: sum last_nm_cc if adi_cons==0
3517 bysort sex_cons: sum last_nm_cc if adi_cons==1
3518
3519 bysort sex_cons: sum last_nm_age if edu_cons_bi==0
3520 bysort sex_cons: sum last_nm_age if edu_cons_bi==1
3521 bysort sex_cons: sum last_nm_age if inc_bimi==0
3522 bysort sex_cons: sum last_nm_age if inc_bimi==1
3523 bysort sex_cons: sum last_nm_age if adi_cons==0
3524 bysort sex_cons: sum last_nm_age if adi_cons==1
3525 clear
3526
3527 use intermediate2.dta

```

```

3528  * Baseline characteristics of participants who remained in, versus withdrew from, the study
3529  (Supplementary Table S2)
3529  bysort STUDYID(intbloc): replace permrefuser = permrefuser[_n+1] if permrefuser==.
3530  keep if intbloc==0
3531  tab permrefuser sex_cons, chi2
3532  tab permrefuser eth_bi, chi2
3533  tab permrefuser eth_bi, exact
3534  tab permrefuser edu_cons_bi, chi2
3535  tab permrefuser inc_bimi, chi2
3536  tab permrefuser adi_cons, chi2
3537  tab permrefuser alc_cons, chi2
3538  tab permrefuser last_nm_smok, chi2
3539  tab permrefuser last_nm_pa, chi2
3540  tab permrefuser last_nm_ob, chi2
3541
3542  swilk last_nm_age if permrefuser==0
3543  swilk last_nm_age if permrefuser==1
3544  swilk last_nm_cc if permrefuser==0
3545  swilk last_nm_cc if permrefuser==1
3546  bysort permrefuser: sum last_nm_age
3547  bysort permrefuser: sum last_nm_cc
3548  ttest last_nm_age, by(permrefuser) welch
3549  ttest last_nm_cc, by(permrefuser) welch
3550  clear
3551
3552  * Baseline characteristics of participants by sex and/or indicator of SES (Table 1), complete
3553  breakdown of race or ethnicity
3553  * Use updated master dataset
3554  use comprehensive_analyses.dta
3555  keep if interval==1
3556  bysort sex_cons: tab eth_cons
3557  bysort sex_cons: tab edu_cons_bi eth_cons
3558  bysort sex_cons: tab inc_bimi eth_cons
3559  bysort sex_cons: tab adi_cons eth_cons
3560  clear
3561
3562  * Run discrete-time multistate modeling code from relevant R scripts in RStudio

```